nature portfolio

Peer Review File

Machine learning-driven circulating metabolomics of retinal nerve fibre layer to characterize risks of mortality and cardiometabolic diseases

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
NCOMMS Review 555255 Machine-learning Enhanced Plasma Metabolomics of Retinal Nerve Fiber Layer to Characterize Risks of Mortality and Cardiometabolic Diseases: AMulti-Cohort Study

General Comments: The objective of this well designed and clearly written manuscript is to elucidate relationships between metabolites associated with RNFLT and cardiovascular metabolic diseases (CMD). The manuscript provides a comprehensive assessment of both metabolites associated with macula RNFLT and CMD in distinct ethnic populations of varying genetic susceptibility. Their results suggest that there is a relationship between RNFLT and CMD metabolic fingerprints and that including RNFLT associated metabolic information into existing CMD models significantly improved their discriminatory power of the models. One concern is that the metabolic fingerprints were identified based on their association with RNFLT – which is affected by many ocular diseases; adjustment for these ocular conditions (glaucoma for example) is encouraged and mention of this issue in the discussion section of the manuscript. Although the authors used external datasets, the population was generally of White and Chinese ethnicity.

Specific Comments:
Abstract: This concluding sentence in the abstract may be overstating the results; the authors should consider revising a bit. “Our integrated approach, validated across cohorts of diverse ethnic backgrounds and genetic risks, highlights the possibility that RNFLT metabolic fingerprints as a share basis underlying retinal–cardiometabolic connections and as early indicators that advance equitable cardiometabolic disease management, positioning the retina as a valuable window into cardiometabolic health.”
Methods:
1) RNFLT metabolic states are based on associations with RNFLT. However numerous eye diseases can also affect RNFLT. Were these eye diseases considered in the models?
2) More detail is needed in describing the OCT methods used:
a. For the UKB dataset, RNFLT was measured from macula volume scans centered on the fovea
i. Which specific scans were acquired in the independent cohort and how was RNFLT measured?

3) The authors are commended for using several population groups in their analysis. The datasets include mainly Chinese and White individuals. The lack of representation of Black individuals should be mentioned in the discussion as a limitation to this study.
Results
4) Results / tables β and HR should indicate the specific amount of change related to the coefficient of interest; “per 1-SD increment” as well as the value of the 1-SD increment should be included in the tables.
5) It should be clearly stated in tables and figures which datasets the results represent.

Reviewer #2

(Remarks to the Author)
The manuscript titled "Machine-learning Enhanced Plasma Metabolomics of Retinal Nerve Fiber Layer to Characterize Risks of Mortality and Cardiometabolic Diseases: A Multi-Cohort Study" investigates the relationship between retinal nerve fiber layer thickness (RNFLT) and cardiometabolic diseases (CMDs) using advanced metabolomics and machine learning techniques. The study identifies RNFLT metabolic fingerprints as biomarkers of CMD risk and integrates them into predictive models, demonstrating improved risk prediction across multiple CMD outcomes. These findings are validated in both the UK Biobank and an independent, ethnically distinct cohort, highlighting the potential for RNFLT as a non-invasive tool for systemic health profiling and CMD management.

The study has several strengths. It leverages large, well-characterized cohorts with advanced imaging and metabolomics data, providing robust validation across populations. The integration of machine learning significantly enhances the predictive power of traditional models, showcasing the utility of RNFLT as a bridge between ophthalmology and cardiometabolic health. The study also highlights key use cases, including personalized risk prediction, early detection of CMDs, and potential applications in addressing health inequities. The framework paves the way for translational applications, particularly in integrating retinal imaging into routine primary care for systemic health assessments.

Suggestions for Improvement
1) The study's biggest limitation is lack of reproducibility. Without data and source code, the value of this study is significantly diminished and evaluation of the methodology is impossible. It's understandable that UKBB data can only be accessed through the established protocols. The validation data appears to be not accessible at all. At the very least, a working source code and simulated data should be made available.

2) Given the class imbalance in CMD outcomes, including Area Under the Precision-Recall Curve (AUPRC) as a complementary metric (and associated baselines including a completely random/no-skill predictor) is critical.

3) How should the magnitude of the improvement in predictive power (compared to baseline) be interpreted from an application/clinical standpoint? Are these improvements clinically relevant? A simulation study of benefits from such an improvement would be very valuable.

4) Future steps are focused on making the model more accurate and analyzing other diseases. But what is next for this particular model? An RCT? A clinical dashboard / risk predictor? I am struggling with understanding the algorithm's real-world potential.

5) [minor] Please Elaborate on the criteria used to select retinal scans and how imaging quality issues were addressed to minimize bias.

Reviewer #3

(Remarks to the Author)
This manuscript details a study of metabolomics from UK Biobank to find associations between metabolites and retinal nerve fiber layer thickness and then see if these metabolites are associated with incident cardiometabolic diseases (RNFLT; N=7824 with OCT and metabolomics; N=86000 for just metabolomics to look at incident cardiometabolic diseases). This is based on the premise that RNFLT may have implications for cardiometabolic diseases. They find: (1) 26 metabolites were associated with RNFLT after adjustment for multiple comparisons; (2) majority of metabolites were also associated with incident cardiometabolic diseases; (2) metabolites mediate a significant portion of RNFLT-CMD association (up to 35.1%); (3) metabolite “fingerprints” added to clinical models of risk for many of the cardiometabolic diseases as evidenced by a statistically (and often clinically) meaningful increase in the c-statistic; (4) they do analyses of “health disparities” looking at sex- and social determinants of health stratified models which suggest that the metabolite “fingerprint” improves risk prediction even more so in the group in whom the clinical model performs less well; (5) they do similar analyses in a more ethnically diverse cohort (Guangzhou Diabetes Eye Study) with a more comprehensive metabolomics platform and show that most RNFLT associated metabolites are also associated with incident CVD and mortality.

Overall, the concept of linking ocular findings with biology through circulating analytes/omics is of interest and the link to incident cardiometabolic diseases in that context is interesting. The strengths of the study are the good use of the UKB dataset, the incorporation of genetic risk to show that the metabolite signature adds to clinical risk prediction even in high genetic risk, and the inclusion of an external validation cohort (GDES).

Enthusiasm for the manuscript is significantly tempered due to the following major concerns. Firstly, the impact of the manuscript is significantly decreased due to an already published manuscript (PMID: 39581638, including the senior author of this manuscript) detailing these associations; the fact that a large proportion of the metabolites converge on HDL cholesterol which is a known risk factor; and relatedly, that the clinical implications of this metabolite score are unclear given that the clinical models that were used are not robust and the analyses were not adjusted for HDL cholesterol; and that many of the models other than DM have a very small increase in the C-statistic (0.02) for most outcomes. Secondly, there are validity concerns as detailed below, the primary of which is the lack of accounting for confounders in the original analyses, thus any link between RNFLT metabolites and incident CMD more likely reflects residual confounding related to the presence of CMD risk factors (i.e. pre-DM), as opposed to a direct link between the metabolite and the incident CMD (relatedly, the mediation analyses do not appear to account for confounders).

Major issues:
(1) Link between RNFLT is tenuous since all metabolites are associated with CMD (and they don't show that RNFLT is associated with incident CMD); this likely reflects confounders that are not accounted for in the models (HTN, BP, individual prevalent CMD, HgA1c, HDL).
(2) Clinical models need to include more granular data like HgA1C that may reflect residual confounding; also need to include traditional cholesterol parameters including HDL); Framingham risk score not a very comprehensive risk score (need to compare to the clinical model from the discovery dataset)
(3) How was prevalent CVD/DM addressed?
(4) Not clear that Cox models and sex/SDoH stratified models are adjusted for covariates?
(5) The majority of the increase in c-statistics are minimal (0.02) except for DM when compared to more robust models like FHS, so not clinically meaningful, but it's not even clear from Fig 2 if they are statistically significant changes?
(6) Not clear how metabolic "fingerprints" were created; would devote more of main text methods to this rather than regurgitating UKB methods for metabolomics and RNFLT measurements that have already been published
(7) How does this study compare/differ with a previously conducted similar study under the same UKB proposal number (PMID: 39581638)? That study does not appear to be referenced in this study?
(8) The biological interpretation in discussion is too speculative; also need to provide more details and context around individual metabolites and talk about prior literature
(9) Nightingale platform is not comprehensive
(10) Need to adjust for robust risk factors in all analyses (not just RNFLT associations) including in mediation analyses

More minor issues:
*Language in abstract is too expansive and not specific enough
*Throughout, need to be more specific about the metabolite score (i.e. shouldn't call them fingerprints…fingerprints don't change over a lifetime like metabolites do)
*spell out all acronyms in main text first time they are used
*should use "European ancestry" instead of "Caucasian ethnicity" (as the latter was not specifically addressed in UKB); shouldn't use "diabetic patients" (instead should use "patients with diabetes")
*data use statement only covers UKB, not GDES
*fig 2: all Y-axes need to be same range of values
*Table 2 should be a supplementary table
**need to adjust for multiple testing for incident CMD including the number of tests at the level of number of CMD tested

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have extensively revised the original submission and have adequately addressed the reviewer comments. The numerous supplementary tables provide critical analyses that strengthen their conclusions.

Reviewer #2

(Remarks to the Author)
The authors have done a reasonable job addressing my comments. However, I am concerned that the new Supplementary Table S13 demonstrates very limited improvements compared to baseline. In fact, in 5 out of 6 diseases listed the baseline performance is within the confidence interval of the proposed models. Even if "significant" I am not sure how meaningful this is in the real world.

Reviewer #3

(Remarks to the Author)
This manuscript details an analysis in UKB and the GDES cohort to evaluate association between metabolites (NMR Nightingale in UKB; more comprehensive MS metabolomics in GDES) and retinal nerve fiber layer thickness, and to determine if metabolites mediate the association between RNFL with incident cardiometabolic events. They find HDL parameters (more granular than composite HDL, as measured by NMR) are independently and incrementally associated with RNFL; mediate the association between RNFL with incident cardiometabolic diseases; and are themselves associated with incident CMD events.

The authors attempted to address this reviewer's concerns but impact and interpretation concerns remain. Specifically:
(1) I appreciate the careful inclusion of potential confounders across all the models, and that they continue to demonstrate statistically significant increases in c-statistic, newly added AUPRC, and in the mediation analyses. These appear to be well done, however, these results do not appear to be included as primary analyses but instead are buried as supplemental tables as sensitivity analyses. This will make it very confusing for the reader and as such, these fully adjusted models should be the primary analyses reported throughout;
(2) I continue to disagree with the interpretation of the robustness of the increase in c-statistics (and the newly included AUPRC)…while it may be true that they are comparable to other papers reporting new findings, that does not mean that they

have clinical utility. The main increase is really around DM and not particularly robust for hard incident CVD outcomes. The addition of the NRI is, however, a major strength of the paper, but again highlight the results primarily around DM and not incident CVD. Further, the clinical models for incident DM discrimination should be compared to DM prediction models (not CVD prediction);
(3) Relatedly, the authors argue that the paper is about metabolic mechanisms linking RNFL and CMD but most of the paper is devoted to assessing clinical utility, and not really digging into these mechanisms (in fact the abstract doesn't even mention that the metabolite states are reporting primarily on HDL subclasses)
(4) Thank you for clarifying the difference between this manuscript and your prior manuscript published in the Br J. Ophtho, with the primary difference between using OCT techniques in this paper. Having "split" the ocular imaging analyses into separate papers also reduces impact of this paper, and further, there is no comparison of the two papers to provide context to the reader.

Version 2:

Reviewer comments:

Reviewer #2

(Remarks to the Author)
The authors have adequately responded to my comments. Refocusing the manuscript on AUPRC as the primary metric has made this a stronger article.

Reviewer #3

(Remarks to the Author)
The authors have addressed many of the previous concerns including having fully adjusted models as part of the primary manuscript, adding more mechanistic insights including pathway analyses, and clarifying differences between this and their prior manuscripts. However, throughout, for at least two reviewers, they double down on the importance of the clinical interpretation of the increase in c-statistic and AUPRC, noting the magnitude of the increase the AUPRC. The addition of the FRS-DM score as the clinical model was vital (not sure why any CVD model was used for T2DM at all), but while they state that that was a good thing because it led to a comparable effect to CVD, it actually significantly decreased the % change in AUPRC to only 9%. To further frame the clinical relevance of the change in models (which really is what the results are focused on despite the argument that this was a mechanistic manuscript), the authors should at the minimum provide the NRI data as primary tables, make sure they include the FRS-DM score as a model for comparison in the NRI (currently not included in Supplement 15), and place these results in careful context with a softening of the interpretations throughout about clinical utility (the NRIs are modest at best).

Responds to the reviewers' comments

We would like to express our sincere gratitude for all your invaluable comments and constructive suggestions. We are truly honoured to have the opportunity to address all your concerns and incorporate your feedback into our manuscript. We firmly believe that the valuable input you offered has significantly enhanced the overall quality of our work. In the following section, we provide a comprehensive point-by-point response to each specific comment, which is highlighted **in BLUE** for your easy reference.

Responds to Reviewer #1

Comment #1. The objective of this well designed and clearly written manuscript is to elucidate relationships between metabolites associated with RNFLT and cardiovascular metabolic diseases (CMD). The manuscript provides a comprehensive assessment of both metabolites associated with macula RNFLT and CMD in distinct ethnic populations of varying genetic susceptibility. Their results suggest that there is a relationship between RNFLT and CMD metabolic fingerprints and that including RNFLT associated metabolic information into existing CMD models significantly improved their discriminatory power of the models. One concern is that the metabolic fingerprints were identified based on their association with RNFLT – which is affected by many ocular diseases; adjustment for these ocular conditions (glaucoma for example) is encouraged and mention of this issue in the discussion section of the manuscript. Although the authors used external datasets, the population was generally of White and Chinese ethnicity.

Response: We sincerely appreciate the time and effort you have dedicated to reviewing our manuscript and for your thoughtful and constructive feedback. We are also truly grateful for your positive assessment of our work. Your insightful comments have been invaluable in refining our study. We have carefully examined all your specific comments, clarified the population eligibility criteria and the OCT scanning methodology, refined the abstract, improved the legends for tables and figures, and revised the manuscript accordingly. We hope that these modifications have strengthened the manuscript and addressed your concerns effectively.

We totally agree with your concern about the potential influence of ocular conditions on RNFLT and the importance of accounting for these confounders. To minimize their impact, we implemented a predefined procedure to exclude individuals with glaucoma, retinal disease (including retinal detachments and breaks, retinal vascular occlusions, and other structural abnormalities), neurodegenerative disease, and other exclusion criteria (see ***Response to Comment #3***), given their potential to confound RNFLT measurements. This approach ensured that our analysis was conducted in a population free from known retinal pathology, thereby allowing us to identify RNFLT-associated biomarkers more reliably. The limited representation of Black individuals has also been acknowledged in the study limitations, in accordance with your suggestions (see ***Response to Comment #5***). A detailed point-by-point response to each of your specific suggestions is provided below. We believe these revisions improve the clarity, accuracy, and overall impact of our manuscript.

Comment #2. Abstract: This concluding sentence in the abstract may be overstating the results; the authors should consider revising a bit. “Our integrated approach, validated across cohorts of diverse ethnic backgrounds and genetic risks, highlights the possibility that RNFLT metabolic fingerprints as a share basis underlying retinal–cardiometabolic connections and as early indicators that advance equitable cardiometabolic disease management, positioning the retina as a valuable window into cardiometabolic health.”

Response: We sincerely apologize for any overstatement caused by our initial phrasing in the ***Abstract*** conclusion. In the revised manuscript, we have removed the extended claim that "the retina is positioned as a valuable window into cardiometabolic health" and instead emphasize that further studies in broader populations are required to validate our findings (***Page 4, Line 98***). In addition, we have reworded the whole sentence to reflect a more cautious interpretation of our results. We believe this revised phrasing more accurately conveys the tentative nature of our conclusions while highlighting the significance of future research to validate these findings in diverse cohorts.

Comment #3. Methods: RNFLT metabolic states are based on associations with RNFLT. However numerous eye diseases can also affect RNFLT. Were these eye diseases considered in the models?

Response: We sincerely appreciate your insightful comment regarding the potential influence of ocular conditions on RNFLT. We fully agree that factors such as glaucoma and other retinal disorders may affect RNFLT and should be accounted for. To address this concern, we implemented a predefined procedure to exclude individuals with known ocular conditions that could confound the RNFLT measurements.

Specifically, when identifying RNFLT-associated biomarkers, we excluded participants without eligible metabolomics profiling (391,763 participants) and OCT scanning (96,068 participants), as well as those with missing thickness values (294 right eyes and 407 left eyes), low signal strength (1,272 right eyes and 1,067 left eyes), poor centration or segmentation (4,291 right eyes and 4,309 left eyes), high refractive error (spherical equivalent > 6 or < −6 dioptres; 341 right eyes and 404 left eyes), visual impairment (> 0.1 logMAR; 2,054 right eyes and 2,014 left eyes), or abnormal intraocular pressure (IOP ≥ 22 or ≤ 5 mmHg; 701 right eyes and 752 left eyes). Furthermore, patients with glaucoma (82 participants), retinal disease (including retinal detachments and breaks, retinal vascular occlusions, and others) (162 participants), and neurodegenerative disease (39 participants) were excluded, given their potential to confound RNFLT measurements. This approach ensured that our analysis was conducted in a population free from known retinal pathology, thereby allowing us to identify RNFLT-associated biomarkers more reliably.

Originally, these detailed eligibility criteria were provided in the Supplementary Material. However, we recognize that it is crucial for readers to have immediate access to this information in the main text. Accordingly, we have revised the ***Methods*** section (***Page 16, Line 459***) to emphasize the exclusion of prevalent conditions that may confound RNFLT measurements in the population used for identifying RNFLT-associated biomarkers, rather than simply referring to the corresponding section in the Supplementary Material. We hope this clarification addresses your concern.

Comment #4. More detail is needed in describing the OCT methods used: a. For the UKB dataset, RNFLT was measured from macula volume scans centered on the fovea. Which specific scans were acquired in the independent cohort and how was RNFLT measured?

Response: We apologize for any lack of clarity regarding the OCT methods used in the independent cohort (GDES). In this population, all participants underwent retinal OCT scanning using a swept-source OCT (SS-OCT; DRI OCT Triton, Topcon, Japan) to obtain structural images of the macula. Each scan was performed with an internal fixation system, and the fixation was monitored by the instrument's built-in fundus camera. The retina was imaged using a 3D Macula Cube 7×7 mm scan mode centred on the fovea, with a scan density of 512 A-scans × 512 B-scans. The built-in SS-OCT software (IMAGEnet 6, Version 1.22) automatically segmented the retinal layers and measured the RNFLT as the distance from the internal limiting membrane to the nerve fibre layer–ganglion cell layer junction. Where segmentation errors were identified, manual adjustments were performed. This information has been added to the ***Methods*** section (***Page 21, Line 588***) and further elaborated in the ***Supplementary Methods*** . We hope this clarification addresses your concern.

Comment #5. The authors are commended for using several population groups in their analysis. The datasets include mainly Chinese and White individuals. The lack of representation of Black individuals should be mentioned in the discussion as a limitation to this study.

Response: We sincerely appreciate your generous comment and your recognition of our efforts to validate our findings across multiple ethnic groups. We agree that the underrepresentation of Black individuals is a limitation of our study. Accordingly, we have now explicitly mentioned this limitation in both the ***Discussion*** section (***Page 15, Line 420***) and the ***Abstract*** conclusion (***Page 4, Line 98***). We hope that this acknowledgment, along with the suggestion for future work to include more diverse populations, adequately addresses your concern.

Comment #6. Results: Results / tables β and HR should indicate the specific amount of change related to the coefficient of interest; "per 1-SD increment" as well as the value of the 1-SD increment should be included in the tables.

Response: We sincerely appreciate your valuable feedback regarding the clarity of our results presentation. We have carefully reviewed the entire manuscript and supplementary materials to ensure that the effect sizes are clearly defined. We have updated the captions of ***Table 1***, ***Supplementary Tables S2–S9***, and ***S20–S22*** to explicitly indicate that the reported β and HR values are per 1-SD increment of the exposures. The corresponding value of the 1-SD increment has been provided where applicable. We hope these revisions improve the interpretability of our results and address your concern.

Comment #7. It should be clearly stated in tables and figures which datasets the results represent.

Response: We apologize for this oversight. We have reviewed and updated the legends of all tables and figures (***Figs. 1–6***, ***Tables 1–3***, ***Supplementary Figs. S1–***

S6, and ***Supplementary Tables S1–S26***) to explicitly state the datasets they represent. These revisions ensure that readers can immediately identify the source dataset for each result.

Finally, we would like to express our deepest gratitude for your kind words and constructive comments. We truly appreciate the time and effort you have dedicated to reviewing our work, and we hope that our revisions have successfully addressed all your concerns. Your valuable feedback has been instrumental in improving the clarity and rigor of our manuscript.

Responds to Reviewer #2

Comment #1. The manuscript titled "Machine-learning Enhanced Plasma Metabolomics of Retinal Nerve Fiber Layer to Characterize Risks of Mortality and Cardiometabolic Diseases: A Multi-Cohort Study" investigates the relationship between retinal nerve fiber layer thickness (RNFLT) and cardiometabolic diseases (CMDs) using advanced metabolomics and machine learning techniques. The study identifies RNFLT metabolic fingerprints as biomarkers of CMD risk and integrates them into predictive models, demonstrating improved risk prediction across multiple CMD outcomes. These findings are validated in both the UK Biobank and an independent, ethnically distinct cohort, highlighting the potential for RNFLT as a non-invasive tool for systemic health profiling and CMD management.

Response: We would like to express our deepest gratitude for your thoughtful comments and positive assessment of our work. Your insightful feedback has been invaluable in refining our manuscript, and we sincerely appreciate the time and effort you dedicated to reviewing our study. We have carefully examined all your specific comments, reviewed relevant literature, conducted additional analyses including AUPRC analysis that addresses data imbalance, and reclassification and decision analyses that simulate clinical scenarios, and made revisions throughout the manuscript. Through comparisons with previous literature and simulation analyses, we have further highlighted the clinical relevance of our findings. We have also provided synthetic data for the validation cohort, along with the accompanying codes used for data analysis. We firmly believe that these improvements have strengthened the clarity, rigor, and overall quality of our work, and we hope that our revisions have adequately addressed your concerns.

Comment #2. The study has several strengths. It leverages large, well-characterized cohorts with advanced imaging and metabolomics data, providing robust validation across populations. The integration of machine learning significantly enhances the predictive power of traditional models, showcasing the utility of RNFLT as a bridge between ophthalmology and cardiometabolic health. The study also highlights key use cases, including personalized risk prediction, early detection of CMDs, and potential applications in addressing health inequities. The framework paves the way for translational applications, particularly in integrating retinal imaging into routine primary care for systemic health assessments.

Response: We sincerely appreciate your generous and encouraging comments. Your recognition of our study's strengths and its potential translational impact is truly gratifying. We are especially grateful for your acknowledgment of our efforts in leveraging large, well-characterized cohorts, integrating machine learning, and highlighting the broader clinical implications of our findings. Your kind words reinforce our commitment to advancing the role of retinal imaging in systemic health assessments, and we are deeply appreciative of your thoughtful review.

Comment #3. Suggestions for Improvement: The study's biggest limitation is lack of reproducibility. Without data and source code, the value of this study is significantly diminished and evaluation of the methodology is impossible. It's understandable that UKBB data can only be accessed through the established protocols. The validation data appears to be not accessible at all. At the very least, a working source code and simulated data should be made available.

Response: We sincerely apologize for not providing the source code and data in our initial submission. To enhance the transparency and reproducibility of our study, we have now made all source code used in our analyses publicly available at a ***GitHub repository*** (github.com/zocskl/RNFLT-metabolic-states-predict-CMD-outcomes). In addition, as per your suggestion, we have generated and provided simulated data for the validation cohort (GDES). The code and data are provided in a .zip file with a readme.txt file along with the revised manuscript.

We appreciate your understanding regarding the accessibility of the UKB data, which can only be obtained through established protocols. The primary dataset used in this study is available via the UKB Research Analysis Platform (RAP) (https://ukbiobank.dnanexus.com/). The UKB operates as a non-profit charity, allowing global access to approved researchers via a secure cloud-based platform enabled by DNAnexus. However, over the last few years, the UKB has progressively restricted the download of raw data, requiring all analyses to be conducted within the UKB RAP platform. As a result, the original dataset cannot be directly shared between researchers. However, any researcher can apply for access by submitting a proposal outlining their study objectives, feasibility, and public health relevance (http://www.ukbiobank.ac.uk). Given that the UKB data is publicly available under these terms and that we have provided the complete source code, all results presented in our manuscript can be independently validated.

Raw data from the GDES cannot be made publicly available due to the presence of sensitive information that could compromise participant privacy and violate local privacy laws. Although the data can be anonymized by removing apparently identifiable features and perturbing them, it may still contain enough information to identify specific individuals once linked to other datasets.[1] To balance data accessibility with privacy protection, we have generated a synthetic dataset that preserves key statistical properties of the original dataset as per your suggestion. This was achieved using the Distribution-Based Random Generation approach, initially introduced by Rubin et al.[2] and later refined by Raghunathan et al.[3], which has been widely adopted for privacy-preserving data sharing.[1,4,5] This approach constructs a synthetic dataset that preserves the statistical distribution of the original data while ensuring individual participant information cannot be inferred. We have also provided the source code for analysing the sample data and the expected output. To further facilitate transparency, we have also provided the source code for analysing the synthetic dataset, along with the expected output. Below, we briefly introduce the methodology used to generate the data, as well as detailed instructions on how to run the code provided.

Specifically, we assume that the original dataset X follows an underlying probability distribution $P(X)$ and generate a synthetic dataset X_{syn} by estimating its parameters. We first extracted the statistical characteristics of each variable from the original dataset, including the mean vector μ and the covariance matrix Σ for continuous variables, while also computing conditional probabilities for categorical variables. To prevent the reconstruction of individual records, Gaussian noise was injected into both the mean vector μ and the covariance matrix Σ to ensure differential privacy compliance. The privacy budget parameters ε and δ were set to 1.0 and 1×10^{-5}, respectively, while the global sensitivity of the estimated parameters is denoted by Δ_μ and Δ_Σ, respectively. The standard deviation of the added noise σ is given by:

$$\sigma = \frac{\Delta\sqrt{2ln(1.25/\delta)}}{\varepsilon}$$

where Δ corresponds to either Δ_μ or Δ_Σ. For each component μ_i in the mean vector, independent Gaussian noise was added:

$$\mu_i' = \mu_i + \eta_i, \qquad \eta_i \sim N\left(0, \sigma_\mu^2\right)$$

where the noise standard deviation for μ is given by $\sigma_\mu = \frac{\Delta_\mu\sqrt{2ln(1.25/\delta)}}{\varepsilon}$. For each element σ_{ij} in the covariance matrix, noise was added as:

$$\sigma_{ij}' = \sigma_{ij} + \zeta_{ij}, \qquad \zeta_{ij} \sim N(0, \sigma_\Sigma^2)$$

where the noise standard deviation for Σ is $\sigma_\Sigma = \frac{\Delta_\Sigma\sqrt{2ln(1.25/\delta)}}{\varepsilon}$.

For continuous variables (e.g., physical measurements, laboratory tests, and metabolite concentrations), a multivariate normal distribution was assumed:

$$X_{syn} \sim N(\mu', \Sigma')$$

where X_{syn} represents the synthetic data matrix, and N denotes the multivariate normal distribution, which preserves both the marginal distributions and inter-variable dependencies observed in the original dataset. For variables exhibiting skewed distributions, we used a log-normal distribution instead to better retain their statistical properties:

$$X_{syn} \sim Lognormal(\mu', \sigma'^2)$$

where μ' and σ'^2 represent the log-transformed mean and variance estimated from the original dataset. To ensure the realism of the generated variables, a post-processing step was applied to constrain all values within the observed ranges of the original dataset. All numeric variables have been normalised to a range of 0–1. For categorical variables (e.g., sex and disease status), synthetic values were generated by sampling from the observed frequency distribution in the original data:

$$P(Y = k) = n_k / n$$

where n_k represents the number of occurrences of category k in the original dataset, and n is the total sample size. For outcome labels, rather than directly assigning deterministic values, we employed a probabilistic approach to capture the dependencies between predictor variables and outcomes. Specifically, we first trained a model to estimate the conditional probability of the outcome given the predictors:

$$P(Y = 1|X) = f(X)$$

where $f(X)$ represents the estimated probability function derived from the model. Synthetic outcome values were then sampled from a Bernoulli distribution:

$$Y_{syn} \sim Bernoulli\big(P(Y = 1|X)\big)$$

where the synthetic label Y_{syn} was generated probabilistically based on the learned conditional probabilities, ensuring that the synthetic outcomes realistically reflect their dependence on the predictor variables while incorporating inherent variability to avoid overfitting and determinism.

To validate the robustness of our analytical approach and key findings, we repeated the analyses from the original study using the synthetic datasets (***as detailed below***). Our results show that RNFL metabolic states significantly outperform traditional predictors, including systolic blood pressure, HbA1c, BMI, and other biomarkers, in predicting cardiovascular disease. This observation aligns with the findings observed in the original dataset. Further comparisons of model performance show that integrating RNFLT metabolic states into multiple baseline clinical models consistently leads to significant improvements. Specifically, the C-statistic for the Age&Sex model improved from 0.664 to 0.741, with a gain of 0.077 after integrating RNFLT metabolic states; the FGCRS model improved from 0.677 to 0.753, with a gain of 0.076; the Wan model improved from 0.674 to 0.750, with a gain of 0.076; the UKPDS model improved from 0.678 to 0.754, with a gain of 0.076; and the DCS model improved from 0.680 to 0.754, with a gain of 0.074. All improvements were statistically significant (all $P < 0.05$), reaffirming the findings from the original dataset.

It is important to highlight that the primary purpose of providing synthetic data is not to assess the absolute predictive performance of the models, but rather to validate the robustness of our analytical methodology and to compare the relative performance of different models. Our analysis of the synthetic dataset effectively supports the findings derived from the original data. To further enhance transparency and reproducibility, we have made available our code for the analyses in a ***GitHub repository*** (https://github.com/zocskl/RNFLT-metabolic-states-predict-CMD-outcomes). In addition, we have included a Jupyter notebook and a sample dataset as a .zip file in this submission. Below, we provide step-by-step instructions for running the analysis script.

1. Setup

Before executing the analysis workflows, the ***Setup*** section should be run first, as it contains the necessary library imports and function definitions:

(1) Import essential libraries such as \`pandas’, \`numpy’, \`sklearn’, and \`scipy’ for data processing and model training; and \`matplotlib’ and \`seaborn’ for visualization.

(2) Define custom functions for data pre-processing and model evaluation.

2. Comparison of individual predictors

This section evaluates the predictive power of individual predictors in the synthetic dataset:

(1) Load the synthetic dataset using \`pd.read_csv(path)’. The \`path’ should be the directory on your local file system.

(2) Define the outcome label \`event = 'cvd'’.

(3) Specify a list \`convar’ of predictors of 13 variables, including age, sex, smoking status, diabetes duration, systolic blood pressure, diastolic blood pressure, HbA1c, total cholesterol, HDL cholesterol, eGFR, insulin use, BMI, and RNFLT metabolic states.

(4) Initialize an empty DataFrame to store the results.

(5) Iterate through each predictor and perform the following steps:

① Shuffle the dataset: \`sample(frac = 1, random_state = 100)’ to ensure reproducibility.

② Split the dataset into training and test sets in an 8:2 ratio.

③ Extract data for each predictor, ensuring that RNFLT metabolic states (which involve multiple columns) are handled correctly.

④ Convert the data into NumPy arrays for model training.

⑤ Train a \`RandomForestRegressor’ model using: \`RandomForestRegressor(n_estimators = 200, max_depth = 10, oob_score = True, random_state = 100)’. This process sets 200 decision trees with a maximum depth of 10, initializes the out-of-bag (OOB) error estimation and sets the random seed to 100.

⑥ Fit the model to the training set.

⑦ Make predictions on the test set.

⑧ Compute the C-statistic and 95% confidence intervals using the \`delong_roc_ci' function (note that the C-statistic is equivalent to the AUC in binary classification tasks).

⑨ Store the results in the initialized DataFrame .

⑩ Save predictions and true labels together using \`np.save' for further analysis.

(6) Export the final results as a .csv file.

3. Comparison of model performance

This section evaluates how the integration of RNFLT metabolic states improves the predictive performance of multiple clinical models:

(1) Define the \`prepare_model_data' function, which specifies relevant clinical predictors for each of the models, including AgeSex, FGCRS, UKPDS, DCS, and Wan.

(2) Create a list of models and initialize an empty DataFrame to store the results.

(3) For each model, perform the following steps:

① Prepare the dataset with \`prepare_model_data()'.

② Shuffle the dataset with the same random seed and split it into training and test sets in an 8:2 ratio.

③ Extract features and labels for both the baseline model (clinical predictors only) and the integrated model (clinical predictors + RNFLT metabolic states).

④ Convert the data into NumPy arrays.

⑤ Train a \`RandomForestRegressor' model with the same parameters as above: \`RandomForestRegressor(n_estimators = 200, max_depth = 10, oob_score = True, random_state = 100)'.

⑥ Fit the model to the training set.

⑦ Make predictions on the test set.

⑧ Repeat the same process for each baseline clinical model.

⑨ Statistically compare the performance of the baseline clinical model and the model incorporating RNFLT metabolic states.

⑩ Compute 95% confidence intervals for performance differences.

⑪ Store the results in the initialized DataFrame.

⑫ Save predictions and true labels together using `np.save'.

(4) Export the final comparison results as a .csv file.

To ensure full reproducibility, we have fixed the random seed `random_state = 100' across all analyses. Consequently, our results can be fully replicated by running the provided Jupyter notebook with the same parameters and random seed. These steps further strengthen the transparency and reproducibility of our manuscript.

Furthermore, we are also open to facilitating access for interested researchers through a formal request process for applying the GDES data. This information has been added to the ***Data Availability*** section under ***Methods***, as per *Nature Communication*'s policy. All requests for access to in-house data should be addressed to the corresponding authors, Dr. Wei Wang (email: wangwei@gzzoc.com), and will be processed in accordance with Zhongshan Ophthalmic Centre policies. The GDES group will evaluate all requests based on the purpose of the data request. A material-transfer and data-usage agreement will be required between Zhongshan Ophthalmic Centre and the receiving organization, and the requesting organization must state the intended purpose of the data transfer and provide assurances that the transferred data will only be used for non-commercial academic and educational purposes in compliance with Zhongshan Ophthalmic Centre institutional policies.

By implementing these measures, we aim to ensure both data privacy protection and ethical research conduct while enabling transparency and reproducibility of the findings in our manuscript. We hope this addresses your concerns.

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Comment #4. Given the class imbalance in CMD outcomes, including Area Under the Precision-Recall Curve (AUPRC) as a complementary metric (and associated baselines including a completely random/no-skill predictor) is critical.

Response: We fully agree with your suggestion regarding the importance of including AUPRC as a complementary evaluation metric, given the class imbalance in CMD outcomes. We have taken your suggestion and conducted AUPRC analyses for all CMD outcomes, including comparisons with a random/no-skill predictor. Our results consistently show an improvement in AUPRC after integrating RNFLT metabolic states across all outcomes, indicating that our conclusions are robust and not biased by class imbalance.

Class imbalance is an important consideration in our manuscript, with positive sample rates ranging from 0.018 for stroke to 0.074 for all-cause mortality. While the C-statistic provides an overall measure of discrimination, it risks overestimating performance in imbalanced datasets as it gives equal weight to false positives and false negatives. In contrast, the AUPRC is more informative in unbalanced datasets, as it reflects the model's ability to identify true positives while accounting for the high rates of negative cases. For each CMD outcome, we calculated the AUPRC for a random/no-skill predictor, which corresponds to the proportion of positives in the dataset (0.070 for T2D; 0.033 for MI; 0.031 for HF; 0.018 for stroke; 0.074 for all-cause mortality; and 0.019 for cardiovascular mortality). These serve as the baseline values that any useful predictive model should exceed.

Our results show that all models, including clinical baseline models and those integrated with RNFLT metabolic states, outperformed the no-skill predictor across all outcomes (***Supplementary Table S13***, shown below for your easy reference). For T2D, the AUPRC of the model integrating RNFLT metabolic states was 0.533, which is 7.6 times higher than the no-skill predictor. For cardiovascular mortality, the AUPRC for the integrated model was 0.070, which is 3.7 times higher than the no-skill predictor. The gaps between the models integrating RNFLT metabolic states and the no-skill predictor for other outcomes are similarly substantial, ranging from 2.5 times for stroke to 3.2 times for myocardial infarction. Furthermore, all clinical baseline models also showed superior performance compared to the no-skill predictor, demonstrating their robust ability to identify true positives in the minority class.

When compared to the clinical baseline, integrating RNFLT metabolic states consistently improved AUPRC across all CMD outcomes (***Supplementary Table S13***). For T2D, the integration of RNFLT metabolic states improved the AUPRC of the clinical baseline model from 0.480 to 0.533 (ΔAUPRC = +0.053 [+11.0%]; $P <$ 0.001). For stroke, the AUPRC of the integrated model improved from 0.036 to 0.044 (ΔAUPRC = +0.009 [+25.0%]; $P < 0.001$). The improvement was also significant for other outcomes, ranging from +6.2% for myocardial infarction to +21.3% for cardiovascular mortality (all $P < 0.001$). These improvements can be interpreted from a Bayesian perspective as an enhancement in the model's ability to translate prior probabilities into posterior probabilities for identifying high-risk individuals.[1] For instance, for all-cause mortality, setting a threshold λ to divide the population into high-risk and low-risk groups, we are interested in $P(Mortality \mid prediction > \lambda)$.[2] The posterior probability for the traditional Framingham model is 2.3 times higher than the prior (0.170/0.074), while the posterior probability for model integrating RNFLT metabolic states is 2.5 times higher (0.185/0.074). This represents an 8.7% relative improvement [(2.5 − 2.3)/2.3] in the model's capacity to stratify risk at the same recall rate. The improvement in AUPRC aligns with the improvement in C-statistics, reinforcing the robustness of the model across different evaluation metrics.

In terms of clinical decision-making, the practical significance of these improvements becomes evident. In resource-limited screening scenarios, models with higher AUPRC can allocate resources more efficiently by reducing the proportion of false positives among those flagged for intervention. For instance, for cardiovascular mortality, the integrated RNFLT metabolic states model achieved an AUPRC of 0.069, compared to 0.019 for the no-skill predictor. At the same recall rate, the no-skill predictor would misclassify 9,814 false positives in order to identifying 186 true positives out of 10,000 samples (precision = 186 / [186 + FP] = 0.0186). In contrast, integrating RNFLT metabolic states would avoid 7,312 of these false positives, leading to a 74.5% reduction in unnecessary interventions. Furthermore, compared to the traditional Framingham model, the integrated RNFLT metabolic states model would reduce false positives from 3,077 to 2,502 cases (ΔFP = 575), thereby reducing unnecessary interventions by 18.7%. These findings align with and complement our decision curve analysis, which quantifies the net benefit for clinical decision-making, while AUPRC reflects the model's efficiency in optimizing resource allocation by improving the identification of true positive cases.

In summary, our AUPRC analysis confirms that, even in the presence of significant class imbalance, integrating RNFLT metabolic states significantly improves the prediction of all CMD outcomes studied. This consistency of improvement observed in both AUPRC, C-statistic and area under the decision curve further supports the clinical relevance of integrating RNFLT metabolic states into CMD prediction. We hope this adequately addresses your concerns.

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Supplementary Table S13

Improvement in the AUPRC of integrating RNFLT metabolic states for CMD outcome prediction beyond established predictors in the UKB population-II.

Outcomes	**AUPRC**			**Performance gaps ‡**	P_{FDR} **§**
	No-skill predictor	**Integrated models ***	**Basic models †**		
Type 2 diabetes	0.070	0.533 (0.503–0.562)	0.480 (0.448–0.512)	+0.053	<0.001
Myocardial infarction	0.033	0.106 (0.091–0.123)	0.100 (0.085–0.115)	+0.006	<0.001
Heart failure	0.031	0.094 (0.080–0.111)	0.088 (0.075–0.102)	+0.007	<0.001
Stroke	0.018	0.044 (0.036–0.056)	0.036 (0.029–0.043)	+0.009	<0.001
All-cause mortality	0.074	0.185 (0.166–0.203)	0.170 (0.154–0.185)	+0.015	<0.001
Cardiovascular mortality	0.019	0.069 (0.055–0.087)	0.057 (0.046–0.070)	+0.012	<0.001

* Models that integrated RNFLT metabolic states.

† Basic models were based on Framingham General Cardiovascular Risk Score (FGCRS).

‡ Performance gaps between integrated models and basic models.

§ *P* values were calculated through two-sided non-parametric bootstrap tests with 1,000 iterations followed by controlling FDR for multiple testing.

AUPRC = area under the precision-recall curve; CMD = cardiometabolic disease; UKB = UK Biobank; FDR = false discovery rate.

Comment #5. How should the magnitude of the improvement in predictive power (compared to baseline) be interpreted from an application/clinical standpoint? Are these improvements clinically relevant? A simulation study of benefits from such an improvement would be very valuable.

Response: We greatly appreciate your insightful comments regarding the clinical significance of our findings, particularly your observation that the improvement in the C-statistic was generally less than 0.05 (average 0.031). To better interpret the clinical relevance of our study, we: (1) reviewed the literature and compared our improvement in the C-statistic to similar studies published in top-tier journals; and (2) conducted additional simulation studies to assess the clinical benefits of such an improvement in the C-statistic.

It is, however, important to note that our improvement in the C-statistic is **NOT** modest. First, the baseline models employed in our study are well-established, extensively validated, and endorsed by international guidelines from the American Heart Association (AHA),[1] the European Society of Cardiology (ESC),[2] and the World Health Organization (WHO).[3] These models already incorporate a comprehensive set of robust risk factors. In fact, a recent study[4] published in ***Nature Medicine*** incorporated nine novel factors into these same robust baseline models, achieving C-statistic improvements that were slightly lower than those observed in this manuscript (***as detailed below***). Second, rather than directly applying the original coefficients from these established models to our population, we extracted the predictors from each baseline model and re-fitted them. This approach ensured that the baseline models were appropriately calibrated to our population's characteristics, preventing both overly pessimistic baseline estimates and artificially inflated improvements when integrating RNFLT metabolic states. Third, several landmark prediction studies with designs similar to ours have reported comparable performance improvements (***as detailed below***), which are widely considered clinically meaningful. Finally, to further evaluate the clinical relevance of our findings, we performed two additional analyses that simulate risk stratification and decision in clinical scenarios (***as detailed below***), both of which confirmed the clinical utility of integrating RNFLT metabolic states.

(1) Comparison with recent studies published in top-tier journals

Recent landmark studies in cardiometabolic prediction have achieved performance improvements comparable to those observed in our study. Hippisley-Cox et al.[4] (***Nature Medicine***, 2024) developed and validated a new algorithm for cardiovascular risk prediction. Since the AHA/ASCVD model (recommended by the AHA) and the SCORE2 model (recommended by the ESC) are widely adopted for cardiovascular risk assessment, they compared their new algorithm against both models — similar to the comparisons we performed in this manuscript. To facilitate comparison with our study, we focus on the subgroup of participants aged ≥ 40 years (see Hippisley-Cox et al.[44], ***Nature Medicine***, 2024, *Extended Data Table 4*, shown below as ***Appendix Table A1*** for your easy reference). In this subgroup, their algorithm improved the C-statistic for ASCVD from 0.727 to 0.741 (ΔC-statistic = 0.014) and for SCORE2 from 0.728 to 0.741 (ΔC-statistic = 0.013) in men. Similarly, in women, the C-statistic improved from 0.767 to 0.781 (ΔC-statistic = 0.014). When stratified by ethnicity, they reported similar improvements among White and Chinese participants, with improvements of 0.010–0.014 for White participants and 0.007–0.024 for Chinese participants — both comparable to the improvements observed in our study.

Prediction studies beyond cardiometabolic diseases have also reported performance improvements consistent with ours. Perry et al.[5] (***Nature Medicine***, 2024) developed a cardiorespiratory fitness (CRF)-based proteomic score (CRF-Pro) to predict cardiovascular diseases, metabolic diseases, cancers, and neurological conditions (analogous to the RNFLT-based metabolomic score developed in this manuscript for CMD prediction). Their improvements ranged from < 0.005 (still significant) for hypertension, atrial fibrillation, and obesity, to a maximum of 0.03 for T2D, liver disease, and peripheral vascular disease. For cancers and neurological diseases, the improvements remained < 0.01 yet were still considered clinically meaningful. Similarly, Liu et al.[6] (***Nature Aging***, 2024) incorporated genomic and gut metagenomic risk assessment into conventional risk models for predicting coronary artery disease, T2D, Alzheimer's disease, and prostate cancer. Their integrated approach improved the C-statistic by 0.024 for coronary artery disease, 0.014 for T2D, 0.017 for Alzheimer's disease, and 0.031 for prostate cancer. Placido et al.[7] (***Nature Medicine***, 2023) conducted another landmark study that developed sequential neural network models (GRU and Transformer) to predict pancreatic cancer risk using multi-timepoint data. Their innovative modelling approach achieved C-statistic improvements of 0.007 (for GRU) and 0.034 (for Transformer) over the traditional deep learning model (Multilayer Perceptron).

(2) Simulation analyses in reclassification and decision confirm clinical utility

To further evaluate the clinical relevance of our findings, we performed reclassification analysis and decision analysis that simulate risk stratification and decision in clinical scenarios to examine the impact of integrating RNFLT metabolic states on clinical management. Although a model with better discrimination and calibration should theoretically be a better guide to clinical management, they fall short when assessing whether the risk model improves real-world application.[8] In our reclassification analysis, we categorized participants into risk groups (< 5%, 5–10%, and > 10% risk) and assessed how integrating RNFLT metabolic states improved patient reclassification relative to the clinical baseline models (***Supplementary Table S16***, provided below for your easy reference). Across multiple CMD outcomes, integrating RNFLT metabolic states yielded notable improvements in reclassification, particularly for T2D and when compared to the simplest Age&Sex model. This pattern was consistent with the observed improvements in discrimination. For instance, in the prediction of myocardial infarction, integrating RNFLT metabolic states over SCORE2 correctly reclassified 1,610 non-cases (previously misclassified as high risk) into the correct low-risk category, while 1,284 non-cases were incorrectly reclassified as high risk. This resulted in a net improvement of 326 participants (2.0%) correctly reclassified as low risk. Meanwhile, among participants with events, 90 cases were correctly reclassified as high risk, while 75 cases were incorrectly reclassified as low risk, resulting in a net gain of 15 cases (2.1%) correctly reclassified as high risk. Overall, these results led to an additive net reclassification improvement of 4.1%, demonstrating the potential of RNFLT metabolic states to refine CMD risk stratification.

Our decision analysis further demonstrated that the observed gains in discrimination translated to improved clinical utility (***Fig. 4***, provided below for your easy reference). The integration of RNFLT metabolic states either reduced or eradicated the discrepancy between the Age&Sex model and clinically established models. Furthermore, the integration of these RNFL metabolic states into the established models contributed to additional improvements in clinical utility across all six critical CMD outcomes, spanning a wide range of clinical decision thresholds. For instance, integrating RNFLT metabolic states resulted in 656 correctly intervened mortality cases and 3,207 unnecessary interventions at a risk threshold T of 10%. Then, 656 correctly intervened cases minus $[T \div (1 - T)] \times 3{,}207$ unnecessary intervened gives 300 "net" cases. This is equivalent to correctly intervene 300 cases with zero unnecessary intervention. For comparison, the clinical baseline model has 621 correctly intervened heart failure cases and 3,248 unnecessary interventions at the same risk threshold, which is equivalent to correctly intervene 260 cases with zero unnecessary intervention. This means that integrating RNFLT metabolic states provides a 15.4% increase in participants benefiting from potential timely intervention.

These findings align with those reported by Hippisley-Cox et al.[4] (***Nature Medicine***, 2024), reinforcing the clinical relevance of our findings.

(3) Summary of response to Comment #5

Taken together, our results demonstrate that the improvements in C-statistic are consistent with those achieved in recent disease prediction studies published in top-tier journals, such as ***Nature Medicine*** 2024 (#1) & 2024 (#2) & 2023 & ***Nature Aging*** 2024. Furthermore, our reclassification and decision analyses provide compelling evidence that RNFLT metabolic states enhance risk stratification and decision-making. These results highlight the clinical relevance of RNFLT metabolic states in refining risk prediction, improving clinical decision-making, and supporting better CMD outcomes. We greatly appreciate your thoughtful feedback, and we hope that these additional analyses and clarifications address your concerns.

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Appendix Table A1

C statistic (95% CI) for QR4, ASCVD and SCORE2 in people aged 40 and older in the England validation cohort overall and by ethnic group, from Hippisley-Cox et al.[1], ***Nature Medicine*** (2024), *Extended Data Table 4*.

Subgroups	QR4 *	ASCVD	SCORE2
Women			
Overall	0.781 (0.778–0.784)	0.767 (0.764–0.770)	0.767 (0.764–0.770)
White	0.777 (0.773–0.780)	0.764 (0.761–0.767)	0.763 (0.760–0.767)
Indian	0.802 (0.785–0.820)	0.792 (0.774–0.810)	0.792 (0.773–0.811)
Pakistani	0.779 (0.755–0.803)	0.751 (0.724–0.778)	0.754 (0.727–0.780)
Bangladeshi	0.739 (0.698–0.779)	0.714 (0.669–0.759)	0.710 (0.664–0.757)
Other Asian	0.800 (0.777–0.824)	0.791 (0.766–0.816)	0.794 (0.770–0.819)
Caribbean	0.771 (0.751–0.791)	0.754 (0.732–0.775)	0.750 (0.728–0.773)
Black African	0.790 (0.766–0.813)	0.772 (0.747–0.797)	0.762 (0.735–0.789)
Chinese	0.863 (0.822–0.905)	0.856 (0.814–0.898)	0.849 (0.807–0.891)
Other	0.771 (0.750–0.791)	0.747 (0.724–0.769)	0.751 (0.728–0.773)
Men			
Overall	0.741 (0.739–0.744)	0.727 (0.725–0.730)	0.728 (0.725–0.730)

White	0.737 (0.734–0.740)	0.727 (0.724–0.730)	0.725 (0.722–0.728)
Indian	0.736 (0.720–0.753)	0.727 (0.710–0.743)	0.724 (0.708–0.741)
Pakistani	0.735 (0.714–0.756)	0.721 (0.699–0.742)	0.718 (0.696–0.740)
Bangladeshi	0.701 (0.674–0.729)	0.692 (0.664–0.721)	0.687 (0.659–0.716)
Other Asian	0.729 (0.706–0.752)	0.712 (0.688–0.736)	0.711 (0.686–0.735)
Caribbean	0.735 (0.714–0.756)	0.730 (0.708–0.751)	0.725 (0.703–0.746)
Black African	0.745 (0.724–0.767)	0.733 (0.712–0.755)	0.710 (0.686–0.733)
Chinese	0.759 (0.704–0.814)	0.741 (0.685–0.798)	0.735 (0.677–0.793)
Other	0.743 (0.727–0.760)	0.725 (0.708–0.742)	0.721 (0.703–0.738)

* QR4 is their new model, and ASCVD and SCORE2 are the baseline models for comparison.

Reference

1. Hippisley-Cox, J. et al. Development and validation of a new algorithm for improved cardiovascular risk prediction. *Nat. Med.* **30**, 1440-1447 (2024).

Supplementary Table S16

Reclassification improvement of integrating RNFLT metabolic states in the UKB.

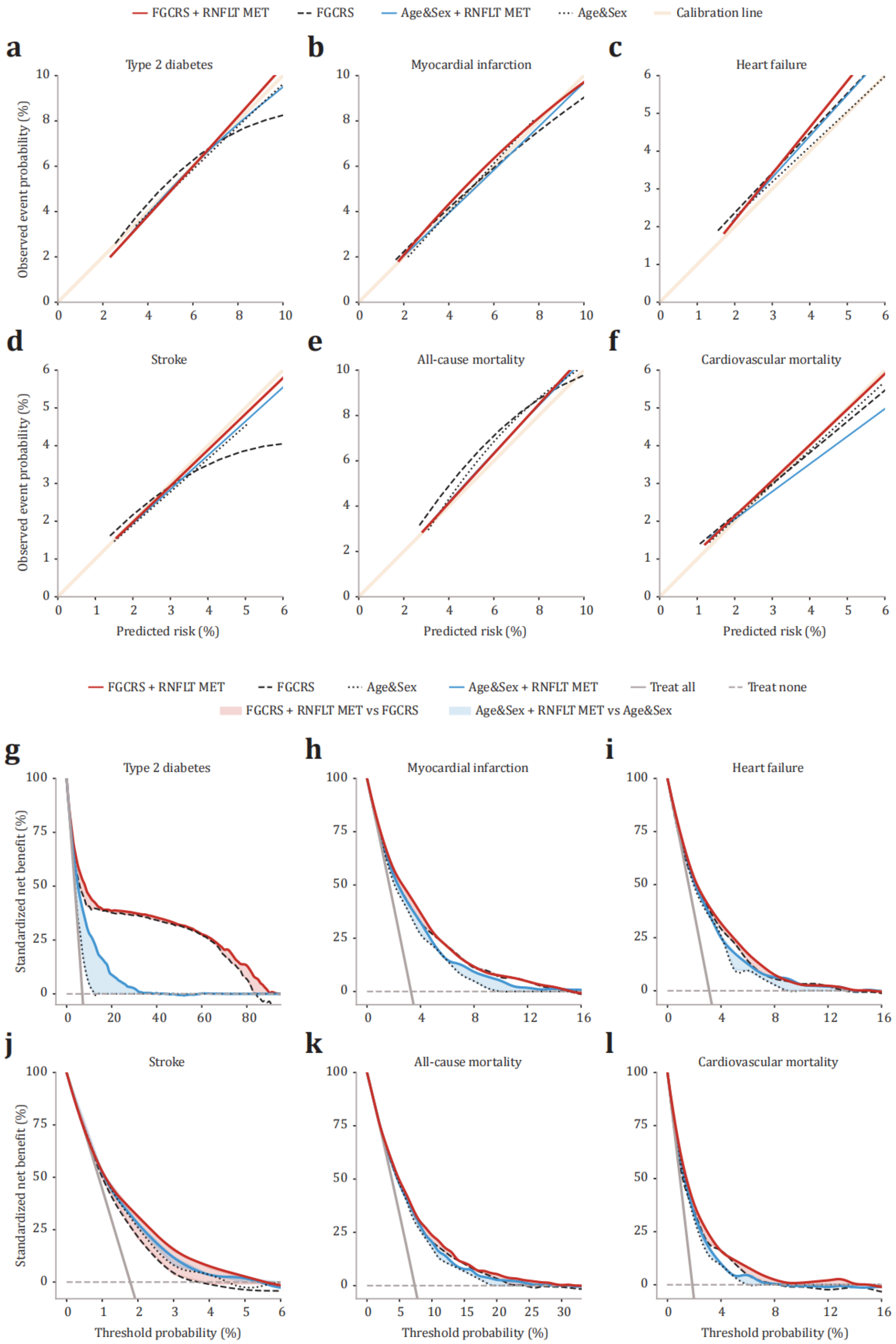
Baseline models *	NRI (case)	NRI (non-case)	Additive NRI
Type 2 diabetes			
Age&Sex	0.286	0.243	0.529
FGCRS	0.080	−0.010	0.070
SCORE2	0.204	0.095	0.299
AHA/ASCVD	0.117	0.006	0.123
WHO-CVD	0.094	−0.022	0.072
Myocardial infarction			
Age&Sex	0.089	−0.015	0.074
FGCRS	0.011	0.001	0.011
SCORE2	0.021	0.020	0.041
AHA/ASCVD	0.005	0.048	0.053
WHO-CVD	0.011	0.007	0.017
Heart failure			
Age&Sex	0.054	0.034	0.089
FGCRS	0.006	0.010	0.015
SCORE2	0.004	0.028	0.032
AHA/ASCVD	0.009	0.020	0.030
WHO-CVD	0.006	0.010	0.016
Stroke			
Age&Sex	0.095	0.047	0.142
FGCRS	0.003	0.020	0.024
SCORE2	0.010	0.020	0.030
AHA/ASCVD	0.042	0.023	0.066
WHO-CVD	0.039	0.027	0.066
All-cause mortality			
Age&Sex	0.028	−0.010	0.017

FGCRS	0.001	0.006	0.007
SCORE2	0.016	0.018	0.033
AHA/ASCVD	0.025	0.045	0.069
WHO-CVD	0.007	0.003	0.010
Cardiovascular mortality			
Age&Sex	0.072	0.031	0.103
FGCRS	0.022	0.010	0.032
SCORE2	0.050	0.021	0.071
AHA/ASCVD	0.072	0.022	0.093
WHO-CVD	0.016	0.012	0.027

* Basic models for CMD outcomes include minimal demographic characteristics (Age&Sex), Framingham General Cardiovascular Risk Score (FGCRS), Systematic Coronary Risk Evaluation 2 (SCORE2) algorithms, American Heart Association/Atherosclerotic Cardiovascular Disease (AHA/ASCVD) algorithms, and World Health Organization Cardiovascular Disease (WHO-CVD) risk charts. RNFLT = retinal nerve fibre layer thickness; UKB = UK Biobank; NRI = net classification improvement.

Fig. 4

Model calibration and clinical utility in the UKB.



Comment #6. Future steps are focused on making the model more accurate and analyzing other diseases. But what is next for this particular model? An RCT? A clinical dashboard / risk predictor? I am struggling with understanding the algorithm's real-world potential.

Response: We greatly appreciate your insightful comment regarding the next steps for our model. While our findings demonstrate the potential of RNFLT metabolic states in predicting cardiometabolic risk, further work is required to translate this model into clinical practice. The necessary direction for our efforts has been illuminated and exemplified in several landmark studies.[1,2] Before discussing the future applications of this manuscript, we would like to highlight that the major innovation of this manuscript is the proposal of a common metabolic basis for retinal–cardiometabolic pathophysiology, revealing potential mechanistic connections across distant systems. Given that RNFLT metabolic states capture CMD risk information comparable to a full metabolomic panel, metabolites identified based on RNFLT thinning (a robust biomarker indicative of CMD risk over a decade prior to clinical manifestation) could provide valuable clues towards the key pathophysiological drivers of multi-system metabolic dysfunction. This not only links retinal and cardiovascular damage as an integral whole but also presents new possibility for CMD risk prediction beyond cardiovascular insights.

First, it is foreseeable that the RNFLT metabolic states will benefit from a broader coverage of metabolomics assays, which is expected to further improve model performance. Second, based on our initial hypothesis, a reasonable assumption is that systemic metabolic insults will extend beyond the retinal and cardiovascular metabolic systems. Third, the natural trajectory of RNFLT metabolic states and their dynamic changes under lifestyle and clinical interventions remain unknown. Fourth, it is reasonable to develop and validate a clinical risk prediction dashboard based on RNFLT metabolic states. Fifth, systematic biases in the model's predictions for specific individuals are worth noting, as they may reflect differential CMD pathogenesis. For example, individuals with healthy RNFLT metabolic states but early onset may suggest mechanisms beyond common metabolic disruption, while those with adverse RNFLT metabolic states but remaining healthy may suggest heterogeneity in susceptibility to common metabolic disruption. At-risk individuals and patients with differential RNFLT metabolic states may respond differentially to the same intervention, which is the basis for precision interventions.

The next steps of the model include (1) making the model more accurate; (2) generalizing it to a broader range of diseases; (3) dynamic monitoring; (4) developing and validating a clinical risk prediction dashboard; and (5) identifying differential susceptibility to common metabolic insults in different patient subgroups (may hint differential subtypes) and comparing intervention outcomes in patients with different metabolic states to identify suitable populations for targeted intervention. Specifically:

(1) The priority is to improve model accuracy while maintaining cost-effectiveness: Accurate models are fundamental to guiding clinical decision-making and management. While we have demonstrated metabolomics' potential to capture multi-system metabolic dysfunction using NMR assay, further expanding the metabolomics coverage is expected to improve model performance. Additional research will involve screening for more metabolomic biomarkers that contribute to the model and designing customized panels of key metabolites. Targeted, low-throughput assays for a few dozen metabolites can significantly reduce costs while maintaining model performance. Additionally, the choice of modelling algorithms will require further exploration. While this manuscript confirms the superiority of random forests in modelling metabolomic data, the effectiveness and efficiency of other machine learning algorithms should be reassessed and compared after expanding data dimensions. Testing and optimizing the model across more diverse populations is also crucial.

(2) Generalized the model to include broader disease coverage: Our initial hypothesis suggests that while the retina may serve as an early window and the cardiovascular system as a key target of common metabolic insults, these insults extend beyond these systems. Other systems vulnerable to metabolic disturbances may also be included in this framework. Recently, the American Heart Association proposed the concept of cardiovascular-kidney-metabolic syndrome, which integrates chronic kidney disease into the CMD framework.[3,4] In line with this, consistent associations of renal function with RNFL and GC-IPL (neuronal cytosol of RNFL) thinning have been noted by others and us.[5-7] Future studies should assess the role of RNFLT metabolic states in kidney disease and other conditions potentially involved in common metabolic disturbances.

(3) Understanding natural trajectory and dynamic changes under interventions: The current framework proposed in this manuscript is based on a single time-point assay, which does not provide insight into the dynamic changes of metabolic states. Future studies should expand data collection to multiple time points to better understand the natural trajectory of RNFLT metabolic states. This will also help clarify whether the current model provides static predictions or supports dynamic tracking. Moreover, future research should evaluate whether the model can capture changes before and after lifestyle and clinical interventions.

(4) Conducting randomized controlled trials (RCTs) to evaluate clinical value: After conceptual development, the next step is to evaluate the model's value in real-world clinical settings. A clinical risk prediction dashboard based on RNFLT metabolic states should be developed and used to randomize traditionally high-risk and low-risk individuals into model-guided care and standard care groups. High-risk patients with adverse metabolic scores should receive intensive management, in order to assess whether early identification of higher-risk populations can guide targeted prevention and intervention, improving clinical outcomes. It is also important to assess

differences in patients' intentions and actions regarding self-health management between the model-guided care and standard care groups.

(5) Identifying subtypes with differential susceptibility to common metabolic injury and treatment response: Cardiometabolic diseases inherently exhibit heterogeneity.[8,9] Although integrating RNFLT metabolic states significantly improved reclassification performance compared to traditional models, many patients were still misclassified. For example, for myocardial infarction, integrating RNFLT metabolic states correctly reclassified 1,610 non-cases (previously misclassified as high-risk) into the correct low-risk category, while 1,284 non-cases were misclassified as high-risk. Careful assessment of noise and modelling artifacts is essential to explore residual biases in specific populations. These biases may indicate differential susceptibility to common metabolic injury, which is the foundation for precision interventions. Comparing treatment responses across subtypes with different metabolic states could lead to more personalized and effective interventions, particularly for targeting metabolic therapies that may show better responses in specific subtypes.

In summary, future directions of the model include exploring mechanisms, improving risk prediction, facilitating early diagnosis, enabling precision interventions and targeted therapies, and advancing interdisciplinary collaboration. For the general public, the model can serve as a tool for promoting healthy lifestyles and self-management while providing dynamic monitoring of overall metabolic health. For preventive medicine experts, the model plays a crucial role in predicting future diseases, guiding risk stratification, and informing public health policy. For clinicians, it can assist in early diagnosis, clinical decision-making, and identifying patients needing intervention. It could also empower ophthalmology care providers to become "sentinels" for overall health. For laboratory researchers, the model provides an integrated perspective on multi-system metabolic dysfunction, offering new insights into the common pathophysiology of multi-system diseases. For clinical researchers, it can be used to identify suitable individuals for clinical trials. Given the slow progression of metabolic diseases, selecting the right patients for trials is especially critical to shortening study timelines. While these goals are ambitious, we emphasize that they will not be achieved overnight. This manuscript represents the first step in the comprehensive evaluation and management of systemic metabolic dysfunction underlying retinal and cardiometabolic systems. We are confident that our findings will have broad impacts and significant value. We hope this clarification adequately addresses your concerns.

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Comment #7. Please Elaborate on the criteria used to select retinal scans and how imaging quality issues were addressed to minimize bias.

Response: We appreciate your pointing out this important issue. For the UKB cohort, given the large sample size, it was not feasible to manually review all images for retinal morphological abnormalities or segmentation errors. Instead, we implemented predefined quality control measures for OCT scanning to ensure accurate RNFLT quantification, as described in previous studies. These measures included the image quality score (IQS), internal limiting membrane (ILM) indicator, validity count, and motion indicators. The IQS evaluates the overall signal strength, while the ILM indicator detects blinks and segmentation errors. The validity count identifies significant clipping in the OCT scan's z-axis dimension. The motion indicators utilize Pearson correlations and absolute differences of the RNFL and full retinal thicknesses from each set of consecutive B-scans to identify blinks, eye-motion artifacts, and segmentation errors.

Participants were excluded from the analysis if they withdrew consent, had poor signal strength and missing thickness values from any Early Treatment Diabetic Retinopathy Study subfield, IQS < 45, poor centration, or poor segmentation according to TABS software (poorest 20% of images excluded based on each of the indicators). This led to the identification of the subset of patients with good-quality, well-centred images and central, stable fixation during the OCT scan. In addition, participants with high refractive error (spherical equivalent > 6 or < −6 dioptres, visual impairment (> 0.1 logMAR), and abnormal intraocular pressure (≥ 22 or ≤ 5 mmHg) were excluded. Patients with self-reported or health record-documented glaucoma, retinal disease (including retinal detachments, retinal vascular occlusions, and other structural abnormalities), or neurodegenerative disease were also excluded, given their potential to confound RNFLT measurements.

For the GDES cohort, automated segmentation results were manually reviewed by an image expert (WW). If segmentation errors existed, manual adjustments were performed. Only high-quality images were included in the analysis, and images were excluded if they met any of the following predefined criteria: (1) IQS < 45, (2) defocus, (3) motion artifacts or blink artifacts, (4) poor centration, (5) poor contrast due to refractive media opacity (local signal missing, blurred images, masking), and (6) uncorrectable segmentation errors. Similar to the UKB, participants with high refractive errors (> +6 D or < −6 D), visual impairment (> 0.1 logMAR), abnormal intraocular pressure (≥ 22 mmHg or ≤ 5 mmHg), and self-reported glaucoma, retinal disease, or neurodegenerative disease were excluded. For both cohorts, in cases in which both eyes of a participant met the eligibility criteria, the right eyes were selected for further analysis. These details have been outlined in the ***Methods*** section (***Page 16, Line 459*** and ***Page 18, Line 494***) and ***Supplementary Methods***.

Finally, we would like to express our most sincere gratitude for your constructive comments and kind words. We are truly grateful for the time and effort you have invested in reviewing our work, and we hope that our revisions have successfully addressed all your concerns. Your feedback has been instrumental in enhancing the clarity and rigour of our manuscript.

Responds to Reviewer #3

Comment #1. This manuscript details a study of metabolomics from UK Biobank to find associations between metabolites and retinal nerve fiber layer thickness and then see if these metabolites are associated with incident cardiometabolic diseases (RNFLT; N=7824 with OCT and metabolomics; N=86000 for just metabolomics to look at incident cardiometabolic diseases). This is based on the premise that RNFLT may have implications for cardiometabolic diseases. They find: (1) 26 metabolites were associated with RNFLT after adjustment for multiple comparisons; (2) majority of metabolites were also associated with incident cardiometabolic diseases; (2) metabolites mediate a significant portion of RNFLT-CMD association (up to 35.1%); (3) metabolite "fingerprints" added to clinical models of risk for many of the cardiometabolic diseases as evidenced by a statistically (and often clinically) meaningful increase in the c-statistic; (4) they do analyses of "health disparities" looking at sex- and social determinants of health stratified models which suggest that the metabolite "fingerprint" improves risk prediction even more so in the group in whom the clinical model performs less well; (5) they do similar analyses in a more ethnically diverse cohort (Guangzhou Diabetes Eye Study) with a more comprehensive metabolomics platform and show that most RNFLT associated metabolites are also associated with incident CVD and mortality.

Response: We sincerely appreciate your accurate summary of this manuscript and your recognition of its key findings. Your insightful feedback has been invaluable in refining our work, and we are truly grateful for the time and effort you dedicated to reviewing it. In response to your comments, we have carefully examined each of your suggestions, reviewed relevant literature, and conducted additional analyses including: (1) further adjusting for key confounders across all analyses; (2) incorporating more granular predictors into all baseline models; (3) performing reclassification and decision analyses to demonstrate clinical relevance; and (4) integrating LC-MS metabolomics to expand metabolite coverage and provide deeper biological insights. Our results show strong consistency across these additional analyses, further reinforcing the robustness of our findings. Regarding the previously published study you highlighted, we have explicitly discussed how this manuscript differs from and significantly exceeds those prior results. By incorporating reclassification and decision analyses, we have further strengthened the clinical relevance. In addition, we have reviewed the entire manuscript and implemented revisions based on your guidance to address all other suggested refinements. We firmly believe that these improvements have strengthened the clarity, rigor, and overall quality of our work, and we hope that our revisions have adequately addressed your concerns.

Comment #2. Overall, the concept of linking ocular findings with biology through circulating analytes/omics is of interest and the link to incident cardiometabolic diseases in that context is interesting. The strengths of the study are the good use of the UKB dataset, the incorporation of genetic risk to show that the metabolite signature adds to clinical risk prediction even in high genetic risk, and the inclusion of an external validation cohort (GDES).

Response: We sincerely appreciate your thoughtful comments and your recognition of the strengths of our study. We are particularly grateful for your acknowledgment of our approach in leveraging the UKB dataset, integrating genetic risk assessment, and incorporating an independent validation cohort to enhance the robustness and generalizability of our findings. The intersection of ocular biomarkers, systemic metabolism, and cardiometabolic disease risk represents an exciting area of research, and we are encouraged by your appreciation of our efforts to explore this link. We hope our study contributes to a broader understanding of how retinal imaging and circulating metabolomics can complement traditional risk assessment models and pave the way for future translational applications. Thank you once again for your insightful review and for highlighting these key aspects of our work.

Comment #3. Enthusiasm for the manuscript is significantly tempered due to the following major concerns. Firstly, the impact of the manuscript is significantly decreased due to an already published manuscript (PMID: 39581638, including the senior author of this manuscript) detailing these associations; the fact that a large proportion of the metabolites converge on HDL cholesterol which is a known risk factor; and relatedly, that the clinical implications of this metabolite score are unclear given that the clinical models that were used are not robust and the analyses were not adjusted for HDL cholesterol; and that many of the models other than DM have a very small increase in the C-statistic (0.02) for most outcomes. Secondly, there are validity concerns as detailed below, the primary of which is the lack of accounting for confounders in the original analyses, thus any link between RNFLT metabolites and incident CMD more likely reflects residual confounding related to the presence of CMD risk factors (i.e. pre-DM), as opposed to a direct link between the metabolite and the incident CMD (relatedly, the mediation analyses do not appear to account for confounders).

Response: We sincerely appreciate your insightful comments on several critical points. In response to your concerns, we have reviewed previous literature, performed additional analyses, and revised the manuscript accordingly. First, we provide a systematic comparison between this manuscript and our previously published study in *Br J Ophthalmol* (2024).[1] Despite utilizing a similar biomarker discovery approach, the two studies are fundamentally different in terms of objectives, theoretical rationale, research questions, hypotheses, study designs, key findings (including metabolites identified), interpretation, and clinical implications. (***as detailed below***).

Then, we performed a series of sensitivity analyses by sequentially adjusting for additional confounders in the main analyses, including enzymatic assay-based HDL, systolic blood pressure, HbA1c levels, and prevalent CMD. These analyses not only addressed potential confounding but also reinforced the complementary nature of NMR-based metabolomics and enzymatic assays in capturing distinct facets of HDL cholesterol. Notably, although the associations between NMR metabolites and RNFLT attenuated after adjusting for enzymatic HDL, all metabolites remained statistically significant with consistent directionality (***as detailed below***). This suggests that the physical properties of lipoprotein particles captured by the NMR assay (beyond traditional enzymatic assays) may play a key role in the association with RNFLT. We further incorporated these covariates into the mediation analyses. Although effect sizes weakened, the primary results remained robust, suggesting that RNFLT metabolic states mediate the RNFLT–CMD association beyond enzymatic HDL contributions (***as detailed below***).

While our study suggests the value of integrating RNFLT metabolic states for improving CMD risk prediction, we emphasize that the main research question addressed in this manuscript is whether metabolomics mediate the RNFLT–CMD association, i.e. serving as the basis of retinal–cardiometabolic connections. Nevertheless, we assessed the performance of adding more granular variables to all baseline models and additionally evaluated a comprehensive model integrating all predictors from each clinical model (***as detailed below***). The predictive improvement from incorporating RNFLT metabolic states remained stable. Finally, we reviewed recent prediction studies published in top-tier journals and conducted additional analyses to demonstrate the clinical relevance of our findings (***as detailed below***).

(1) Systematic comparison to our previous work

We thank you for raising this important point. While both studies apply a biomarker discovery approach, we emphasize that they are fundamentally different in terms of objectives, theoretical rationale, research questions, hypotheses, study designs, key findings (including metabolites identified), interpretation, and clinical implications. Our prior study in *Br J Ophthalmol* (2024) served as a preliminary exploration into the ocular-systemic health connection, providing a broad direction for future research rather than offering immediate clinical applications. In contrast, the present study represents a deeper and more targeted investigation into the specific metabolic basis of the RNFLT–CMD relationship, with actionable insights for clinical decision-making.

Another important difference between these two studies is the imaging modality and the metabolites identified. While both studies explored the linkage between ocular phenotypes and systemic health, *Br J Ophthalmol* (2024) was based on an unvalidated colour fundus photography-derived parameter, whereas the present manuscript focuses on optical coherence tomography (OCT)-derived RNFLT — a structural biomarker that bridges the well-established link between the retina and the heart established by us and others,[2-7] beyond the reach of fundus photography. This distinction naturally led to the minimal overlap in the metabolic biomarkers identified (only 7 shared out of 47), which is expected given that OCT and fundus photography capture distinct facets of health information.

Furthermore, the findings of the present manuscript are significantly more robust and clinically promising than those in *Br J Ophthalmol* (2024), where the association strengths and predictive improvements are approximately 10 times greater than those reported in our previous work. It is noteworthy that the predictive improvement achieved using only RNFLT-associated metabolites was comparable to that of the full NMR panel, suggesting that these metabolites represent key biological processes underlying the RNFLT–CMD link, rather than merely being a random subset of CMD-associated metabolites.

Specifically, this manuscript differs from *Br J Ophthalmol* (2024) in multiple aspects:

① Study aims and main exposures: Although both studies investigated the linkage between ocular phenotypes and systemic diseases, *Br J Ophthalmol* (2024) was based on colour fundus photography-derived parameters, whereas this manuscript focuses on OCT-derived RNFLT, providing a higher-dimensional, layer-specific assessment of retinal structure beyond what is possible with fundus photography (tomographic vs. frontal view). Specifically, colour fundus photography captures frontal retinal morphology using a digital camera, while OCT produces cross-sectional images based on low-coherence near-infrared interferometry, allowing for precise layer-specific retinal measurements. Therefore, both studies share CMDs as the main outcomes, their exposures are fundamentally different. Furthermore, *Br J Ophthalmol* (2024) took a preliminary exploratory approach by broadly examining the association between a fundus photography-derived parameter and a wide range of systemic health outcomes without a clear motivation. As a result, its findings lack immediate clinical relevance and serve primarily as a foundation for future hypothesis-driven research. In contrast, the present manuscript is a focused, hypothesis-driven investigation that builds upon previously established RNFLT–CMD associations while focusing on clinical decision making and health disparities in socially vulnerable populations, with the aim of providing direct actionable insights.

② Theoretical rationale and hypotheses: Since the association between the parameter used in *Br J Ophthalmol* (2024) and systemic health outcomes was previously unknown, the analyses and discussion regarding metabolic associations with systemic outcomes lacked a clear theoretical rationale. Moreover, that study did not establish a direct link between the fundus photography-derived parameter and incident CMDs. Conversely, the present manuscript builds upon prior evidence of RNFLT–CMD associations, integrating both RNFLT–metabolite and metabolite–CMD links to provide a mechanistic exploration of the retinal–cardiometabolic connection. We hypothesize that a metabolic basis may underlie the connection between RNFLT changes and CMDs, while the susceptibility of the retina may allow for early manifestation of these subtle disturbances. This extends the concept of retinal-systemic metabolic linkage from fundus photography to OCT, providing in-depth investigation and robust evidence for the metabolic basis of RNFLT–CMD connections.

③ Study designs: While *Br J Ophthalmol* (2024) hypothesized that fundus photography-derived metabolic biomarkers might stratify and predict multisystem disease risk, it did not specify studied outcomes using a predefined process, making it inherently a data-driven rather than hypothesis-driven study. As discussed above, the lack of a clear motivation and biological rationale further limits its interpretability. In contrast, while the present manuscript also examines metabolite-based CMD prediction, it is entirely hypothesis-driven to test whether RNFLT metabolic states mediate the RNFLT–CMD relationship. Findings from mediation and multiple downstream analyses support our initial hypothesis, suggesting that the RNFLT–CMD association is at least partially explained by shared metabolic pathways.

④ Key findings and interpretation: While both studies have investigated metabolomic associations with ocular phenotypes, the key findings are different. *Br J Ophthalmol* (2024) identified 28 metabolic biomarkers associated with a fundus photography-derived parameter, and the present study identified 26 metabolic biomarkers associated with OCT-derived RNFLT. It is noteworthy that only 7 out of 47 metabolic biomarkers overlapped between the two studies, highlighting the distinct nature of the information captured by fundus photography versus OCT. More importantly, the metabolic biomarkers associated with OCT-derived RNFLT played a mediating role in the RNFLT–CMD association, whereas those linked to fundus photography-derived parameters are likely to represent incidental statistical correlations. The contribution of fundus photography-derived metabolic biomarkers to CMD prediction in *Br J Ophthalmol* (2024) was modest, suggesting that these biomarkers represent a random subset of CMD-associated metabolites rather than capturing key processes of biological basis. In contrast, the integration of only the RNFLT-associated subset in the present manuscript achieved predictive improvements comparable to the full NMR panel (~10 times higher than those in *Br J Ophthalmol*, 2024). This finding strongly suggests that these metabolites serve as key processes underlying the RNFLT–CMD link, rather than merely being a random subset of CMD-related biomarkers.

⑤ Clinical relevance: Moving beyond *Br J Ophthalmol* (2024), we have performed reclassification and decision analysis that reveal clinical relevance of integrating RNFLT metabolic states into CMD prediction. Although a model with better discrimination and calibration should theoretically be a better guide to clinical management, these measures fall short when assessing whether the risk model improves clinical decision making.[8] By summarizing the performance of the model in reclassifying risk categories and aiding decision making, we found that the observed discriminatory gains translated to clinical utility gains. The integration of RNFLT metabolic states either reduced or eradicated the discrepancy between the Age&Sex model and clinically established models. Furthermore, the integration of these RNFLT metabolic states into the established models contributed to additional improvements in risk reclassification and clinical decision making for all six CMD outcomes. These findings further support the clinical relevance of this manuscript.

In summary, this manuscript and our previously published work in *Br J Ophthalmol* (2024) differ in multiple aspects, including objectives, theoretical rationale, research questions, hypotheses, study designs, key findings, interpretation, and clinical implications. Our study focuses on OCT-derived RNFLT (a distinct imaging modality from fundus photography), identified distinct metabolic biomarkers, and provides the first evidence of a metabolic basis for the longitudinal RNFLT–CMD association, improving risk prediction and clinical utility while addressing health disparities in women and socially vulnerable populations. Thus, our conclusions remain novel and independent, and the findings in this manuscript are not diminished by our previous work in *Br J Ophthalmol* (2024).

(2) Sensitivity analyses adjusting for key confounders

We fully agree with your concerns regarding the key confounders that were not addressed in our initial submission. To address this, we performed a series of sensitivity analyses by sequentially adjusting for additional covariates to ensure the robustness of our findings: (1) further adjusting for enzymatic assay-based HDL, systolic blood pressure, and HbA1c levels in the identification of RNFLT metabolic states; (2) additionally adjusting model (1) for prevalent CMD; (3) adjusting for enzymatic assay-based HDL, systolic blood pressure, and HbA1c levels in the association between metabolic biomarkers and incident CMD outcomes; and (4) adjusting for the above in all mediation analyses. It is important to note that we did not adjust for prevalent CMD in: (1) in the association between metabolic biomarkers and incident CMD outcomes; and (2) all mediation analysis, since all individuals with prevalent CMD at baseline had already been excluded in the incident CMD analysis.

As anticipated, the associations between NMR metabolites and RNFLT were attenuated after adjusting for key confounders as suggested. Nevertheless, all originally identified biomarkers (including HDL-related ones) remained robust. Specifically, when adjusting for enzymatic assay-based HDL, systolic blood pressure, and HbA1c levels, the attenuated yet consistent associations suggest that only part of the associations were confounded by these factors (***Supplementary Table S3***, provided below for your easy reference). When further adjusting for prevalent CMDs, the associations remain virtually unchanged, suggesting that the metabolite–RNFLT associations were not likely to be confounded by CMD status (***Supplementary Table S4***, provided below for your easy reference). Overall, the consistent results remained after adjusting for enzymatic assay-based HDL and others are to be expected and align with the known biological distinction between NMR-based and enzymatic HDL assays. While enzymatic HDL assays measure concentration, NMR-derived HDL assay captures unique particle properties that extend beyond the traditional enzymatic assay.

Specifically, the enzymatic HDL assay separates HDL particles via chemical precipitation and measures their concentration. However, this method does not differentiate between HDL subtypes or account for particle composition. In contrast, the NMR-based HDL assay applies deconvolution of diffusion and relaxation properties, integrating both physical properties (e.g., particle size, lipid surface-to-core ratio) and lipid composition (e.g., phospholipids, cholesterol esters, free cholesterol, and apolipoprotein A1). For instance, the higher molecular mobility of cholesterol in small HDL particles extends their T2 relaxation times, resulting in reduced linewidth attenuation in the presaturated proton spectra and thus stronger signals than their larger counterparts. Thus, while both assays share some variation (concentration), the NMR-based assays capture additional dimensions of HDL structure and composition that are likely relevant to the RNFLT–CMD link. Additionally, non-cholesterol components such as phosphatidylcholine and phosphoglycerides also contribute to the RNFLT metabolic state, further highlighting the expanded biological insights captured by the NMR assay. Overall, our results suggest that the association between RNFLT and NMR-identified metabolic biomarkers is unlikely to be explained by the confounding from known risk factors, including HDL concentration.

We also incorporated the same set of covariates into: (1) the association between metabolic biomarkers and incident CMD outcomes; and (2) all mediation analyses. As noted earlier, prevalent CMD was not adjusted in these analyses, since individuals with prevalent CMD had already been excluded from the incident CMD analysis. Although effect sizes were attenuated after these adjustments, the primary results remained robust (***Supplementary Table S6*** and

Table 1, provided below for your easy reference). Interestingly, after adjusting for the suggested key confounders (including enzymatic HDL), the mediating effect of RNFLT metabolic states in the RNFLT–CMD association became even stronger (***Table 1***). This finding suggests that RNFLT metabolic states predominantly mediate the RNFLT–CMD association beyond enzymatic HDL concentration and other traditional risk factors.

Collectively, these sensitivity analyses effectively addressed concerns about residual confounding and reinforced the robustness of our primary findings. The consistent results across multiple adjusted models underscore the biological relevance of RNFLT metabolic states in explaining the RNFLT–CMD association. Importantly, these findings emphasize that RNFLT metabolic states capture variation that extends beyond known CMD risk factors, strengthening their potential as meaningful biomarkers for CMD risk assessment.

(3) Model performance of incorporating more granular predictors

We appreciate your guidance on incorporating more granular data including HbA1c and traditional HDL into the baseline models. While the baseline models we employed are well-established and widely endorsed by international guidelines from the American Heart Association (AHA),[9] the European Society of Cardiology (ESC),[10] and the World Health Organization (WHO),[11] we acknowledge that some known predictors were not fully leveraged. To address this, we incorporated additional predictors and re-ran all analyses, including: (1) integrating HbA1c and enzymatic HDL into all baseline clinical models; and (2) constructing a comprehensive model that incorporates all predictors used across all considered models. We then compared the performance of models integrating RNFLT metabolic states with all baseline models. These additional analyses provide a more rigorous assessment of the independent predictive value of RNFLT metabolic states beyond conventional risk factors. Our results demonstrate that even after enhancing the clinical baseline with more granular predictors, the predictive improvement gained from integrating RNFLT metabolic states remains robust.

Incorporating HbA1c and enzymatic HDL led to a numerical improvement in the performance of all clinical baseline models (***Supplementary Table S25***, provided below for your easy reference). Among all CMD outcomes, the largest gain from adding HbA1c and enzymatic HDL was observed in the prediction of T2D, which is expected given that HbA1c is one of the most important risk factors for T2D. Specifically, the baseline C-statistic for the FGCRS model in predicting T2D increased from 0.810 to 0.840 when incorporating HbA1c and enzymatic HDL, approaching the performance of the FRS-for-T2D model, which was specifically designed for T2D risk assessment.[12,13] Similarly, among all baseline models considered, the WHO-CVD model exhibited the greatest improvement after incorporating these additional predictors. This is likely because WHO-CVD is the only model that does not originally include enzymatic HDL. Although the absolute improvement from integrating RNFLT metabolic states was slightly attenuated after incorporating HbA1c and enzymatic HDL (e.g., ΔC-statistic decreased from 0.044 to 0.028 for T2D over FGCRS; from 0.020 to 0.018 for myocardial infarction over FGCRS), the improvement remained statistically significant and consistent across different baseline models (***Supplementary Table S25***). This finding underscores that RNFLT metabolic states capture risk information beyond that provided by traditional predictors, even in the context of an enhanced clinical baseline.

To further evaluate the incremental predictive value of RNFLT metabolic states, we constructed a comprehensive model that integrates all individual predictors from each baseline clinical model considered (***Supplementary Table S26***, provided below for your easy reference). While this model achieved the highest overall predictive performance, adding additional conventional risk factors beyond those in the validated baseline models yielded marginal improvements.

This further supports the effectiveness and robustness of the predictor sets originally used in this manuscript. Notably, the improvement in predicting stroke did not reach the significance threshold for multiple comparison. However, for all other CMD outcomes, the predictive improvement of integrating RNFLT metabolic states over the comprehensive model remained significant and robust. We hope this additional analysis adequately address your concerns. While these improvements are statistically significant, one might question their clinical relevance. In the next section, we will address this by comparing our findings to similar prediction studies recently published in top-tier journals and conducting additional analyses that further establish the clinical value of integrating RNFLT metabolic states into CMD risk prediction.

(4) How this study is clinically relevance

We greatly appreciate your insightful comments regarding the clinical significance of our findings, particularly your observation that the improvement in the C-statistic was generally less than 0.05 (average 0.031). However, it is important to note that, even when excluding for diabetes, the improvement is **NOT** modest. First, the baseline models employed in our study are well-established, extensively validated, and endorsed by international guidelines from the American Heart Association (AHA),[9] the European Society of Cardiology (ESC),[10] and the World Health Organization (WHO).[11] These models already incorporate a comprehensive set of robust risk factors. In fact, a recent study[14] published in ***Nature Medicine*** incorporated nine novel factors into these same robust baseline models, achieving C-statistic improvements that were slightly lower than those observed in this manuscript (***as detailed below***). Second, rather than directly applying the original coefficients from these established models to our population, we extracted the predictors from each baseline model and re-fitted them. This approach ensured that the baseline models were appropriately calibrated to our population's characteristics, preventing both overly pessimistic baseline estimates and artificially inflated improvements when integrating RNFLT metabolic states. Third, several landmark prediction studies with designs similar to ours have reported comparable performance improvements (***as detailed below***), which are widely considered clinically meaningful. Finally, to further evaluate the clinical relevance of our findings, we performed additional reclassification analysis and decision analysis (***as detailed below***), both of which confirmed the clinical utility of integrating RNFLT metabolic states.

Recent landmark studies in cardiometabolic prediction have achieved performance improvements comparable to those observed in our study. Hippisley-Cox et al.[14] (***Nature Medicine***, 2024) developed and validated a new algorithm for cardiovascular risk prediction. Since the AHA/ASCVD model (recommended by the AHA) and the SCORE2 model (recommended by the ESC) are widely adopted for cardiovascular risk assessment, they compared their new algorithm against both models — similar to the comparisons we performed in this manuscript. To facilitate comparison with our study, we focus on the subgroup of participants aged ≥ 40 years (see Hippisley-Cox et al.[1414], *Nature Medicine*, 2024, *Extended Data Table 4*, shown below as ***Appendix Table A1*** for your easy reference). In this subgroup, their algorithm improved the C-statistic for AHA/ASCVD from 0.727 to 0.741 (ΔC-statistic = 0.014) and for SCORE2 from 0.728 to 0.741 (ΔC-statistic = 0.013) in men. Similarly, in women, the C-statistic improved from 0.767 to 0.781 (ΔC-statistic = 0.014). When stratified by ethnicity, they reported similar improvements among White and Chinese participants, with improvements of 0.010–0.014 for White participants and 0.007–0.024 for Chinese participants — both comparable to the improvements observed in our study.

Prediction studies beyond cardiometabolic diseases have also reported performance improvements consistent with ours. Perry et al.[15] (***Nature Medicine***, 2024) developed a cardiorespiratory fitness (CRF)-based proteomic score (CRF-Pro) to predict cardiovascular diseases, metabolic diseases, cancers, and neurological conditions (analogous to the RNFLT-based metabolomic score developed in this manuscript for CMD prediction). Their improvements ranged from < 0.005 (still significant) for hypertension, atrial fibrillation, and obesity, to a maximum of 0.03 for T2D, liver disease, and peripheral vascular disease. For cancers and neurological diseases, the improvements remained < 0.01 yet were still considered clinically meaningful. Similarly, Liu et al.[16] (***Nature Aging***, 2024) incorporated genomic and gut metagenomic risk assessment into conventional risk models for predicting coronary artery disease, T2D, Alzheimer's disease, and prostate cancer. Their integrated approach improved the C-statistic by 0.024 for coronary artery disease, 0.014 for T2D, 0.017 for Alzheimer's disease, and 0.031 for prostate cancer. Placido et al.[17] (***Nature Medicine***, 2023) conducted another landmark study that developed sequential neural network models (GRU and Transformer) to predict pancreatic cancer risk using multi-timepoint data. Their innovative modelling approach achieved C-statistic improvements of 0.007 (for GRU) and 0.034 (for Transformer) over the traditional deep learning model (Multilayer Perceptron).

To further evaluate the clinical relevance of our findings, we performed reclassification analysis and decision analysis to examine the impact of integrating RNFLT metabolic states on risk prediction and clinical management. Although a model with better discrimination and calibration should theoretically be a better guide to clinical management, they fall short when assessing whether the risk model improves real-world application.[8] In our reclassification analysis, we categorized participants into risk groups (< 5%, 5–10%, and > 10% risk) and assessed how integrating RNFLT metabolic states improved patient reclassification relative to the baseline models (***Supplementary Table S16***, provided below for your easy reference). Across multiple CMD outcomes, integrating RNFLT metabolic states yielded notable improvements in reclassification, particularly for T2D and when compared to the simplest

Age&Sex model. This pattern was consistent with the observed improvements in discrimination. For instance, in the prediction of myocardial infarction, integrating RNFLT metabolic states over SCORE2 correctly reclassified 1,610 non-cases (previously misclassified as high risk) into the correct low-risk category, while 1,284 non-cases were incorrectly reclassified as high risk. This resulted in a net improvement of 326 participants (2.0%) correctly reclassified as low risk. Meanwhile, among participants with events, 90 cases were correctly reclassified as high risk, while 75 cases were incorrectly reclassified as low risk, resulting in a net gain of 15 cases (2.1%) correctly reclassified as high risk. Overall, these results led to an additive net reclassification improvement of 4.1%, demonstrating the potential of RNFLT metabolic states to refine CMD risk stratification.

Our decision analysis further demonstrated that the observed gains in discrimination translated to improved clinical utility (***Fig. 4***, provided below for your easy reference). The integration of RNFLT metabolic states either reduced or eradicated the discrepancy between the Age&Sex model and clinically established models. Furthermore, the integration of these RNFL metabolic states into the established models contributed to additional improvements in clinical utility across all six critical CMD outcomes, spanning a wide range of clinical decision thresholds. For instance, integrating RNFLT metabolic states resulted in 656 correctly intervened mortality cases and 3,207 unnecessary interventions at a risk threshold T of 10%. Then, 656 correctly intervened cases minus $[T \div (1 - T)] \times 3{,}207$ unnecessary intervened gives 300 "net" cases. This is equivalent to correctly intervene 300 cases with zero unnecessary intervention. For comparison, the clinical baseline model has 621 correctly intervened heart failure cases and 3,248 unnecessary interventions at the same risk threshold, which is equivalent to correctly intervene 260 cases with zero unnecessary intervention. This means that integrating RNFLT metabolic states has 15.4% additional participants benefit from potential timely intervention. These improvements were slightly superior to those reported by Hippisley-Cox et al.[14] (***Nature Medicine***, 2024), reinforcing the clinical relevance of our findings.

Taken together, our results demonstrate that the improvements in C-statistic are consistent with those achieved in recent disease prediction studies published in top-tier journals, such as ***Nature Medicine*** 2024 (#1) & 2024 (#2) & 2023 & ***Nature Aging*** 2024. The complementary findings from our reclassification and decision analyses further confirm the clinical utility of integrating RNFLT metabolic states into established models. These findings highlight the potential of RNFLT metabolic states to refine risk prediction, improve clinical decision-making, and support better CMD outcomes.

(5) Summary of response to Comment #3

Overall, we are deeply grateful for the series of thoughtful and constructive comments you provided. We first conducted a systematic comparison between this manuscript and our published article (*Br J Ophthalmol.*, 2024), highlighting key differences in imaging modalities, metabolic biomarkers of interest, study design, and clinical implications. We then incorporated HDL concentration and other key confounders into all analyses to confirm the robustness of our results. This suggests that the findings in this manuscript are unlikely to reflect residual confounding by known risk factors. Lastly, we reviewed recent prediction studies published in top-tier journals and performed additional reclassification analysis and decision analysis, to demonstrate that our findings extend beyond statistical significance and are clinical relevance. We hope that these additional analyses and clarifications adequately address your concerns.

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Supplementary Table S3

Identification of RNFLT-associated metabolic biomarkers in the UKB population-I, additionally adjusting for enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

Metabolic biomarkers *	**β †**	**95% CI**		**P_{FDR} ‡**
HDL Cholesterol	−0.163	−0.277	−0.049	0.007
Total Phospholipids in Lipoprotein Particles	−0.117	−0.229	−0.005	0.041
Phospholipids in HDL	−0.167	−0.277	−0.058	0.005
Cholesteryl Esters in HDL	−0.168	−0.282	−0.055	0.006
Free Cholesterol in HDL	−0.142	−0.254	−0.029	0.017
Total Lipids in HDL	−0.163	−0.274	−0.052	0.006
Total Concentration of Lipoprotein Particles	−0.183	−0.291	−0.076	0.002
Concentration of HDL Particles	−0.190	−0.295	−0.085	0.002
Phosphoglycerides	−0.129	−0.242	−0.016	0.027
Total Cholines	−0.128	−0.244	−0.012	0.032
Phosphatidylcholines	−0.132	−0.245	−0.018	0.027
Apolipoprotein A1	−0.177	−0.285	−0.068	0.003
Concentration of Medium HDL Particles	−0.192	−0.298	−0.085	0.002
Total Lipids in Medium HDL	−0.190	−0.296	−0.083	0.002
Phospholipids in Medium HDL	−0.191	−0.296	−0.085	0.002
Cholesterol in Medium HDL	−0.197	−0.305	−0.089	0.002
Cholesteryl Esters in Medium HDL	−0.201	−0.309	−0.093	0.002
Free Cholesterol in Medium HDL	−0.183	−0.291	−0.074	0.002
Concentration of Small HDL Particles	−0.163	−0.264	−0.062	0.003
Total Lipids in Small HDL	−0.170	−0.271	−0.069	0.002
Phospholipids in Small HDL	−0.177	−0.278	−0.076	0.002
Cholesterol in Small HDL	−0.169	−0.269	−0.069	0.002

Cholesteryl Esters in Small HDL	−0.166	−0.265	−0.067	0.002
Free Cholesterol in Small HDL	−0.155	−0.259	−0.050	0.006
Phospholipids to total lipids ratio in very small VLDL	0.120	0.015	0.224	0.027
Phospholipids to total lipids ratio in IDL	0.126	0.026	0.227	0.017

* Adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

† Data shown as per-SD change of RNFLT (4.24 μm).

‡ *P* values were calculated through two-sided Student's t-tests followed by controlling FDR for multiple testing.

HDL = high-density lipoprotein; UKB = UK Biobank; CI = confidence interval; FDR = false discovery rate; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Supplementary Table S4

Identification of RNFLT-associated metabolic biomarkers in the UKB population-I, additionally adjusting for prevalent CMDs.

Metabolic biomarkers *	**β †**	**95% CI**		***P*_FDR ‡**
HDL Cholesterol	−0.164	−0.279	−0.050	0.007
Total Phospholipids in Lipoprotein Particles	−0.114	−0.227	−0.001	0.048
Phospholipids in HDL	−0.166	−0.276	−0.056	0.005
Cholesteryl Esters in HDL	−0.170	−0.284	−0.056	0.006
Free Cholesterol in HDL	−0.142	−0.255	−0.029	0.018
Total Lipids in HDL	−0.162	−0.274	−0.051	0.007
Total Concentration of Lipoprotein Particles	−0.181	−0.289	−0.073	0.003
Concentration of HDL Particles	−0.188	−0.294	−0.082	0.003
Phosphoglycerides	−0.125	−0.239	−0.012	0.033
Total Cholines	−0.125	−0.241	−0.008	0.038
Phosphatidylcholines	−0.129	−0.243	−0.014	0.031
Apolipoprotein A1	−0.175	−0.284	−0.066	0.003
Concentration of Medium HDL Particles	−0.190	−0.297	−0.083	0.003
Total Lipids in Medium HDL	−0.187	−0.294	−0.081	0.003
Phospholipids in Medium HDL	−0.188	−0.294	−0.082	0.003
Cholesterol in Medium HDL	−0.196	−0.305	−0.087	0.003
Cholesteryl Esters in Medium HDL	−0.200	−0.309	−0.091	0.003
Free Cholesterol in Medium HDL	−0.181	−0.290	−0.072	0.003
Concentration of Small HDL Particles	−0.158	−0.260	−0.057	0.004
Total Lipids in Small HDL	−0.165	−0.266	−0.063	0.003
Phospholipids in Small HDL	−0.171	−0.273	−0.070	0.003
Cholesterol in Small HDL	−0.165	−0.266	−0.064	0.003
Cholesteryl Esters in Small HDL	−0.162	−0.262	−0.063	0.003

Free Cholesterol in Small HDL	−0.150	−0.255	−0.045	0.007
Phospholipids to total lipids ratio in very small VLDL	0.123	0.018	0.228	0.026
Phospholipids to total lipids ratio in IDL	0.126	0.025	0.226	0.018

* Adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, HbA1c, and prevalent CMDs.

† Data shown as per-SD change of RNFLT (4.24 μm).

‡ *P* values were calculated through two-sided Student's t-tests followed by controlling FDR for multiple testing.

CMD = cardiometabolic disease; UKB = UK Biobank; CI = confidence interval; FDR = false discovery rate; HDL = high-density lipoprotein; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Supplementary Table S6

Associations of RNFLT-associated metabolic biomarkers with CMD outcomes in the UKB population-II, additionally adjusting for enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

Metabolic biomarkers *	Type 2 diabetes		Myocardial infarction		Heart failure		Stroke		All-cause mortality		Cardiovascular mortality	
	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡
HDL Cholesterol	0.706	<0.001	0.762	<0.001	0.929	0.007	0.918	0.020	0.984	0.373	0.904	0.005
Total Phospholipids in Lipoprotein Particles	0.919	<0.001	0.914	<0.001	0.870	<0.001	0.994	0.840	0.925	<0.001	0.902	0.001
Phospholipids in HDL	0.889	<0.001	0.812	<0.001	0.919	0.001	0.933	0.034	0.998	0.887	0.923	0.013
Cholesteryl Esters in HDL	0.688	<0.001	0.753	<0.001	0.927	0.006	0.909	0.017	0.978	0.205	0.895	0.002
Free Cholesterol in HDL	0.784	<0.001	0.807	<0.001	0.948	0.048	0.953	0.190	1.017	0.328	0.948	0.120
Total Lipids in HDL	0.834	<0.001	0.796	<0.001	0.916	0.001	0.929	0.029	0.990	0.569	0.917	0.009
Total Concentration of Lipoprotein Particles	0.845	<0.001	0.814	<0.001	0.838	<0.001	0.925	0.020	0.900	<0.001	0.843	<0.001
Concentration of HDL Particles	0.854	<0.001	0.797	<0.001	0.846	<0.001	0.913	0.017	0.912	<0.001	0.846	<0.001
Phosphoglycerides	0.977	0.109	0.905	<0.001	0.874	<0.001	0.986	0.634	0.938	<0.001	0.913	0.002

Total Cholines	0.920	<0.001	0.893	<0.001	0.870	<0.001	0.982	0.564	0.931	<0.001	0.903	0.001
Phosphatidylcholines	0.950	0.001	0.898	<0.001	0.866	<0.001	0.978	0.498	0.926	<0.001	0.894	<0.001
Apolipoprotein A1	0.861	<0.001	0.796	<0.001	0.886	<0.001	0.920	0.020	0.960	0.014	0.890	<0.001
Concentration of Medium HDL Particles	0.874	<0.001	0.790	<0.001	0.893	<0.001	0.913	0.017	0.968	0.050	0.892	0.001
Total Lipids in Medium HDL	0.921	<0.001	0.809	<0.001	0.890	<0.001	0.919	0.017	0.965	0.023	0.894	<0.001
Phospholipids in Medium HDL	0.969	0.036	0.827	<0.001	0.895	<0.001	0.926	0.020	0.972	0.072	0.905	0.001
Cholesterol in Medium HDL	0.820	<0.001	0.772	<0.001	0.894	<0.001	0.905	0.016	0.960	0.014	0.878	<0.001
Cholesteryl Esters in Medium HDL	0.811	<0.001	0.768	<0.001	0.895	<0.001	0.900	0.016	0.958	0.011	0.876	<0.001
Free Cholesterol in Medium HDL	0.858	<0.001	0.794	<0.001	0.901	<0.001	0.926	0.026	0.977	0.168	0.900	0.002
Concentration of Small HDL Particles	0.968	0.026	0.873	<0.001	0.810	<0.001	0.931	0.020	0.852	<0.001	0.819	<0.001
Total Lipids in Small HDL	1.048	0.001	0.886	<0.001	0.825	<0.001	0.939	0.029	0.881	<0.001	0.847	<0.001
Phospholipids in Small HDL	1.063	<0.001	0.884	<0.001	0.837	<0.001	0.938	0.029	0.900	<0.001	0.859	<0.001
Cholesterol in Small HDL	0.940	<0.001	0.860	<0.001	0.811	<0.001	0.926	0.018	0.851	<0.001	0.815	<0.001
Cholesteryl Esters in Small HDL	0.918	<0.001	0.855	<0.001	0.810	<0.001	0.918	0.016	0.842	<0.001	0.804	<0.001

Free Cholesterol in Small HDL	1.015	0.278	0.898	<0.001	0.844	<0.001	0.963	0.204	0.905	<0.001	0.881	<0.001
Phospholipids to total lipids ratio in very small VLDL	1.345	<0.001	1.225	<0.001	1.222	<0.001	1.075	0.020	1.194	<0.001	1.286	<0.001
Phospholipids to total lipids ratio in IDL	1.085	<0.001	1.146	<0.001	1.168	<0.001	1.037	0.199	1.134	<0.001	1.172	<0.001

* Adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

† Data shown as per-SD change of metabolic biomarkers.

‡ *P* values were calculated through two-sided Wald tests followed by controlling FDR for multiple testing.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB= UK Biobank; HR = hazard ratio; FDR = false discovery rate; HDL = high-density lipoprotein; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Supplementary Table S25

Incremental predictability of integrating RNFLT metabolic states for CMD outcome prediction beyond established models that integrated with enzymatic HDL and HbA1c, in the UKB population-II.

Basic models + enzymatic HDL + HbA1c *	C-statistics			P_{FDR} §
	Integrated models †	Basic models	Benefits ‡	
Type 2 diabetes				
FGCRS	0.868 (0.857–0.880)	0.840 (0.826–0.853)	0.028 (0.008–0.048)	<0.001
SCORE2	0.829 (0.816–0.842)	0.765 (0.749–0.781)	0.064 (0.045–0.083)	<0.001
AHS/ASCVD	0.871 (0.859–0.882)	0.847 (0.833–0.860)	0.024 (0.006–0.043)	<0.001
WHO-CVD	0.858 (0.846–0.871)	0.813 (0.798–0.828)	0.045 (0.030–0.060)	<0.001
Myocardial infarction				
FGCRS	0.779 (0.761-0.797)	0.760 (0.741–0.779)	0.018 (0.003–0.034)	0.029
SCORE2	0.762 (0.743–0.780)	0.735 (0.715–0.756)	0.026 (0.011–0.042)	<0.001
AHS/ASCVD	0.780 (0.762–0.798)	0.762 (0.743–0.781)	0.018 (0.002–0.035)	0.033
WHO-CVD	0.767 (0.748–0.786)	0.736 (0.716–0.757)	0.030 (0.016–0.044)	<0.001
Heart failure				
FGCRS	0.763 (0.744–0.782)	0.747 (0.727–0.767)	0.016 (0.001–0.032)	0.029

SCORE2	0.750 (0.730–0.769)	0.724 (0.703–0.745)	0.026 (0.009–0.044)	<0.001
AHS/ASCVD	0.766 (0.747–0.785)	0.750 (0.730–0.770)	0.016 (0.002–0.031)	0.033
WHO-CVD	0.758 (0.738–0.777)	0.735 (0.714–0.756)	0.023 (0.006–0.040)	0.005
Stroke				
FGCRS	0.727 (0.699–0.755)	0.703 (0.676–0.731)	0.023 (0.005–0.042)	0.029
SCORE2	0.723 (0.696–0.750)	0.696 (0.667–0.724)	0.027 (0.008–0.047)	0.029
AHS/ASCVD	0.726 (0.698–0.754)	0.697 (0.670–0.724)	0.029 (0.012–0.047)	0.029
WHO-CVD	0.732 (0.700–0.755)	0.712 (0.684–0.740)	0.016 (−0.004–0.036)	0.140
All-cause mortality				
FGCRS	0.737 (0.723–0.751)	0.728 (0.714–0.742)	0.010 (0.001–0.020)	0.024
SCORE2	0.736 (0.723–0.750)	0.720 (0.706–0.735)	0.016 (0.006–0.026)	<0.001
AHS/ASCVD	0.738 (0.724–0.751)	0.729 (0.716–0.743)	0.008 (0.001–0.018)	0.044
WHO-CVD	0.742 (0.728–0.756)	0.730 (0.715–0.744)	0.012 (0.003–0.021)	0.005
Cardiovascular mortality				
FGCRS	0.777 (0.753–0.801)	0.756 (0.730–0.782)	0.021 (0.006–0.037)	0.007
SCORE2	0.769 (0.745–0.793)	0.737 (0.709–0.766)	0.032 (0.017–0.048)	0.014
AHS/ASCVD	0.778 (0.754–0.802)	0.754 (0.728–0.779)	0.024 (0.008–0.041)	0.045

WHO-CVD	0.773 (0.749–0.797)	0.742 (0.712–0.771)	0.031 (0.016–0.046)	0.015

* Basic models for CMD outcomes include minimal demographic characteristics (Age&Sex), Framingham General Cardiovascular Risk Score (FGCRS), Systematic Coronary Risk Evaluation 2 (SCORE2) algorithms, American Heart Association/Atherosclerotic Cardiovascular Disease (AHA/ASCVD) algorithms, and World Health Organization Cardiovascular Disease (WHO-CVD) risk charts. All models were integrated with enzymatic HDL and HbA1c.

† Models that integrated RNFLT metabolic states.

‡ Confidential intervals were calculated by bootstrapping with 1,000 iterations followed by controlling FDR for multiple testing.

§ *P* values were calculated through two-sided non-parametric bootstrap tests with 1,000 iterations followed by controlling FDR for multiple testing. Adjusted *P* were set identical when their step-up thresholds were truncated to a shared minimum.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB = UK Biobank; HDL = high-density lipoprotein; FDR = false discovery rate.

Supplementary Table S26

Incremental predictability of integrating RNFLT metabolic states for CMD outcome prediction beyond comprehensive models combining all predictors from each clinical model, in the UKB population-II.

Outcomes	C-statistics			P_{FDR} §
	Integrated models *	Basic models †	Benefits ‡	
Type 2 diabetes	0.872 (0.860–0.883)	0.851 (0.838–0.864)	0.021 (0.013–0.028)	<0.001
Myocardial infarction	0.782 (0.764–0.800)	0.765 (0.746–0.783)	0.017 (0.005–0.029)	0.036
Heart failure	0.768 (0.749–0.787)	0.753 (0.733–0.772)	0.016 (0.003–0.029)	0.029
Stroke	0.728 (0.700–0.756)	0.712 (0.685–0.739)	0.016 (−0.002–0.034)	0.106
All-cause mortality	0.739 (0.725–0.753)	0.731 (0.717–0.744)	0.009 (0.002–0.019)	0.019
Cardiovascular mortality	0.780 (0.757–0.804)	0.756 (0.731–0.783)	0.024 (0.011–0.037)	0.032

* Models that integrated RNFLT metabolic states.

† Comprehensive baseline models that combined all predictors from each clinical model.

‡ Confidential intervals were calculated by bootstrapping with 1,000 iterations followed by controlling FDR for multiple testing.

§ *P* values were calculated through two-sided non-parametric bootstrap tests with 1,000 iterations followed by controlling FDR for multiple testing. Adjusted *P* were set identical when their step-up thresholds were truncated to a shared minimum.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB = UK Biobank; FDR = false discovery rate.

Appendix Table A1

C statistic (95% CI) for QR4, ASCVD and SCORE2 in people aged 40 and older in the England validation cohort overall and by ethnic group, from Hippisley-Cox et al.[1], ***Nature Medicine*** (2024), *Extended Data Table 4*.

Subgroups	QR4 *	ASCVD	SCORE2
Women			
Overall	0.781 (0.778–0.784)	0.767 (0.764–0.770)	0.767 (0.764–0.770)
White	0.777 (0.773–0.780)	0.764 (0.761–0.767)	0.763 (0.760–0.767)
Indian	0.802 (0.785–0.820)	0.792 (0.774–0.810)	0.792 (0.773–0.811)
Pakistani	0.779 (0.755–0.803)	0.751 (0.724–0.778)	0.754 (0.727–0.780)
Bangladeshi	0.739 (0.698–0.779)	0.714 (0.669–0.759)	0.710 (0.664–0.757)
Other Asian	0.800 (0.777–0.824)	0.791 (0.766–0.816)	0.794 (0.770–0.819)
Caribbean	0.771 (0.751–0.791)	0.754 (0.732–0.775)	0.750 (0.728–0.773)
Black African	0.790 (0.766–0.813)	0.772 (0.747–0.797)	0.762 (0.735–0.789)
Chinese	0.863 (0.822–0.905)	0.856 (0.814–0.898)	0.849 (0.807–0.891)
Other	0.771 (0.750–0.791)	0.747 (0.724–0.769)	0.751 (0.728–0.773)
Men			
Overall	0.741 (0.739–0.744)	0.727 (0.725–0.730)	0.728 (0.725–0.730)

White	0.737 (0.734–0.740)	0.727 (0.724–0.730)	0.725 (0.722–0.728)
Indian	0.736 (0.720–0.753)	0.727 (0.710–0.743)	0.724 (0.708–0.741)
Pakistani	0.735 (0.714–0.756)	0.721 (0.699–0.742)	0.718 (0.696–0.740)
Bangladeshi	0.701 (0.674–0.729)	0.692 (0.664–0.721)	0.687 (0.659–0.716)
Other Asian	0.729 (0.706–0.752)	0.712 (0.688–0.736)	0.711 (0.686–0.735)
Caribbean	0.735 (0.714–0.756)	0.730 (0.708–0.751)	0.725 (0.703–0.746)
Black African	0.745 (0.724–0.767)	0.733 (0.712–0.755)	0.710 (0.686–0.733)
Chinese	0.759 (0.704–0.814)	0.741 (0.685–0.798)	0.735 (0.677–0.793)
Other	0.743 (0.727–0.760)	0.725 (0.708–0.742)	0.721 (0.703–0.738)

* QR4 is their new model, and ASCVD and SCORE2 are the baseline models for comparison.

Reference

1. Hippisley-Cox, J. et al. Development and validation of a new algorithm for improved cardiovascular risk prediction. *Nat. Med.* **30**, 1440-1447 (2024).

Supplementary Table S16

Reclassification improvement of integrating RNFLT metabolic states in the UKB.

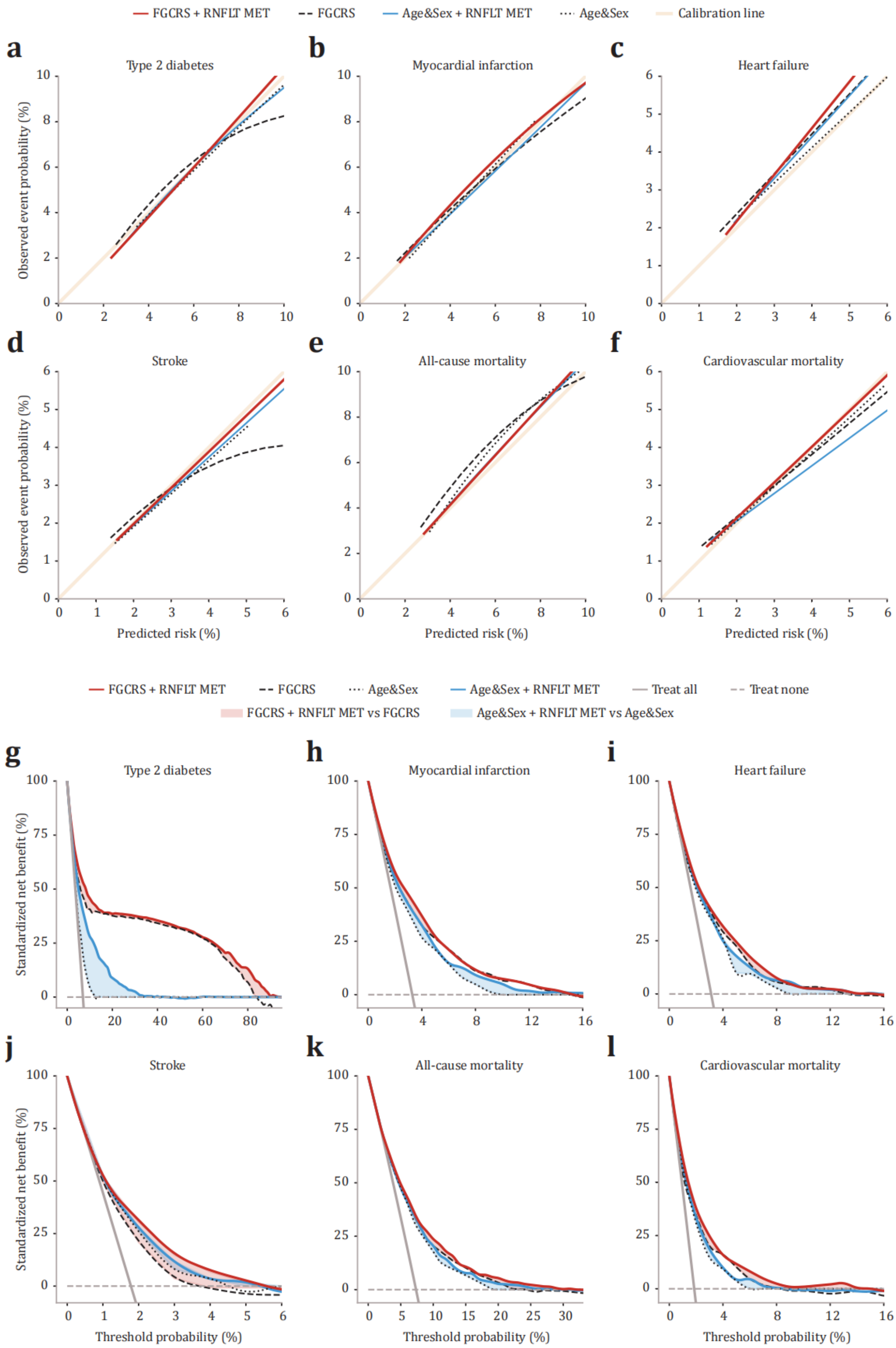
Baseline models *	NRI (case)	NRI (non-case)	Additive NRI
Type 2 diabetes			
Age&Sex	0.286	0.243	0.529
FGCRS	0.080	−0.010	0.070
SCORE2	0.204	0.095	0.299
AHA/ASCVD	0.117	0.006	0.123
WHO-CVD	0.094	−0.022	0.072
Myocardial infarction			
Age&Sex	0.089	−0.015	0.074
FGCRS	0.011	0.001	0.011
SCORE2	0.021	0.020	0.041
AHA/ASCVD	0.005	0.048	0.053
WHO-CVD	0.011	0.007	0.017
Heart failure			
Age&Sex	0.054	0.034	0.089
FGCRS	0.006	0.010	0.015
SCORE2	0.004	0.028	0.032
AHA/ASCVD	0.009	0.020	0.030
WHO-CVD	0.006	0.010	0.016
Stroke			
Age&Sex	0.095	0.047	0.142
FGCRS	0.003	0.020	0.024
SCORE2	0.010	0.020	0.030
AHA/ASCVD	0.042	0.023	0.066
WHO-CVD	0.039	0.027	0.066
All-cause mortality			
Age&Sex	0.028	−0.010	0.017

FGCRS	0.001	0.006	0.007
SCORE2	0.016	0.018	0.033
AHA/ASCVD	0.025	0.045	0.069
WHO-CVD	0.007	0.003	0.010
Cardiovascular mortality			
Age&Sex	0.072	0.031	0.103
FGCRS	0.022	0.010	0.032
SCORE2	0.050	0.021	0.071
AHA/ASCVD	0.072	0.022	0.093
WHO-CVD	0.016	0.012	0.027

* Basic models for CMD outcomes include minimal demographic characteristics (Age&Sex), Framingham General Cardiovascular Risk Score (FGCRS), Systematic Coronary Risk Evaluation 2 (SCORE2) algorithms, American Heart Association/Atherosclerotic Cardiovascular Disease (AHA/ASCVD) algorithms, and World Health Organization Cardiovascular Disease (WHO-CVD) risk charts. RNFLT = retinal nerve fibre layer thickness; UKB = UK Biobank; NRI = net classification improvement.

Fig. 4

Model calibration and clinical utility in the UKB.



Comment #4. Link between RNFLT is tenuous since all metabolites are associated with CMD (and they don't show that RNFLT is associated with incident CMD); this likely reflects confounders that are not accounted for in the models (HTN, BP, individual prevalent CMD, HgA1c, HDL).

Response: We sincerely appreciate your thoughtful comment. We acknowledge that some key confounders were not adequately addressed in our initial submission. To address this concern, we conducted a series of sensitivity analyses to sequentially adjust for additional covariates in our primary analyses, including: (1) further adjusted for HDL, systolic blood pressure, and HbA1c in the identification of RNFLT-associated biomarkers; and (2) additionally adjusted for prevalent CMD in these models. Although these adjustments attenuated the observed associations, all associations remained robust and directionally consistent, further reinforcing the validity of our findings (***as detailed below***). In addition to these sensitivity analyses, we recognize the importance of establishing the direct association between RNFLT and incident CMD outcomes to support the interpretation of our findings. While the link between RNFLT and incident CMD has already been reported in prior studies (as referenced in the ***Introduction*** section [***Page 5, Line 110***]), we have now explicitly presented these associations in ***Table 1*** to provide further clarity (provided below for your easy reference). After adjusting for the same covariates mentioned above, we found that for each SD thinning in RNFLT corresponds to future risk reductions of 15% for T2D, 14% for myocardial infarction, 21% for heart failure, 9% for stroke, 13% for all-cause mortality, and 17% for cardiovascular mortality. These results align with previously reported evidence and further support the role of RNFLT as a relevant biomarker in incident cardiometabolic disease.

Despite the expected attenuation in the associations between NMR metabolic biomarkers and RNFLT after adjusting for all suggested key confounders, all associations remained statistically significant and directionally consistent. Specifically, when adjusting for HDL, systolic blood pressure, and HbA1c, the attenuated yet consistent associations suggest that only part of the associations were confounded by these factors (***Supplementary Table S3,*** provided below for your easy reference). Notably, further adjustment for prevalent CMD produced minimal change in these associations, suggesting that the RNFLT–metabolite associations are unlikely to be driven by residual confounding from CMD status (***Supplementary Table S4,*** provided below for your easy reference). These robust findings reinforce the notion that these metabolic biomarkers reflect distinct pathways rather than mere confounding from known risk factors.

Importantly, while adjusting for HDL weakened the associations, they remained robust. This aligns with the known biological distinction between NMR-based and enzymatic HDL assays. While enzymatic HDL assays measure concentration, NMR-derived HDL assay captures unique particle properties that extend beyond the traditional enzymatic assay. Specifically, the NMR-based HDL assay applies deconvolution of diffusion and relaxation properties, integrating both physical properties (e.g., particle size, lipid surface-to-core ratio) and lipid composition (e.g., phospholipids, cholesterol esters, free cholesterol, and apolipoprotein A1) — characteristics that are increasingly recognized as key determinants of HDL function. For instance, the higher molecular mobility of cholesterol in small HDL particles extends their T2 relaxation times, resulting in reduced linewidth attenuation in the presaturated proton spectra and thus stronger signals than their larger counterparts. Thus, while both assays share some variation (concentration), the NMR-based assays capture additional facets of HDL structure and composition that are likely relevant to the RNFLT–CMD link.

Overall, these sensitivity analyses effectively addressed concerns about residual confounding. While the mechanistic distinction explains why NMR-based HDL captures unique biological variability that extends beyond traditional enzymatic HDL measurements, the robust results strongly suggests that our findings are unlikely to be explained by the confounding from known risk factors, including

HDL concentration. We hope these additional analyses and clarifications have adequately addressed your concerns.

Supplementary Table S3

Identification of RNFLT-associated metabolic biomarkers in the UKB population-I, additionally adjusting for enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

Metabolic biomarkers *	**β †**	**95% CI**		***P_{FDR}* ‡**
HDL Cholesterol	−0.163	−0.277	−0.049	0.007
Total Phospholipids in Lipoprotein Particles	−0.117	−0.229	−0.005	0.041
Phospholipids in HDL	−0.167	−0.277	−0.058	0.005
Cholesteryl Esters in HDL	−0.168	−0.282	−0.055	0.006
Free Cholesterol in HDL	−0.142	−0.254	−0.029	0.017
Total Lipids in HDL	−0.163	−0.274	−0.052	0.006
Total Concentration of Lipoprotein Particles	−0.183	−0.291	−0.076	0.002
Concentration of HDL Particles	−0.190	−0.295	−0.085	0.002
Phosphoglycerides	−0.129	−0.242	−0.016	0.027
Total Cholines	−0.128	−0.244	−0.012	0.032
Phosphatidylcholines	−0.132	−0.245	−0.018	0.027
Apolipoprotein A1	−0.177	−0.285	−0.068	0.003
Concentration of Medium HDL Particles	−0.192	−0.298	−0.085	0.002
Total Lipids in Medium HDL	−0.190	−0.296	−0.083	0.002
Phospholipids in Medium HDL	−0.191	−0.296	−0.085	0.002
Cholesterol in Medium HDL	−0.197	−0.305	−0.089	0.002
Cholesteryl Esters in Medium HDL	−0.201	−0.309	−0.093	0.002
Free Cholesterol in Medium HDL	−0.183	−0.291	−0.074	0.002
Concentration of Small HDL Particles	−0.163	−0.264	−0.062	0.003
Total Lipids in Small HDL	−0.170	−0.271	−0.069	0.002
Phospholipids in Small HDL	−0.177	−0.278	−0.076	0.002
Cholesterol in Small HDL	−0.169	−0.269	−0.069	0.002

Cholesteryl Esters in Small HDL	−0.166	−0.265	−0.067	0.002
Free Cholesterol in Small HDL	−0.155	−0.259	−0.050	0.006
Phospholipids to total lipids ratio in very small VLDL	0.120	0.015	0.224	0.027
Phospholipids to total lipids ratio in IDL	0.126	0.026	0.227	0.017

* Adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

† Data shown as per-SD change of RNFLT (4.24 µm).

‡ *P* values were calculated through two-sided Student's t-tests followed by controlling FDR for multiple testing.

HDL = high-density lipoprotein; UKB = UK Biobank; CI = confidence interval; FDR = false discovery rate; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Supplementary Table S4

Identification of RNFLT-associated metabolic biomarkers in the UKB population-I, additionally adjusting for prevalent CMDs.

Metabolic biomarkers *	**β** †	**95% CI**		***P**FDR* ‡
HDL Cholesterol	−0.164	−0.279	−0.050	0.007
Total Phospholipids in Lipoprotein Particles	−0.114	−0.227	−0.001	0.048
Phospholipids in HDL	−0.166	−0.276	−0.056	0.005
Cholesteryl Esters in HDL	−0.170	−0.284	−0.056	0.006
Free Cholesterol in HDL	−0.142	−0.255	−0.029	0.018
Total Lipids in HDL	−0.162	−0.274	−0.051	0.007
Total Concentration of Lipoprotein Particles	−0.181	−0.289	−0.073	0.003
Concentration of HDL Particles	−0.188	−0.294	−0.082	0.003
Phosphoglycerides	−0.125	−0.239	−0.012	0.033
Total Cholines	−0.125	−0.241	−0.008	0.038
Phosphatidylcholines	−0.129	−0.243	−0.014	0.031
Apolipoprotein A1	−0.175	−0.284	−0.066	0.003
Concentration of Medium HDL Particles	−0.190	−0.297	−0.083	0.003
Total Lipids in Medium HDL	−0.187	−0.294	−0.081	0.003
Phospholipids in Medium HDL	−0.188	−0.294	−0.082	0.003
Cholesterol in Medium HDL	−0.196	−0.305	−0.087	0.003
Cholesteryl Esters in Medium HDL	−0.200	−0.309	−0.091	0.003
Free Cholesterol in Medium HDL	−0.181	−0.290	−0.072	0.003
Concentration of Small HDL Particles	−0.158	−0.260	−0.057	0.004
Total Lipids in Small HDL	−0.165	−0.266	−0.063	0.003
Phospholipids in Small HDL	−0.171	−0.273	−0.070	0.003
Cholesterol in Small HDL	−0.165	−0.266	−0.064	0.003
Cholesteryl Esters in Small HDL	−0.162	−0.262	−0.063	0.003

Free Cholesterol in Small HDL	−0.150	−0.255	−0.045	0.007
Phospholipids to total lipids ratio in very small VLDL	0.123	0.018	0.228	0.026
Phospholipids to total lipids ratio in IDL	0.126	0.025	0.226	0.018

* Adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, HbA1c, and prevalent CMDs.

† Data shown as per-SD change of RNFLT (4.24 μm).

‡ *P* values were calculated through two-sided Student's t-tests followed by controlling FDR for multiple testing.

CMD = cardiometabolic disease; UKB = UK Biobank; CI = confidence interval; FDR = false discovery rate; HDL = high-density lipoprotein; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Table 1. The mediation effect of RNFLT metabolic states on the association between RNFLT and CMD risks in the UKB.

Outcomes *	**Base models †**		**Plus RNFLT metabolic states**		**Percentage §**
	HR (95% CI)	***P*$_{FDR}$ ‡**	**HR (95% CI)**	***P*$_{FDR}$ ‡**	
Type 2 diabetes	0.85 (0.81 to 0.89)	**2.73×10^{-10}**	0.94 (0.83 to 1.06)	0.391	63.69%
Myocardial infarction	0.86 (0.80 to 0.93)	**3.02×10^{-4}**	0.90 (0.77 to 1.06)	0.391	31.39%
Heart failure	0.79 (0.72 to 0.85)	**3.83×10^{-8}**	0.83 (0.70 to 0.99)	0.208	24.07%
Stroke	0.91 (0.82 to 1.01)	0.074	0.93 (0.75 to 1.15)	0.482	18.06%
All-cause mortality	0.87 (0.82 to 0.92)	**7.50×10^{-6}**	0.89 (0.79 to 1.00)	0.208	16.97%
Cardiovascular mortality	0.83 (0.73 to 0.95)	**9.42×10^{-3}**	0.87 (0.68 to 1.12)	0.391	23.10%

* Data are presented as per-SD change in RNFLT (4.24 μm).

† Base model was adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

‡ *P* values were calculated through two-sided Wald tests followed by controlling FDR for multiple testing. Bold indicates significance. Adjusted *P* were set identical when their step-up thresholds were truncated to a shared minimum.

§ Percent change in HR coefficient.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB = UK Biobank; FDR = false positive rate.

Comment #5. Clinical models need to include more granular data like HgA1C that may reflect residual confounding; also need to include traditional cholesterol parameters including HDL); Framingham risk score not a very comprehensive risk score (need to compare to the clinical model from the discovery dataset).

Response: We appreciate your guidance on incorporating more granular data including HbA1c and HDL into the baseline models. Even with the exception of the FGCRS, we acknowledge that certain known predictors were not fully leveraged in these models (SCORE2, AHA/ASCVD, and WHO-CVD), despite the fact that they are well-established and widely endorsed by international guidelines from the American Heart Association (AHA),[1] the European Society of Cardiology (ESC),[2] and the World Health Organization (WHO).[33] To address this, we incorporated additional predictors and re-ran all analyses, including: (1) integrating HbA1c and enzymatic HDL into all baseline clinical models; and (2) constructing a comprehensive model that incorporates all predictors used across all considered models. We then compared the performance of models integrating RNFLT metabolic states with all baseline models. These additional analyses provide a more rigorous assessment of the independent predictive value of RNFLT metabolic states beyond conventional risk factors. Our results demonstrate that even after enhancing the clinical baseline with more granular predictors, the predictive improvement gained from integrating RNFLT metabolic states remains robust.

Incorporating HbA1c and enzymatic HDL led to a numerical improvement in the performance of all clinical baseline models (***Supplementary Table S25***, provided below for your easy reference). Among all CMD outcomes, the largest gain from adding HbA1c and enzymatic HDL was observed in the prediction of T2D, which is expected given that HbA1c is one of the most important risk factors for T2D. Specifically, the baseline C-statistic for the FGCRS model in predicting T2D increased from 0.810 to 0.840 when incorporating HbA1c and enzymatic HDL, approaching the performance of the FRS-for-T2D model, which was specifically designed for T2D risk assessment.[4,5] Similarly, among all baseline models considered, the WHO-CVD model exhibited the greatest improvement after incorporating these additional predictors. This is likely because WHO-CVD is the only model that does not originally include enzymatic HDL. Although the absolute improvement from integrating RNFLT metabolic states was slightly attenuated after incorporating HbA1c and enzymatic HDL (e.g., ΔC-statistic decreased from 0.044 to 0.028 for T2D over FGCRS; from 0.020 to 0.018 for myocardial infarction over FGCRS), the improvement remained statistically significant and consistent across different baseline models (***Supplementary Table S25***). This finding underscores that RNFLT metabolic states capture risk information beyond that provided by traditional predictors, even in the context of an enhanced clinical baseline.

To further evaluate the incremental predictive value of RNFLT metabolic states, we constructed a comprehensive model that integrates all individual predictors from each baseline clinical model considered (***Supplementary Table S26***, provided below for your easy reference). While this model achieved the highest overall predictive performance, adding additional conventional risk factors beyond those in the validated baseline models yielded marginal improvements. This further supports the effectiveness and robustness of the predictor sets originally used in this manuscript. Notably, the improvement in predicting stroke did not reach the significance threshold for multiple comparison. However, for all other CMD outcomes, the predictive improvement of integrating RNFLT metabolic states over the comprehensive model remained significant and robust. We hope this additional analysis adequately address your concerns.

References

1. Goff, D. C. et al. 2013 ACC/AHA Guideline on the Assessment of Cardiovascular Risk. *J. Am. Coll. Cardiol.* **63**, 2935-2959 (2014).
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estimate 10-year risk of cardiovascular disease in Europe. *Eur. Heart J.* **42**, 2439-2454 (2021).

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4. Bragg, F. et al. Predictive value of circulating NMR metabolic biomarkers for type 2 diabetes risk in the UK Biobank study. *BMC Med.* **20**, 159 (2022).
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Supplementary Table S25

Incremental predictability of integrating RNFLT metabolic states for CMD outcome prediction beyond established models that integrated with enzymatic HDL and HbA1c, in the UKB population-II.

Basic models + enzymatic HDL + HbA1c *	C-statistics			P_{FDR} §
	Integrated models †	Basic models	Benefits ‡	
Type 2 diabetes				
FGCRS	0.868 (0.857–0.880)	0.840 (0.826–0.853)	0.028 (0.008–0.048)	<0.001
SCORE2	0.829 (0.816–0.842)	0.765 (0.749–0.781)	0.064 (0.045–0.083)	<0.001
AHS/ASCVD	0.871 (0.859–0.882)	0.847 (0.833–0.860)	0.024 (0.006–0.043)	<0.001
WHO-CVD	0.858 (0.846–0.871)	0.813 (0.798–0.828)	0.045 (0.030–0.060)	<0.001
Myocardial infarction				
FGCRS	0.779 (0.761-0.797)	0.760 (0.741–0.779)	0.018 (0.003–0.034)	0.029
SCORE2	0.762 (0.743–0.780)	0.735 (0.715–0.756)	0.026 (0.011–0.042)	<0.001
AHS/ASCVD	0.780 (0.762–0.798)	0.762 (0.743–0.781)	0.018 (0.002–0.035)	0.033
WHO-CVD	0.767 (0.748–0.786)	0.736 (0.716–0.757)	0.030 (0.016–0.044)	<0.001
Heart failure				
FGCRS	0.763 (0.744–0.782)	0.747 (0.727–0.767)	0.016 (0.001–0.032)	0.029

SCORE2	0.750 (0.730–0.769)	0.724 (0.703–0.745)	0.026 (0.009–0.044)	<0.001
AHS/ASCVD	0.766 (0.747–0.785)	0.750 (0.730–0.770)	0.016 (0.002–0.031)	0.033
WHO-CVD	0.758 (0.738–0.777)	0.735 (0.714–0.756)	0.023 (0.006–0.040)	0.005
Stroke				
FGCRS	0.727 (0.699–0.755)	0.703 (0.676–0.731)	0.023 (0.005–0.042)	0.029
SCORE2	0.723 (0.696–0.750)	0.696 (0.667–0.724)	0.027 (0.008–0.047)	0.029
AHS/ASCVD	0.726 (0.698–0.754)	0.697 (0.670–0.724)	0.029 (0.012–0.047)	0.029
WHO-CVD	0.732 (0.700–0.755)	0.712 (0.684–0.740)	0.016 (−0.004–0.036)	0.140
All-cause mortality				
FGCRS	0.737 (0.723–0.751)	0.728 (0.714–0.742)	0.010 (0.001–0.020)	0.024
SCORE2	0.736 (0.723–0.750)	0.720 (0.706–0.735)	0.016 (0.006–0.026)	<0.001
AHS/ASCVD	0.738 (0.724–0.751)	0.729 (0.716–0.743)	0.008 (0.001–0.018)	0.044
WHO-CVD	0.742 (0.728–0.756)	0.730 (0.715–0.744)	0.012 (0.003–0.021)	0.005
Cardiovascular mortality				
FGCRS	0.777 (0.753–0.801)	0.756 (0.730–0.782)	0.021 (0.006–0.037)	0.007
SCORE2	0.769 (0.745–0.793)	0.737 (0.709–0.766)	0.032 (0.017–0.048)	0.014
AHS/ASCVD	0.778 (0.754–0.802)	0.754 (0.728–0.779)	0.024 (0.008–0.041)	0.045

WHO-CVD	0.773 (0.749–0.797)	0.742 (0.712–0.771)	0.031 (0.016–0.046)	0.015

* Basic models for CMD outcomes include minimal demographic characteristics (Age&Sex), Framingham General Cardiovascular Risk Score (FGCRS), Systematic Coronary Risk Evaluation 2 (SCORE2) algorithms, American Heart Association/Atherosclerotic Cardiovascular Disease (AHA/ASCVD) algorithms, and World Health Organization Cardiovascular Disease (WHO-CVD) risk charts. All models were integrated with enzymatic HDL and HbA1c.

† Models that integrated RNFLT metabolic states.

‡ Confidential intervals were calculated by bootstrapping with 1,000 iterations followed by controlling FDR for multiple testing.

§ *P* values were calculated through two-sided non-parametric bootstrap tests with 1,000 iterations followed by controlling FDR for multiple testing. Adjusted *P* were set identical when their step-up thresholds were truncated to a shared minimum.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB = UK Biobank; HDL = high-density lipoprotein; FDR = false discovery rate.

Supplementary Table S26

Incremental predictability of integrating RNFLT metabolic states for CMD outcome prediction beyond comprehensive models combining all predictors from each clinical model, in the UKB population-II.

Outcomes	C-statistics			P_{FDR} §
	Integrated models *	Basic models †	Benefits ‡	
Type 2 diabetes	0.872 (0.860–0.883)	0.851 (0.838–0.864)	0.021 (0.013–0.028)	<0.001
Myocardial infarction	0.782 (0.764–0.800)	0.765 (0.746–0.783)	0.017 (0.005–0.029)	0.036
Heart failure	0.768 (0.749–0.787)	0.753 (0.733–0.772)	0.016 (0.003–0.029)	0.029
Stroke	0.728 (0.700–0.756)	0.712 (0.685–0.739)	0.016 (−0.002–0.034)	0.106
All-cause mortality	0.739 (0.725–0.753)	0.731 (0.717–0.744)	0.009 (0.002–0.019)	0.019
Cardiovascular mortality	0.780 (0.757–0.804)	0.756 (0.731–0.783)	0.024 (0.011–0.037)	0.032

* Models that integrated RNFLT metabolic states.

† Comprehensive baseline models that combined all predictors from each clinical model.

‡ Confidential intervals were calculated by bootstrapping with 1,000 iterations followed by controlling FDR for multiple testing.

§ *P* values were calculated through two-sided non-parametric bootstrap tests with 1,000 iterations followed by controlling FDR for multiple testing. Adjusted *P* were set identical when their step-up thresholds were truncated to a shared minimum.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB = UK Biobank; FDR = false discovery rate.

Comment #6. How was prevalent CVD/DM addressed?

Response: We sincerely appreciate you raising this important point. We have carefully addressed prevalent CMD (CVD/DM) status by excluding them in analyses involving incident CMD outcomes, while incorporating prevalent CMD as a covariate in other analyses where appropriate:

(1) For analyses involving incident CMD outcomes, we excluded participants with the corresponding prevalent CMD diagnosis at baseline to ensure that the outcomes reflected true incident events. For example, in the analysis examining the association between RNFLT-associated metabolites and incident T2D, we excluded individuals with T2D at baseline. Similarly, participants with baseline CMD were excluded from all mediation analysis, predictive analysis, reclassification analysis, and decision analysis to ensure that only new-onset events were considered.

(2) For analyses that did not directly involve incident CMD outcomes — such as the identification of RNFLT-associated metabolites (where RNFLT was the exposure and metabolites were the outcome) — participants with prevalent CMD were not excluded. Instead, to account for potential confounding by CMD status, we performed a sensitivity analysis in which prevalent CMD was included as a covariate in the models. This ensured that our findings remained robust and were not influenced by CMD status.

We acknowledge that this important methodological detail was not clearly conveyed in our initial submission. To address this, we have now revised the ***Methods*** section to provide a clearer explanation of how prevalent CMD was handled across different analyses (***Page 19, Line 542*** and ***Page 17, Line 459***). We hope this clarification addresses your concern.

Comment #7. Not clear that Cox models and sex/SDoH stratified models are adjusted for covariates?

Response: We sincerely apologize for not clearly conveying this important information in our initial submission. We employed Cox models to assess the longitudinal associations between RNFLT-associated metabolic biomarkers and incident CMD outcomes, with all covariates adjusted consistently as described in our primary analyses (age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, and use of lipid-lowering medications). To further address your concerns regarding potential confounding factors, we performed additional analyses that further adjusted for HDL, systolic blood pressure, and HbA1c in the identification of RNFLT-associated biomarkers (***Supplementary Table S6***). All findings remained stable, supporting that the observed associations were unlikely to be driven by residual confounding from known risk factors.

In addition, our initial submission did not include sex- and SDoH-stratified Cox models, and we greatly appreciate your suggestion to incorporate these analyses. To address this, we performed three additional stratified analyses, including: (1) sex-stratified analysis (female and male); (2) deprivation-stratified analysis (higher deprivation vs. lower deprivation); and (3) educational attainment-stratified analysis (non-university vs. university). All stratified models were adjusted for the same covariates as in our primary analyses including HDL, systolic blood pressure, and HbA1c. We are pleased to report that the associations observed in these stratified models remained consistent and robust across all subgroups. These results are now presented in ***Supplementary Table S7–S9*** and shown below for your easy reference. We hope that these additional analyses and clarifications adequately address your concerns.

Supplementary Table S6

Associations of RNFLT-associated metabolic biomarkers with CMD outcomes in the UKB population-II, additionally adjusting for enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

Metabolic biomarkers *	Type 2 diabetes		Myocardial infarction		Heart failure		Stroke		All-cause mortality		Cardiovascular mortality	
	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡
HDL Cholesterol	0.706	<0.001	0.762	<0.001	0.929	0.007	0.918	0.020	0.984	0.373	0.904	0.005
Total Phospholipids in Lipoprotein Particles	0.919	<0.001	0.914	<0.001	0.870	<0.001	0.994	0.840	0.925	<0.001	0.902	0.001
Phospholipids in HDL	0.889	<0.001	0.812	<0.001	0.919	0.001	0.933	0.034	0.998	0.887	0.923	0.013
Cholesteryl Esters in HDL	0.688	<0.001	0.753	<0.001	0.927	0.006	0.909	0.017	0.978	0.205	0.895	0.002
Free Cholesterol in HDL	0.784	<0.001	0.807	<0.001	0.948	0.048	0.953	0.190	1.017	0.328	0.948	0.120
Total Lipids in HDL	0.834	<0.001	0.796	<0.001	0.916	0.001	0.929	0.029	0.990	0.569	0.917	0.009
Total Concentration of Lipoprotein Particles	0.845	<0.001	0.814	<0.001	0.838	<0.001	0.925	0.020	0.900	<0.001	0.843	<0.001
Concentration of HDL Particles	0.854	<0.001	0.797	<0.001	0.846	<0.001	0.913	0.017	0.912	<0.001	0.846	<0.001
Phosphoglycerides	0.977	0.109	0.905	<0.001	0.874	<0.001	0.986	0.634	0.938	<0.001	0.913	0.002

Total Cholines	0.920	<0.001	0.893	<0.001	0.870	<0.001	0.982	0.564	0.931	<0.001	0.903	0.001
Phosphatidylcholines	0.950	0.001	0.898	<0.001	0.866	<0.001	0.978	0.498	0.926	<0.001	0.894	<0.001
Apolipoprotein A1	0.861	<0.001	0.796	<0.001	0.886	<0.001	0.920	0.020	0.960	0.014	0.890	<0.001
Concentration of Medium HDL Particles	0.874	<0.001	0.790	<0.001	0.893	<0.001	0.913	0.017	0.968	0.050	0.892	0.001
Total Lipids in Medium HDL	0.921	<0.001	0.809	<0.001	0.890	<0.001	0.919	0.017	0.965	0.023	0.894	<0.001
Phospholipids in Medium HDL	0.969	0.036	0.827	<0.001	0.895	<0.001	0.926	0.020	0.972	0.072	0.905	0.001
Cholesterol in Medium HDL	0.820	<0.001	0.772	<0.001	0.894	<0.001	0.905	0.016	0.960	0.014	0.878	<0.001
Cholesteryl Esters in Medium HDL	0.811	<0.001	0.768	<0.001	0.895	<0.001	0.900	0.016	0.958	0.011	0.876	<0.001
Free Cholesterol in Medium HDL	0.858	<0.001	0.794	<0.001	0.901	<0.001	0.926	0.026	0.977	0.168	0.900	0.002
Concentration of Small HDL Particles	0.968	0.026	0.873	<0.001	0.810	<0.001	0.931	0.020	0.852	<0.001	0.819	<0.001
Total Lipids in Small HDL	1.048	0.001	0.886	<0.001	0.825	<0.001	0.939	0.029	0.881	<0.001	0.847	<0.001
Phospholipids in Small HDL	1.063	<0.001	0.884	<0.001	0.837	<0.001	0.938	0.029	0.900	<0.001	0.859	<0.001
Cholesterol in Small HDL	0.940	<0.001	0.860	<0.001	0.811	<0.001	0.926	0.018	0.851	<0.001	0.815	<0.001
Cholesteryl Esters in Small HDL	0.918	<0.001	0.855	<0.001	0.810	<0.001	0.918	0.016	0.842	<0.001	0.804	<0.001

Free Cholesterol in Small HDL	1.015	0.278	0.898	<0.001	0.844	<0.001	0.963	0.204	0.905	<0.001	0.881	<0.001
Phospholipids to total lipids ratio in very small VLDL	1.345	<0.001	1.225	<0.001	1.222	<0.001	1.075	0.020	1.194	<0.001	1.286	<0.001
Phospholipids to total lipids ratio in IDL	1.085	<0.001	1.146	<0.001	1.168	<0.001	1.037	0.199	1.134	<0.001	1.172	<0.001

* Adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

† Data shown as per-SD change of metabolic biomarkers.

‡ *P* values were calculated through two-sided Wald tests followed by controlling FDR for multiple testing.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB= UK Biobank; HR = hazard ratio; FDR = false discovery rate; HDL = high-density lipoprotein; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Supplementary Table S7

Associations of RNFLT-associated metabolic biomarkers with CMD outcomes in the UKB population-II, stratified by sex.

Metabolic biomarkers *†	Type 2 diabetes		Myocardial infarction		Heart failure		Stroke		All-cause mortality		Cardiovascular mortality	
	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low
HDL Cholesterol	0.634	0.716	0.701	0.786	0.857	0.968	0.865	0.944	0.958	0.950	0.842	0.888
Total Phospholipids in Lipoprotein Particles	0.875	0.875	0.852	0.893	0.827	0.890	0.890	0.989	0.936	0.906	0.917	0.871
Phospholipids in HDL	0.778	0.872	0.742	0.817	0.862	0.948	0.862	0.972	0.986	0.962	0.908	0.886
Cholesteryl Esters in HDL	0.623	0.702	0.698	0.779	0.856	0.967	0.865	0.932	0.951	0.945	0.830	0.884
Free Cholesterol in HDL	0.695	0.777	0.729	0.817	0.880	0.986	0.887	0.992	0.995	0.982	0.909	0.912
Total Lipids in HDL	0.729	0.823	0.726	0.808	0.854	0.948	0.858	0.965	0.974	0.955	0.887	0.885
Total Concentration of Lipoprotein Particles	0.777	0.804	0.767	0.809	0.799	0.876	0.862	0.960	0.914	0.875	0.828	0.829
Concentration of HDL Particles	0.780	0.814	0.754	0.797	0.807	0.883	0.858	0.945	0.923	0.883	0.830	0.831
Phosphoglycerides	0.923	0.928	0.841	0.886	0.830	0.895	0.890	0.964	0.951	0.915	0.936	0.876
Total Cholines	0.861	0.874	0.826	0.876	0.821	0.895	0.884	0.988	0.940	0.909	0.911	0.870

Phosphatidylcholines	0.888	0.901	0.841	0.877	0.821	0.891	0.889	0.969	0.939	0.903	0.916	0.857
Apolipoprotein A1	0.767	0.834	0.740	0.801	0.838	0.917	0.854	0.956	0.958	0.926	0.871	0.862
Concentration of Medium HDL Particles	0.782	0.844	0.745	0.790	0.848	0.923	0.865	0.944	0.967	0.929	0.882	0.859
Total Lipids in Medium HDL	0.819	0.885	0.759	0.808	0.843	0.917	0.858	0.951	0.963	0.931	0.884	0.864
Phospholipids in Medium HDL	0.871	0.929	0.779	0.820	0.851	0.919	0.865	0.960	0.975	0.938	0.911	0.868
Cholesterol in Medium HDL	0.730	0.803	0.729	0.783	0.841	0.931	0.861	0.930	0.950	0.926	0.841	0.861
Cholesteryl Esters in Medium HDL	0.724	0.795	0.729	0.780	0.841	0.932	0.862	0.923	0.948	0.924	0.836	0.861
Free Cholesterol in Medium HDL	0.763	0.834	0.740	0.797	0.849	0.933	0.871	0.959	0.971	0.938	0.883	0.866
Concentration of Small HDL Particles	0.946	0.877	0.875	0.833	0.791	0.829	0.892	0.947	0.877	0.831	0.810	0.807
Total Lipids in Small HDL	0.998	0.951	0.891	0.844	0.809	0.836	0.889	0.964	0.908	0.856	0.867	0.819
Phospholipids in Small HDL	0.985	0.974	0.883	0.846	0.819	0.849	0.884	0.967	0.926	0.873	0.889	0.826
Cholesterol in Small HDL	0.905	0.858	0.857	0.826	0.788	0.835	0.890	0.940	0.872	0.834	0.795	0.810
Cholesteryl Esters in Small HDL	0.884	0.839	0.861	0.819	0.792	0.832	0.894	0.924	0.863	0.827	0.779	0.803
Free Cholesterol in Small HDL	0.989	0.933	0.871	0.866	0.815	0.862	0.897	0.994	0.931	0.877	0.894	0.851

Phospholipids to total lipids ratio in very small VLDL	1.482	1.340	1.341	1.210	1.329	1.154	1.104	1.084	1.211	1.187	1.370	1.261
Phospholipids to total lipids ratio in IDL	1.150	1.124	1.208	1.158	1.229	1.138	1.051	1.044	1.134	1.149	1.224	1.176

* Adjusted for age, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

† Data shown as the hazard ratios per-SD change of metabolic biomarkers.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB= UK Biobank; HDL = high-density lipoprotein; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Supplementary Table S8

Associations of RNFLT-associated metabolic biomarkers with CMD outcomes in the UKB population-II, stratified by Townsend deprivation.

Metabolic biomarkers *†	**Type 2 diabetes**		**Myocardial infarction**		**Heart failure**		**Stroke**		**All-cause mortality**		**Cardiovascular mortality**	
	High	**Low**	**High**	**Low**	**High**	**Low**	**High**	**Low**	**High**	**Low**	**High**	**Low**
HDL Cholesterol	0.659	0.662	0.661	0.629	0.806	0.822	0.800	0.829	0.860	0.855	0.759	0.731
Total Phospholipids in Lipoprotein Particles	0.835	0.843	0.790	0.764	0.768	0.833	0.916	0.878	0.827	0.863	0.774	0.802
Phospholipids in HDL	0.789	0.790	0.690	0.671	0.808	0.820	0.820	0.845	0.876	0.875	0.787	0.745
Cholesteryl Esters in HDL	0.648	0.650	0.659	0.625	0.810	0.820	0.795	0.827	0.859	0.852	0.758	0.726
Free Cholesterol in HDL	0.703	0.709	0.670	0.642	0.803	0.837	0.825	0.842	0.880	0.877	0.776	0.753
Total Lipids in HDL	0.748	0.749	0.682	0.658	0.802	0.817	0.814	0.838	0.866	0.866	0.775	0.743
Total Concentration of Lipoprotein Particles	0.759	0.769	0.706	0.689	0.760	0.788	0.841	0.840	0.812	0.831	0.742	0.723
Concentration of HDL Particles	0.764	0.774	0.694	0.679	0.771	0.787	0.827	0.837	0.824	0.836	0.750	0.716
Phosphoglycerides	0.877	0.887	0.776	0.761	0.778	0.824	0.898	0.881	0.839	0.867	0.785	0.798

Total Cholines	0.827	0.836	0.765	0.743	0.770	0.820	0.892	0.868	0.830	0.857	0.769	0.789
Phosphatidylcholines	0.847	0.861	0.769	0.744	0.764	0.823	0.884	0.870	0.824	0.858	0.761	0.786
Apolipoprotein A1	0.769	0.772	0.688	0.668	0.791	0.804	0.818	0.838	0.851	0.858	0.769	0.733
Concentration of Medium HDL Particles	0.776	0.780	0.676	0.664	0.800	0.802	0.813	0.837	0.862	0.862	0.776	0.726
Total Lipids in Medium HDL	0.816	0.818	0.700	0.687	0.806	0.805	0.821	0.847	0.863	0.864	0.787	0.736
Phospholipids in Medium HDL	0.856	0.859	0.710	0.702	0.811	0.810	0.831	0.854	0.874	0.873	0.799	0.743
Cholesterol in Medium HDL	0.740	0.741	0.674	0.657	0.806	0.804	0.802	0.837	0.857	0.853	0.770	0.719
Cholesteryl Esters in Medium HDL	0.734	0.735	0.674	0.656	0.810	0.804	0.800	0.836	0.859	0.853	0.772	0.717
Free Cholesterol in Medium HDL	0.762	0.767	0.673	0.659	0.795	0.808	0.817	0.840	0.862	0.862	0.770	0.728
Concentration of Small HDL Particles	0.877	0.901	0.788	0.808	0.772	0.794	0.896	0.890	0.803	0.838	0.769	0.756
Total Lipids in Small HDL	0.946	0.974	0.791	0.814	0.781	0.797	0.898	0.894	0.825	0.857	0.790	0.767
Phospholipids in Small HDL	0.958	0.985	0.781	0.804	0.787	0.801	0.889	0.891	0.837	0.866	0.796	0.765
Cholesterol in Small HDL	0.850	0.870	0.777	0.790	0.776	0.790	0.884	0.884	0.804	0.832	0.766	0.746
Cholesteryl Esters in Small HDL	0.836	0.854	0.784	0.798	0.786	0.793	0.883	0.889	0.805	0.831	0.769	0.745

Free Cholesterol in Small HDL	0.912	0.933	0.783	0.792	0.776	0.808	0.904	0.888	0.829	0.862	0.791	0.783
Phospholipids to total lipids ratio in very small VLDL	1.421	1.441	1.332	1.340	1.309	1.231	1.156	1.135	1.275	1.216	1.394	1.354
Phospholipids to total lipids ratio in IDL	1.138	1.162	1.227	1.180	1.239	1.132	1.070	1.056	1.210	1.101	1.254	1.168

* Adjusted for age, sex, ethnicity, assessment centre, household income, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

† Data shown as the hazard ratios per-SD change of metabolic biomarkers. Participants with high deprivation were determined as the top 50% of the Townsend Deprivation Index and those with low deprivation as the bottom 50%.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB= UK Biobank; HDL = high-density lipoprotein; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Comment #8. The majority of the increase in c-statistics are minimal (0.02) except for DM when compared to more robust models like FHS, so not clinically meaningful, but it's not even clear from Fig 2 if they are statistically significant changes?

Response: We sincerely appreciate your insightful comments regarding the statistical and clinical significance of our findings. First, all stratified results in ***Fig. 2***, where outcomes are analysed by RNFLT metabolic state deciles, are statistically significant, as detailed in the main manuscript (***Page 8, Line 193***). In addition, the results visualized in ***Fig. 3*** (including point estimates, 95% confidence intervals, and pairwise comparison *P*) are reported in ***Supplementary Table S11*** (shown below for your easy reference). All findings were statistically significant, except for predicting stroke in certain cases (after correction for multiple comparisons). Furthermore, to support the clinical relevance of our findings beyond statistical significance, we conducted an extensive literature review and performed additional reclassification and decision analyses, which further support the practical utility of integrating RNFLT metabolic states into CMD prediction.

It is important to highlight that, even when excluding diabetes, the observed improvements of 0.02–0.04 in C-statistics for other CMD outcomes are **NOT** modest. First, the baseline models employed in our study are well-established, extensively validated, and endorsed by international guidelines from the American Heart Association (AHA),[1] the European Society of Cardiology (ESC),[2] and the World Health Organization (WHO).[3] These models already incorporate a comprehensive set of robust risk factors. In fact, a recent study[4] published in ***Nature Medicine*** incorporated nine novel factors into these same robust baseline models, achieving C-statistic improvements that were slightly lower than those observed in this manuscript (***as detailed below***). Second, rather than directly applying the original coefficients from these established models to our population, we extracted the predictors from each baseline model and re-fitted them. This approach ensured that the baseline models were appropriately calibrated to our population's characteristics, preventing both overly pessimistic baseline estimates and artificially inflated improvements when integrating RNFLT metabolic states. Third, several landmark prediction studies with designs similar to ours have reported comparable performance improvements (***as detailed below***), which are widely considered clinically meaningful. Finally, to further evaluate the clinical relevance of our findings, we performed additional reclassification analysis and decision analysis (***as detailed below***), both of which confirmed the clinical utility of integrating RNFLT metabolic states.

(1) Comparison with recent prediction studies published in top-tier journals

Recent landmark studies in cardiometabolic prediction have achieved performance improvements comparable to those observed in our study. Hippisley-Cox et al.[4] (***Nature Medicine***, 2024) developed and validated a new algorithm for cardiovascular risk prediction. Since the AHA/ASCVD model (recommended by the AHA) and the SCORE2 model (recommended by the ESC) are widely adopted for cardiovascular risk assessment, they compared their new algorithm against both models — similar to the comparisons we performed in this manuscript. To facilitate comparison with our study, we focus on the subgroup of participants aged ≥ 40 years (see Hippisley-Cox et al.[4], *Nature Medicine*, 2024, *Extended Data Table 4*, shown below as ***Appendix Table A1*** for your easy reference). In this subgroup, their algorithm improved the C-statistic for AHA/ASCVD from 0.727 to 0.741 (ΔC-statistic = 0.014) and for SCORE2 from 0.728 to 0.741 (ΔC-statistic = 0.013) in men. Similarly, in women, the C-statistic improved from 0.767 to 0.781 (ΔC-statistic = 0.014). When stratified by ethnicity, they reported similar improvements among White and Chinese participants, with improvements of 0.010–0.014 for White participants and 0.007–0.024 for Chinese participants — both comparable to the improvements observed in our study.

Prediction studies beyond cardiometabolic diseases have also reported performance improvements consistent with ours. Perry et al.[5] (***Nature Medicine***, 2024) developed a cardiorespiratory fitness (CRF)-based proteomic score (CRF-Pro) to predict cardiovascular diseases, metabolic diseases, cancers, and neurological conditions (analogous to the RNFLT-based metabolomic score developed in this manuscript for CMD prediction). Their improvements ranged from < 0.005 (still significant) for hypertension, atrial fibrillation, and obesity, to a maximum of 0.03 for T2D, liver disease, and peripheral vascular disease. For cancers and neurological diseases, the improvements remained < 0.01 yet were still considered clinically meaningful. Similarly, Liu et al.[6] (***Nature Aging***, 2024) incorporated genomic and gut metagenomic risk assessment into conventional risk models for predicting coronary artery disease, T2D, Alzheimer's disease, and prostate cancer. Their integrated approach improved the C-statistic by 0.024 for coronary artery disease, 0.014 for T2D, 0.017 for Alzheimer's disease, and 0.031 for prostate cancer. Placido et al.[7] (***Nature Medicine***, 2023) conducted another landmark study that developed sequential neural network models (GRU and Transformer) to predict pancreatic cancer risk using multi-timepoint data. Their innovative modelling approach achieved C-statistic improvements of 0.007 (for GRU) and 0.034 (for Transformer) over the traditional deep learning model (Multilayer Perceptron).

(2) Reclassification and decision analyses confirm clinical utility

To further evaluate the clinical relevance of our findings, we performed reclassification analysis and decision analysis to examine the impact of integrating RNFLT metabolic states on risk prediction and clinical management. Although a model with better discrimination and calibration should theoretically be a better guide to clinical management, they fall short when assessing whether the risk model improves real-world application.[8] In our reclassification analysis, we categorized participants into risk groups (< 5%, 5–10%, and > 10% risk) and assessed how integrating RNFLT metabolic states improved patient reclassification relative to the baseline models (***Supplementary Table S16***, provided below for your easy reference). Across multiple CMD outcomes, integrating RNFLT metabolic states yielded notable improvements in reclassification, particularly for T2D and when compared to the simplest Age&Sex model. This pattern was consistent with the observed improvements in discrimination. For instance, in the prediction of myocardial infarction, integrating RNFLT metabolic states over SCORE2 correctly reclassified 1,610 non-cases (previously misclassified as high risk) into the correct low-risk category, while 1,284 non-cases were incorrectly reclassified as high risk. This resulted in a net improvement of 326 participants (2.0%) correctly reclassified as low risk. Meanwhile, among participants with events, 90 cases were correctly reclassified as high risk, while 75 cases were incorrectly reclassified as low risk, resulting in a net gain of 15 cases (2.1%) correctly reclassified as high risk. Overall, these results led to an additive net reclassification improvement of 4.1%, demonstrating the potential of RNFLT metabolic states to refine CMD risk stratification.

Our decision analysis further demonstrated that the observed gains in discrimination translated to improved clinical utility (***Fig. 4***, provided below for your easy reference). The integration of RNFLT metabolic states either reduced or eradicated the discrepancy between the Age&Sex model and clinically established models. Furthermore, the integration of these RNFL metabolic states into the established models contributed to additional improvements in clinical utility across all six critical CMD outcomes, spanning a wide range of clinical decision thresholds. For instance, integrating RNFLT metabolic states resulted in 656 correctly intervened mortality cases and 3,207 unnecessary interventions at a risk threshold T of 10%. Then, 656 correctly intervened cases minus $[T \div (1 - T)] \times 3{,}207$ unnecessary intervened gives 300 "net" cases. This is equivalent to correctly intervene 300 cases with zero unnecessary intervention. For comparison, the clinical baseline model has 621 correctly intervened heart failure cases and 3,248 unnecessary interventions at the same risk threshold, which is equivalent to correctly intervene 260 cases with zero unnecessary intervention. This means that integrating RNFLT metabolic states has 15.4% additional participants benefit from potential timely intervention. These improvements were slightly superior to those reported by Hippisley-Cox et al.[4] (***Nature Medicine***, 2024), reinforcing the clinical relevance of our findings.

(3) Summary of response to Comment #8

Taken together, our results demonstrate that the improvements in C-statistic are consistent with those achieved in recent disease prediction studies published in top-tier journals, such as ***Nature Medicine*** 2024 (#1) & 2024 (#2) & 2023 & ***Nature Aging*** 2024. Furthermore, our reclassification and decision analyses provide compelling evidence that RNFLT metabolic states enhance risk stratification and decision-making. These results highlight the clinical relevance of RNFLT metabolic states in refining risk prediction, improving clinical decision-making, and supporting better CMD outcomes. We greatly appreciate your thoughtful feedback, and we hope that these additional analyses and clarifications address your concerns.

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Supplementary Table S11

Incremental predictability of integrating RNFLT metabolic states for CMD outcome prediction beyond established predictors in the UKB population-II.

Basic models *	**C-statistics**			***P*$_{FDR}$ §**
	Integrated models †	**Basic models**	**Benefits ‡**	
Type 2 diabetes				
Age&Sex	0.768 (0.755–0.782)	0.631 (0.615–0.647)	0.137 (0.120–0.154)	<0.001
FGCRS	0.854 (0.842–0.866)	0.810 (0.795–0.825)	0.044 (0.033–0.055)	<0.001
SCORE2	0.772 (0.758–0.857)	0.674 (0.658–0.690)	0.098 (0.081–0.114)	<0.001
AHA/ASCVD	0.858 (0.846–0.870)	0.822 (0.808–0.836)	0.036 (0.026–0.046)	<0.001
WHO-CVD	0.844 (0.832–0.857)	0.778 (0.762–0.794)	0.066 (0.053–0.081)	<0.001
FRS for type 2 diabetes	0.883 (0.873–0.892)	0.864 (0.853–0.875)	0.019 (0.013–0.025)	<0.001
Myocardial infarction				
Age&Sex	0.753 (0.734–0.772)	0.725 (0.705–0.745)	0.028 (0.016–0.040)	<0.001
FGCRS	0.778 (0.760-0.796)	0.757 (0.737–0.777)	0.020 (0.007–0.033)	0.007
SCORE2	0.761 (0.742–0.778)	0.733 (0.712–0.754)	0.028 (0.011–0.044)	<0.001
AHA/ASCVD	0.778 (0.760–0.769)	0.760 (0.740–0.779)	0.019 (0.005–0.031)	0.007

WHO-CVD	0.760 (0.741–0.778)	0.731 (0.711–0.752)	0.028 (0.013–0.043)	<0.001
Heart failure				
Age&Sex	0.741 (0.721–0.761)	0.719 (0.699–0.739)	0.022 (0.009–0.035)	<0.001
FGCRS	0.762 (0.743–0.781)	0.743 (0.723–0.764)	0.019 (0.005–0.034)	0.011
SCORE2	0.748 (0.729–0.769)	0.719 (0.697–0.740)	0.029 (0.014–0.045)	<0.001
AHA/ASCVD	0.764 (0.745–0.783)	0.749 (0.729–0.770)	0.014 (0.001–0.029)	0.039
WHO-CVD	0.749 (0.730–0.769)	0.725 (0.703–0.747)	0.024 (0.009–0.040)	0.003
Stroke				
Age&Sex	0.701 (0.673–0.730)	0.692 (0.664–0.720)	0.009 (−0.005–0.024)	0.216
FGCRS	0.723 (0.695–0.750)	0.683 (0.655–0.736)	0.040 (0.015–0.006)	<0.001
SCORE2	0.712 (0.685–0.748)	0.678 (0.647–0.709)	0.034 (0.006–0.063)	0.023
AHA/ASCVD	0.725 (0.697–0.752)	0.691 (0.664–0.719)	0.034 (0.011–0.054)	0.005
WHO-CVD	0.720 (0.639–0.748)	0.705 (0.676–0.734)	0.016 (−0.008–0.039)	0.213
All-cause mortality				
Age&Sex	0.720 (0.706–0.734)	0.706 (0.692–0.720)	0.014 (0.007–0.021)	<0.001
FGCRS	0.736 (0.723–0.750)	0.722 (0.708–0.736)	0.015 (0.006–0.024)	<0.001
SCORE2	0.734 (0.720–0.749)	0.720 (0.706–0.735)	0.014 (0.006–0.023)	0.002

AHA/ASCVD	0.737 (0.723–0.750)	0.725 (0.711–0.740)	0.011 (0.003–0.019)	0.008
WHO-CVD	0.735 (0.721–0.749)	0.725 (0.710–0.739)	0.010 (0.002–0.019)	0.008
Cardiovascular mortality				
Age&Sex	0.750 (0.725–0.775)	0.729 (0.703–0.754)	0.021 (0.005–0.039)	0.021
FGCRS	0.773 (0.749–0.798)	0.738 (0.710–0.766)	0.035 (0.013–0.059)	0.015
SCORE2	0.759 (0.734–0.791)	0.724 (0.693–0.754)	0.035 (0.011–0.062)	0.018
AHA/ASCVD	0.774 (0.750–0.799)	0.749 (0.723–0.776)	0.025 (0.005–0.047)	0.021
WHO-CVD	0.766 (0.741–0.791)	0.736 (0.706–0.765)	0.030 (0.007–0.056)	0.021

* Basic models for CMD outcomes include minimal demographic characteristics (Age&Sex), Framingham General Cardiovascular Risk Score (FGCRS), Systematic Coronary Risk Evaluation 2 (SCORE2) algorithms, American Heart Association/Atherosclerotic Cardiovascular Disease (AHA/ASCVD) algorithms, and World Health Organization Cardiovascular Disease (WHO-CVD) risk charts. Framingham risk scores for type 2 diabetes algorithms were additionally included for type 2 diabetes assessment.

† Models that integrated RNFLT metabolic states.

‡ Confidential intervals were calculated by bootstrapping with 1,000 iterations followed by controlling FDR for multiple testing.

§ *P* values were calculated through two-sided non-parametric bootstrap tests with 1,000 iterations followed by controlling FDR for multiple testing. Adjusted *P* were set identical when their step-up thresholds were truncated to a shared minimum.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB = UK Biobank; FDR = false discovery rate.

Appendix Table A1

C statistic (95% CI) for QR4, ASCVD and SCORE2 in people aged 40 and older in the England validation cohort overall and by ethnic group, from Hippisley-Cox et al.[1], ***Nature Medicine*** (2024), *Extended Data Table 4*.

Subgroups	**QR4 ***	**ASCVD**	**SCORE2**
Women			
Overall	0.781 (0.778–0.784)	0.767 (0.764–0.770)	0.767 (0.764–0.770)
White	0.777 (0.773–0.780)	0.764 (0.761–0.767)	0.763 (0.760–0.767)
Indian	0.802 (0.785–0.820)	0.792 (0.774–0.810)	0.792 (0.773–0.811)
Pakistani	0.779 (0.755–0.803)	0.751 (0.724–0.778)	0.754 (0.727–0.780)
Bangladeshi	0.739 (0.698–0.779)	0.714 (0.669–0.759)	0.710 (0.664–0.757)
Other Asian	0.800 (0.777–0.824)	0.791 (0.766–0.816)	0.794 (0.770–0.819)
Caribbean	0.771 (0.751–0.791)	0.754 (0.732–0.775)	0.750 (0.728–0.773)
Black African	0.790 (0.766–0.813)	0.772 (0.747–0.797)	0.762 (0.735–0.789)
Chinese	0.863 (0.822–0.905)	0.856 (0.814–0.898)	0.849 (0.807–0.891)
Other	0.771 (0.750–0.791)	0.747 (0.724–0.769)	0.751 (0.728–0.773)
Men			
Overall	0.741 (0.739–0.744)	0.727 (0.725–0.730)	0.728 (0.725–0.730)

White	0.737 (0.734–0.740)	0.727 (0.724–0.730)	0.725 (0.722–0.728)
Indian	0.736 (0.720–0.753)	0.727 (0.710–0.743)	0.724 (0.708–0.741)
Pakistani	0.735 (0.714–0.756)	0.721 (0.699–0.742)	0.718 (0.696–0.740)
Bangladeshi	0.701 (0.674–0.729)	0.692 (0.664–0.721)	0.687 (0.659–0.716)
Other Asian	0.729 (0.706–0.752)	0.712 (0.688–0.736)	0.711 (0.686–0.735)
Caribbean	0.735 (0.714–0.756)	0.730 (0.708–0.751)	0.725 (0.703–0.746)
Black African	0.745 (0.724–0.767)	0.733 (0.712–0.755)	0.710 (0.686–0.733)
Chinese	0.759 (0.704–0.814)	0.741 (0.685–0.798)	0.735 (0.677–0.793)
Other	0.743 (0.727–0.760)	0.725 (0.708–0.742)	0.721 (0.703–0.738)

* QR4 is their new model, and ASCVD and SCORE2 are the baseline models for comparison.

Reference

1. Hippisley-Cox, J. et al. Development and validation of a new algorithm for improved cardiovascular risk prediction. *Nat. Med.* **30**, 1440-1447 (2024).

Supplementary Table S16

Reclassification improvement of integrating RNFLT metabolic states in the UKB.

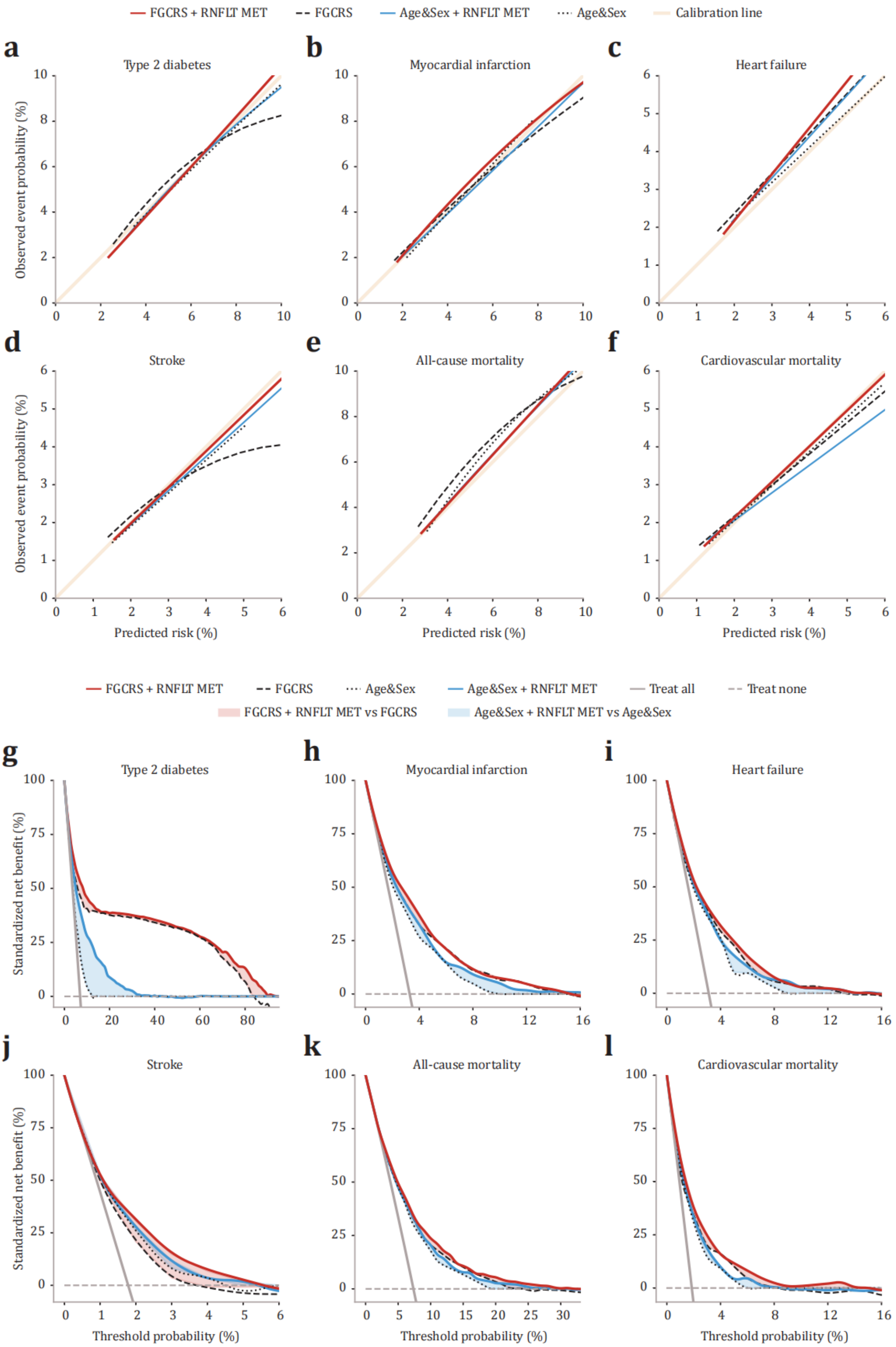
Baseline models *	NRI (case)	NRI (non-case)	Additive NRI
Type 2 diabetes			
Age&Sex	0.286	0.243	0.529
FGCRS	0.080	−0.010	0.070
SCORE2	0.204	0.095	0.299
AHA/ASCVD	0.117	0.006	0.123
WHO-CVD	0.094	−0.022	0.072
Myocardial infarction			
Age&Sex	0.089	−0.015	0.074
FGCRS	0.011	0.001	0.011
SCORE2	0.021	0.020	0.041
AHA/ASCVD	0.005	0.048	0.053
WHO-CVD	0.011	0.007	0.017
Heart failure			
Age&Sex	0.054	0.034	0.089
FGCRS	0.006	0.010	0.015
SCORE2	0.004	0.028	0.032
AHA/ASCVD	0.009	0.020	0.030
WHO-CVD	0.006	0.010	0.016
Stroke			
Age&Sex	0.095	0.047	0.142
FGCRS	0.003	0.020	0.024
SCORE2	0.010	0.020	0.030
AHA/ASCVD	0.042	0.023	0.066
WHO-CVD	0.039	0.027	0.066
All-cause mortality			
Age&Sex	0.028	−0.010	0.017

FGCRS	0.001	0.006	0.007
SCORE2	0.016	0.018	0.033
AHA/ASCVD	0.025	0.045	0.069
WHO-CVD	0.007	0.003	0.010
Cardiovascular mortality			
Age&Sex	0.072	0.031	0.103
FGCRS	0.022	0.010	0.032
SCORE2	0.050	0.021	0.071
AHA/ASCVD	0.072	0.022	0.093
WHO-CVD	0.016	0.012	0.027

* Basic models for CMD outcomes include minimal demographic characteristics (Age&Sex), Framingham General Cardiovascular Risk Score (FGCRS), Systematic Coronary Risk Evaluation 2 (SCORE2) algorithms, American Heart Association/Atherosclerotic Cardiovascular Disease (AHA/ASCVD) algorithms, and World Health Organization Cardiovascular Disease (WHO-CVD) risk charts. RNFLT = retinal nerve fibre layer thickness; UKB = UK Biobank; NRI = net classification improvement.

Fig. 4

Model calibration and clinical utility in the UKB.



Comment #9. Not clear how metabolic “fingerprints” were created; would devote more of main text methods to this rather than regurgitating UKB methods for metabolomics and RNFLT measurements that have already been published.

Response: We sincerely appreciate your insightful comment (which is also closely related to *Comment #15*) and have thoroughly reviewed the manuscript to improve clarity. Recognizing that "fingerprints" imply a lifelong characteristic whereas metabolites fluctuate over time, we have revised all references to "metabolic fingerprints" as "metabolic states" to more accurately reflect that these biomarkers capture a snapshot of health at a single time point. In addition, we have significantly refined the ***Methods*** and ***Supplementary Methods*** sections to provide a more detailed description of how metabolic states were created, while condensing the redundant descriptions of previously published UKB methods to avoid repetition.

The metabolic states were constructed using Random Forest modelling, a widely applied machine-learning approach for disease risk prediction. Specifically, we trained a Random Forest model using RNFLT-associated metabolic biomarkers to predict incident CMD outcomes. Additionally, we compared multiple modelling strategies, including traditional machine-learning methods (Extreme Gradient Boosting, Support Vector Machine, LASSO, and Ridge Regression) and deep-learning approaches (Multilayer Perceptron and Convolutional Neural Network). The datasets were randomly split into training and testing sets at an 8:2 ratio. In the case of deep learning modelling, an additional subset of the training data (20%) was set aside for validation, during which loss reduction monitoring was performed to ensure optimal model performance. Hyperparameters were fine-tuned through a grid search for optimal model configurations.

(1) Random Forest modelling

The Random Forest operates by aggregating predictions from an ensemble of decision trees:[1]

$$\hat{y}_i = \frac{1}{T}\sum_{t=1}^{T} h_t(x;\Theta_t)$$

where $\hat{y}_i$ is the predicted risk for the i^{th} CMD outcome, T is the number of trees (set at 200), x is the input vector of RNFLT-associated metabolites, Θ_t is the parameter for the t^{th} tree, and h_t is the output from an individual tree. To train the ensemble model, we employed bootstrap aggregation (bagging), wherein each decision tree was constructed using bootstrap-sampled training sets:

$$D_{bootstrap} = \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$$

where x_N is the metabolic feature set for the N^{th} participant, and y_N is the corresponding CMD outcome. Through sampling with replacement, a metabolite-CMD mapping space of the same size as the original dataset was created, allowing each tree to learn specific patterns from different perspectives. This bagging strategy mitigates dependency on individual samples by averaging predictions across multiple trees, thereby achieving unbiased estimation and addressing overfitting inherent in single decision trees.[2,3]

A key mechanism of random forest is its random feature selection, where a subset S_m of metabolic features is randomly selected at each node split, from which the optimal splitting feature that maximizes variance reduction is determined:

$$j^* = \arg\max_{j \in S_m} SplitCriterion(j)$$

where j^* is the optimal splitting feature, and the $SplitCriterion(j)$ maximizes variance reduction:

$$SplitCriterion(j) = Var(S) - \left(\frac{|S_L|}{|S|}Var(S_L) + \frac{|S_R|}{|S|}Var(S_R)\right)$$

where the variance at each node is calculated as:

$$Var(S) = \frac{1}{S}\sum_{i \in S}(y_i{}^n - \bar{y})^2$$

Here, S represents the sample set at the current node, S_L and S_R are the left and right child node sample sets after splitting, $|S|$ is the number of samples in set S, and $\bar{y}$ is the mean predicted risk for a given CMD outcome.

To prevent overfitting, we did not prune the trees, but instead controlled tree growth by setting a maximum depth of 10. Model performance was evaluated using out-of-bag (OOB) error estimation, a built-in cross-validation method within Random Forest that enhances generalizability:[4]

$$OOB\ \ Error = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i^{OOB}\right)^2$$

where $\hat{y}_i^{OOB}$ is the prediction made using only trees that did not include the i^{th} sample during training. On average, 36.8% of the samples were not selected in any given bootstrap iteration, forming an independent out-of-bag validation for evaluating the generalization error.

To further optimize performance, grid search cross-validation was performed, selecting the best hyperparameters by minimizing loss across K-fold validation:

$$L(\theta) = \frac{1}{K}\sum_{j=1}^{K} Loss\left(f_\theta, D_j\right)$$

$$\hat{\theta} = \arg\min_{\theta\in\Theta}\ L(\theta)$$

where $L(\theta)$ is the average loss across K fold, f_θ is the trained model with parameter θ, and D_j represents the validation set from the j^{th} fold. Optimized hyperparameters included the number of trees, maximum tree depth, and minimum leaf node size.

(2) Comparison with other modelling strategies

In addition, we compared multiple modelling strategies, including traditional machine-learning (Extreme Gradient Boosting, Support Vector Machine, LASSO, Ridge Regression) and deep-learning approaches (Multilayer Perceptron and Convolutional Neural Network), to assess whether alternative approaches could improve performance. Details of these models are provided in ***Supplementary Methods***. We hope that these clarifications address your concerns.

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Comment #10. How does this study compare/differ with a previously conducted similar study under the same UKB proposal number (PMID: 39581638)? That study does not appear to be referenced in this study.

Response: We greatly appreciate your insightful comment. While we acknowledge the importance of these two studies, we firmly believe that our findings remain novel and independent from the prior one. Below, we provide a detailed systematic comparison between our current study and *Br J Ophthalmol* (2024)[1], highlighting their fundamental differences in terms of objectives, theoretical rationale, research questions, hypotheses, study designs, key findings (including metabolites identified), interpretation, and clinical implications. Our prior study in *Br J Ophthalmol* (2024) served as a preliminary exploration into the ocular-systemic health connection, providing a broad direction for future research rather than offering immediate clinical applications. In contrast, the present study represents a deeper and more targeted investigation into the specific metabolic basis of the RNFLT–CMD relationship, with actionable insights for clinical decision-making. Another important difference between these two studies is the imaging modality and the metabolites identified. While both studies explored the linkage between ocular phenotypes and systemic health, *Br J Ophthalmol* (2024) was based on an unvalidated colour fundus photography-derived parameter, whereas the present manuscript focuses on optical coherence tomography (OCT)-derived RNFLT — a structural biomarker that bridges the well-established link between the retina and the heart established by us and others,[2-7] beyond the reach of fundus photography. This distinction naturally led to the minimal overlap in the metabolic biomarkers identified (only 7 shared out of 47), which is expected given that OCT and fundus photography capture distinct facets of health information. Furthermore, the findings of the present manuscript are significantly more robust and clinically promising than those in *Br J Ophthalmol* (2024), where the association strengths and predictive improvements are approximately 10 times greater than those reported in our previous work. It is noteworthy that the predictive improvement achieved using only RNFLT-associated metabolites was comparable to that of the full NMR panel, suggesting that these metabolites represent key biological processes underlying the RNFLT–CMD link, rather than merely being a random subset of CMD-associated metabolites.

Specifically, this manuscript differs from *Br J Ophthalmol* (2024) in multiple aspects:

(1) Study aims and main exposures

Although both studies investigated the linkage between ocular phenotypes and systemic diseases, *Br J Ophthalmol* (2024) was based on colour fundus photography-derived parameters, whereas this manuscript focuses on OCT-derived RNFLT, providing a higher-dimensional, layer-specific assessment of retinal structure beyond what is possible with fundus photography (tomographic vs. frontal view). Specifically, colour fundus photography captures frontal retinal morphology using a digital camera, while OCT produces cross-sectional images based on low-coherence near-infrared interferometry, allowing for precise layer-specific retinal measurements. Therefore, both studies share CMDs as the main outcomes, their exposures are fundamentally different. Furthermore, *Br J Ophthalmol* (2024) took a preliminary exploratory approach by broadly examining the association between a fundus photography-derived parameter and a wide range of systemic health outcomes without a clear motivation. As a result, its findings lack immediate clinical relevance and serve primarily as a foundation for future hypothesis-driven research. In contrast, the present manuscript is a focused, hypothesis-driven investigation that builds upon previously established RNFLT–CMD associations while focusing on clinical decision making and health disparities in socially vulnerable populations, with the aim of providing direct actionable insights.

(2) Theoretical rationale and hypotheses

Since the association between the parameter used in *Br J Ophthalmol* (2024) and systemic health outcomes was previously unknown, the analyses and discussion regarding metabolic associations with systemic outcomes lacked a clear theoretical rationale. Moreover, that study did not establish a direct link between the fundus photography-derived parameter and incident CMDs. Conversely, the present manuscript builds upon prior evidence of RNFLT–CMD associations, integrating both RNFLT–metabolite and metabolite–CMD links to provide a mechanistic exploration of the retinal–cardiometabolic connection. We hypothesize that a metabolic basis may underlie the connection between RNFLT changes and CMDs, while the susceptibility of the retina may allow for early manifestation of these subtle disturbances. This extends the concept of retinal-systemic metabolic linkage from fundus photography to OCT, providing in-depth investigation and robust evidence for the metabolic basis of RNFLT–CMD connections.

(3) Study designs

While *Br J Ophthalmol* (2024) hypothesized that fundus photography-derived metabolic biomarkers might stratify and predict multisystem disease risk, it did not specify studied outcomes using a predefined process, making it inherently a data-driven rather than hypothesis-driven study. As discussed above, the lack of a clear motivation and biological rationale further limits its interpretability. In contrast, while the present manuscript also examines metabolite-based CMD prediction, it is entirely hypothesis-driven to test whether RNFLT metabolic states mediate the RNFLT–CMD relationship. Findings from mediation and multiple downstream analyses support our initial hypothesis, suggesting that the RNFLT–CMD association is at least partially explained by shared metabolic pathways.

(4) Key findings and interpretation

While both studies have investigated metabolomic associations with ocular phenotypes, the key findings are different. *Br J Ophthalmol* (2024) identified 28 metabolic biomarkers associated with a fundus photography-derived parameter, and the present study identified 26 metabolic biomarkers associated with OCT-derived RNFLT. It is noteworthy that only 7 out of 47 metabolic biomarkers overlapped between the two studies, highlighting the distinct nature of the information captured by fundus photography versus OCT. More importantly, the metabolic biomarkers associated with OCT-derived RNFLT played a mediating role in the RNFLT–CMD association, whereas those linked to fundus photography-derived parameters are likely to represent incidental statistical correlations. The contribution of fundus photography-derived metabolic biomarkers to CMD prediction in *Br J Ophthalmol* (2024) was modest, suggesting that these biomarkers represent a random subset of CMD-associated metabolites rather than capturing key processes of biological basis. In contrast, the integration of only the RNFLT-associated subset in the present manuscript achieved predictive improvements comparable to the full NMR panel (~10 times higher than those in *Br J Ophthalmol*, 2024). This finding strongly suggests that these metabolites serve as key processes underlying the RNFLT–CMD link, rather than merely being a random subset of CMD-related biomarkers.

(5) Clinical relevance

Moving beyond *Br J Ophthalmol* (2024), we have performed reclassification and decision analysis that reveal clinical relevance of integrating RNFLT metabolic states into CMD prediction. Although a model with better discrimination and calibration should theoretically be a better guide to clinical management, these measures fall short when assessing whether the risk model improves clinical decision making.[8] By summarizing the performance of the model in reclassifying risk categories and aiding decision making, we found that the observed discriminatory gains translated to clinical utility gains. The integration of RNFLT metabolic states either reduced or eradicated the discrepancy between the Age&Sex model and clinically established models. Furthermore, the integration of these RNFLT metabolic states into the established models contributed to additional improvements in risk reclassification and clinical decision making for all six CMD outcomes. These findings further support the clinical relevance of this manuscript.

(6) Summary of response to Comment #10

In summary, this manuscript and our previously published work in *Br J Ophthalmol* (2024) differ in multiple aspects, including objectives, theoretical rationale, research questions, hypotheses, study designs, key findings, interpretation, and clinical implications. Our study focuses on OCT-derived RNFLT (a distinct imaging modality from fundus photography), identified distinct metabolic biomarkers, and provides the first evidence of a metabolic basis for the longitudinal RNFLT–CMD association, improving risk prediction and clinical utility while addressing health disparities in women and socially vulnerable populations. Thus, our conclusions remain novel and independent, and our findings are not diminished by our previous work in *Br J Ophthalmol* (2024).

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Comment #11. The biological interpretation in discussion is too speculative; also need to provide more details and context around individual metabolites and talk about prior literature.

Response: We apologise for the over-generalised discussion of the biological interpretation of the RNFLT metabolic states. To provide more biological insights, we reviewed previous literature and re-evaluated the metabolites identified in this manuscript. This included incorporating complementary liquid chromatography-mass spectrometry (LC-MS) assays in the GDES and comparing

NMR assays in the UKB to traditional enzymatic assays. Through pathway enrichment analysis,[1] we identified 20 significantly enriched pathways, including aminoacyl-tRNA biosynthesis, glutathione metabolism, and linoleic acid metabolism (***as detailed below***). Besides, while the Nightingale NMR platform quantifies over 200 metabolites, its limited coverage of small molecules restricts pathway-level biological interpretation. Therefore, we have refined our discussion on NMR findings to focus on how particle characteristics, as captured by NMR, provide unique biological insights that extend beyond traditional enzymatic assays (***as detailed below***).

Previous studies have established the protective role of HDL concentration against both RNFL thinning and CMDs.[2-4] Our findings extend this by demonstrating that, even after accounting for enzymatic assay-based HDL concentration, all NMR-based findings remained robust. This suggests that variability in HDL particle properties (size, density, and chemical composition) may be at least as important as HDL concentration in retinal and cardiometabolic physiology. Unlike traditional enzymatic assays that measure concentration following chemical precipitation separation, NMR assays employ deconvolution based on diffusion and relaxation properties, generating composite metrics that integrate both particle properties and lipid concentration.[5] For instance, the higher molecular mobility of cholesterol in small HDL particles extends their T2 relaxation times, resulting in reduced linewidth attenuation in presaturated proton spectra and thus stronger signals than their larger counterparts. Given that these biomarkers that integrate both physic properties and chemical composition independently mediate the RNFLT–CMD connection and enhance CMD prediction, our findings highlight the added value of NMR-based assays over traditional HDL measurements.

However, NMR-derived metabolic markers are not directly compatible with classical pathway analysis, thus limiting biological interpretation. To address this, we incorporated LC-MS-based metabolic profiling in the GDES, which enables high-sensitivity detection of a broader range of metabolites, including amino acids, benzene derivatives, alcohols, amines, coenzymes, nucleotides, carbohydrates, organic acids, fatty acids, and heterocyclic compounds.[6,7] Detailed methodological and quality control information is previously described[8,9] and provided in the ***Supplementary Methods***. This further reinforces the potential of RNFLT metabolic states to capture the common metabolic basis from distinct facets complementary to NMR assays. Using the same biomarker identification framework, we additionally identified 673 LC-MS-based metabolic biomarkers significantly associated with RNFLT (see ***Supplementary Table S20***), of which 457 were also associated with incident cardiovascular disease and 279 with all-cause mortality (see ***Supplementary Table S21–S22***).

Pathway enrichment analysis of RNFLT metabolic states revealed 20 significantly enriched pathways, including 19 pathways linked to cardiovascular diseases and 16 linked to mortality. These pathways were predominantly involved in amino acid metabolism and biosynthetic processes, including: aminoacyl-tRNA biosynthesis, arginine biosynthesis, caffeine metabolism, nicotinate and nicotinamide metabolism, arginine and proline metabolism, histidine metabolism, D-Glutamine and D-glutamate metabolism, glycine, serine and threonine metabolism, pantothenate and CoA biosynthesis, alanine, aspartate and glutamate metabolism, glutathione metabolism, valine, leucine and isoleucine biosynthesis, taurine and hypotaurine metabolism, pyrimidine metabolism, cysteine and methionine metabolism, linoleic acid metabolism, β-Alanine metabolism, glyoxylate and dicarboxylate metabolism, biosynthesis of unsaturated fatty acids, and phenylalanine, tyrosine and tryptophan biosynthesis (all $P_{FDR} < 0.05$).

Although LC-MS and NMR assays yielded overlapping findings (e.g., phospholipids, phosphoglycerides, cholines, and phosphatidylcholines), the higher sensitivity of LC-MS enabled the detection of low-abundance metabolites beyond the reach of NMR. Together, these findings provide complementary perspectives on systemic metabolic dysregulation that links RNFLT and contributes to CMD risk:[10]

(1) Aminoacyl-tRNA biosynthesis: This pathway is essential for protein translation, including the synthesis of apolipoproteins involved in HDL assembly and reverse cholesterol transport.[11] Dysregulation may impair HDL biogenesis and cholesterol efflux, as observed in apoA-I knockout mice.[12] In addition, the enrichment of this pathway in the common metabolic basis also aligns with evidence suggesting that impaired aminoacyl-tRNA biosynthesis inhibits angiogenesis and reduces capillary density by modulating endothelial cell migration and proliferation.[13-19]

(2) Phenylalanine and arginine metabolism: Disruptions in phenylalanine metabolism have been associated with reduced apolipoprotein expression and impaired HDL particle formation.[20] Arginine, a precursor for nitric oxide, plays a critical role in endothelial function, modulating lipoprotein transport and HDL's anti-inflammatory function.[21] Dysregulation in both pathways has also been observed in apoA-I knockout mice.[12]

(3) Glutathione metabolism: As a major intracellular antioxidant, glutathione supports the activity of HDL-associated antioxidant enzymes.[22] Glutathione depletion accelerates HDL oxidation, impairing its anti-inflammatory and cholesterol transport functions. In addition, a glutathione-dependent glyoxalase system has been implicated in HDL instability, leading to structural contraction and decreased particle size,[23] which also aligns with the NMR findings in this manuscript.

(4) Linoleic acid (LA) metabolism: LA is a key ω-6 polyunsaturated fatty acid that plays a crucial role in phospholipid membrane composition and HDL stability, influencing membrane fluidity and cholesterol transport.[24-26] In addition, trials on the effect of LA administration has confirmed its effects on lipid

profile, including improved HDL and increased concentrations of triglycerides and apolipoprotein.[27]

(5) Vitamin & coenzyme metabolism: Nicotinate (Vitamin B3) metabolism is a well-recognized regulator of HDL cholesterol, although clinical trials have reported mixed results regarding its cardiovascular benefits.[28] Nonetheless, its role supports the complementary metabolic insights identified by NMR and LC-MS. Pantothenate (Vitamin B5) metabolism contributes to Coenzyme A production, which is essential for cholesterol esterification and HDL remodelling.[29,30]

In summary, NMR-based metabolic findings and LC-MS-derived enrichment pathway jointly reveal a network of metabolic perturbations underlying the RNFLT–CMD connection. RNFLT metabolic states capture a dual profile of lipid metabolism and amino acid/redox imbalances, which collectively contribute to CMD risk. Although our LC-MS-based enrichment analysis provides biological insights, we acknowledge that these interpretations remain exploratory. Future research should focus on: (1) targeted metabolomics and functional assays to validate these findings; (2) longitudinal studies to assess metabolic dynamics over time; and (3) mechanistic studies to establish causal links. We have incorporated these discussions into the ***Discussion*** (***Page 14, Line 396***) and ***Supplementary Discussion*** sections of our revised manuscript and expanded the ***Limitation*** to acknowledge that our findings remain exploratory (***Page 15, Line 428***). We hope this adequately addresses your concerns.

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Comment #12. Nightingale platform is not comprehensive.

Response: We appreciate you raising this important point. While Nightingale proton nuclear magnetic resonance (1H-NMR) has exhibited commendable robustness and reliability demanding minimal sample preparation and requirements of expensive reagents while offering a high degree of automation and reproducibility at low cost, it is not without its limitations.[1] Foremost among these is its inherent sensitivity constraints, especially when compared to mass spectrometry (MS) techniques.[2] To address this, we have not only highlighted the metabolomics platform in study limitation but also sought to augment our findings by incorporating complementary liquid chromatograph (LC)-MS assays in the GDES. This additional assay allows further exploration into the common metabolic basis of RNFLT–CMD connections and its potential for guiding equitable CMD management.

A validated LC-ESI-MS/MS platform (ExionLC AD UPLC coupled with QTRAP® system) was additionally employed in the GDES using following standardized protocols. Serum samples underwent extraction with methanol/acetonitrile containing internal standards, followed by sequential centrifugation to obtain supernatants. Chromatographic separation was performed on a Waters HSS T3 C18 column (1.8 μm, 2.1 × 100 mm) at 40°C, with a mobile phase of 0.1% formic acid in water/acetonitrile and a 14-min gradient program (5% to 90% organic phase over 11 min, followed by re-equilibration). The flow rate was set at 0.4 mL/min, with a 2 μL injection volume. The QTRAP mass spectrometer operated in both positive and negative ionization modes (IS +5500 V and −4500 V) with optimized source parameters (GSI 55 psi, GSII 60 psi, CUR 25 psi, 500°C). Quality control processes included analysis of QC samples (1 per 10 specimens), process blanks, and internal standards added during extraction with CV <5%. Instrument calibration was performed using polypropylene glycol solutions (10 and 100 μmol/L) for QQQ and LIT scan modes, and specific multiple reaction monitoring (MRM) transitions were monitored for each period according to the metabolites eluted within this period. Detailed experimental procedures and QC measures are detailed in the ***Supplementary Methods***.

Our LC-MS assay identified 673 RNFLT-associated metabolites (see ***Supplementary Table S20***), identified 673 RNFLT-associated metabolites (see ***Supplementary Tables S21–S22***). These results not only replicated the findings of 1H-NMR (e.g. phospholipids, phosphoglycerides, cholines, phosphatidylcholines), but also captured a wider range of metabolites, including amino acids, benzene derivatives, alcohols, amines, coenzymes, nucleotides, carbohydrates, organic acids, fatty acids, and heterocyclic compounds, owing to its high sensitivity. Overall, 1H-NMR and LC-MS assays provide complementary metabolite coverage, offering a more comprehensive view of the RNFLT metabolic landscape. While 1H-NMR excels in quantifying lipid-related metabolites and lipoprotein particle properties, LC-MS enables high-sensitivity detection of small molecules, particularly low-abundance metabolites involved in amino acid metabolism and redox homeostasis.

We then assessed the ability of LC-MS-based RNFLT metabolic states in predicting incident CMD outcomes in the GDES. Our results show that RNFLT metabolic states remained among the most powerful predictors of incident CVD and mortality, with their integration yielding significant C-statistic improvements. These findings suggest that, although our primary results were based on Nightingale 1H-NMR platform, the metabolic basis of the RNFLT–CMD association is platform agnostic. More extensive metabolite coverage combining multiple platforms would be expected to further improve the model, as each assay captures distinct facets of the common metabolic basis underlie RNFLT–CMD connections. Despite the current inability to achieve comprehensive metabolite characterization, we nevertheless demonstrate the applicability of our analytical framework across at least two independent platforms. Future studies should focus on combining multi-platform metabolomics integration and broader metabolite coverage to further validate our framework and complement our findings. We have highlighted these key points in the ***Abstract*** (***Page 4, Line 98***), ***Discussion*** (***Page 15, Line 415***) and ***Supplementary Discussion***. We hope this addresses your concerns.

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metabolite analysis in mass spectrometry and NMR-based metabolomics. *Curr. Opin. Chem. Biol.* **80**, 102466 (2024).

Comment #13. Need to adjust for robust risk factors in all analyses (not just RNFLT associations) including in mediation analyses.

Response: We sincerely appreciate your valuable suggestion. We acknowledge that certain important confounders were not addressed in our initial submission. To address this, we performed a series of sensitivity analyses by sequentially adjusting for additional covariates to ensure the robustness of our findings. Specifically, we: (1) further adjusted for enzymatic assay-based HDL, systolic blood pressure, and HbA1c levels in the identification of RNFLT metabolic states; (2) additionally adjusted model (1) for prevalent CMD; (3) adjusted for enzymatic assay-based HDL, systolic blood pressure, and HbA1c levels in the association between metabolic biomarkers and incident CMD outcomes; and (4) adjusted for the above covariates in all mediation analyses. It is important to note that we did not adjust for prevalent CMD in: (1) in the association between metabolic biomarkers and incident CMD outcomes; and (2) all mediation analysis, since all individuals with prevalent CMD at baseline had already been excluded in the incident CMD analysis. Although most of these associations were attenuated after these additional adjustments, all findings remain robust. The results of these analyses can be found in ***Supplementary Tables S3–S4***, ***S6***, and ***Table 1***, and are shown below for your easy reference. These additional sensitivity analyses further bolster our confidence in the robustness of the results in this manuscript.

Supplementary Table S3

Identification of RNFLT-associated metabolic biomarkers in the UKB population-I, additionally adjusting for enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

Metabolic biomarkers *	**β** †	**95% CI**		P_{FDR} ‡
HDL Cholesterol	−0.163	−0.277	−0.049	0.007
Total Phospholipids in Lipoprotein Particles	−0.117	−0.229	−0.005	0.041
Phospholipids in HDL	−0.167	−0.277	−0.058	0.005
Cholesteryl Esters in HDL	−0.168	−0.282	−0.055	0.006
Free Cholesterol in HDL	−0.142	−0.254	−0.029	0.017
Total Lipids in HDL	−0.163	−0.274	−0.052	0.006
Total Concentration of Lipoprotein Particles	−0.183	−0.291	−0.076	0.002
Concentration of HDL Particles	−0.190	−0.295	−0.085	0.002
Phosphoglycerides	−0.129	−0.242	−0.016	0.027
Total Cholines	−0.128	−0.244	−0.012	0.032
Phosphatidylcholines	−0.132	−0.245	−0.018	0.027
Apolipoprotein A1	−0.177	−0.285	−0.068	0.003
Concentration of Medium HDL Particles	−0.192	−0.298	−0.085	0.002
Total Lipids in Medium HDL	−0.190	−0.296	−0.083	0.002
Phospholipids in Medium HDL	−0.191	−0.296	−0.085	0.002
Cholesterol in Medium HDL	−0.197	−0.305	−0.089	0.002
Cholesteryl Esters in Medium HDL	−0.201	−0.309	−0.093	0.002
Free Cholesterol in Medium HDL	−0.183	−0.291	−0.074	0.002
Concentration of Small HDL Particles	−0.163	−0.264	−0.062	0.003
Total Lipids in Small HDL	−0.170	−0.271	−0.069	0.002
Phospholipids in Small HDL	−0.177	−0.278	−0.076	0.002
Cholesterol in Small HDL	−0.169	−0.269	−0.069	0.002

Cholesteryl Esters in Small HDL	−0.166	−0.265	−0.067	0.002
Free Cholesterol in Small HDL	−0.155	−0.259	−0.050	0.006
Phospholipids to total lipids ratio in very small VLDL	0.120	0.015	0.224	0.027
Phospholipids to total lipids ratio in IDL	0.126	0.026	0.227	0.017

* Adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

† Data shown as per-SD change of RNFLT (4.24 μm).

‡ *P* values were calculated through two-sided Student's t-tests followed by controlling FDR for multiple testing.

HDL = high-density lipoprotein; UKB = UK Biobank; CI = confidence interval; FDR = false discovery rate; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Supplementary Table S4

Identification of RNFLT-associated metabolic biomarkers in the UKB population-I, additionally adjusting for prevalent CMDs.

Metabolic biomarkers *	**β †**	**95% CI**		***P*** $_{FDR}$ **‡**
HDL Cholesterol	−0.164	−0.279	−0.050	0.007
Total Phospholipids in Lipoprotein Particles	−0.114	−0.227	−0.001	0.048
Phospholipids in HDL	−0.166	−0.276	−0.056	0.005
Cholesteryl Esters in HDL	−0.170	−0.284	−0.056	0.006
Free Cholesterol in HDL	−0.142	−0.255	−0.029	0.018
Total Lipids in HDL	−0.162	−0.274	−0.051	0.007
Total Concentration of Lipoprotein Particles	−0.181	−0.289	−0.073	0.003
Concentration of HDL Particles	−0.188	−0.294	−0.082	0.003
Phosphoglycerides	−0.125	−0.239	−0.012	0.033
Total Cholines	−0.125	−0.241	−0.008	0.038
Phosphatidylcholines	−0.129	−0.243	−0.014	0.031
Apolipoprotein A1	−0.175	−0.284	−0.066	0.003
Concentration of Medium HDL Particles	−0.190	−0.297	−0.083	0.003
Total Lipids in Medium HDL	−0.187	−0.294	−0.081	0.003
Phospholipids in Medium HDL	−0.188	−0.294	−0.082	0.003
Cholesterol in Medium HDL	−0.196	−0.305	−0.087	0.003
Cholesteryl Esters in Medium HDL	−0.200	−0.309	−0.091	0.003
Free Cholesterol in Medium HDL	−0.181	−0.290	−0.072	0.003
Concentration of Small HDL Particles	−0.158	−0.260	−0.057	0.004
Total Lipids in Small HDL	−0.165	−0.266	−0.063	0.003
Phospholipids in Small HDL	−0.171	−0.273	−0.070	0.003
Cholesterol in Small HDL	−0.165	−0.266	−0.064	0.003
Cholesteryl Esters in Small HDL	−0.162	−0.262	−0.063	0.003

Free Cholesterol in Small HDL	−0.150	−0.255	−0.045	0.007
Phospholipids to total lipids ratio in very small VLDL	0.123	0.018	0.228	0.026
Phospholipids to total lipids ratio in IDL	0.126	0.025	0.226	0.018

* Adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, HbA1c, and prevalent CMDs.

† Data shown as per-SD change of RNFLT (4.24 μm).

‡ *P* values were calculated through two-sided Student's t-tests followed by controlling FDR for multiple testing.

CMD = cardiometabolic disease; UKB = UK Biobank; CI = confidence interval; FDR = false discovery rate; HDL = high-density lipoprotein; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Supplementary Table S6

Associations of RNFLT-associated metabolic biomarkers with CMD outcomes in the UKB population-II, additionally adjusting for enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

Metabolic biomarkers *	Type 2 diabetes		Myocardial infarction		Heart failure		Stroke		All-cause mortality		Cardiovascular mortality	
	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡	HR †	P_{FDR} ‡
HDL Cholesterol	0.706	<0.001	0.762	<0.001	0.929	0.007	0.918	0.020	0.984	0.373	0.904	0.005
Total Phospholipids in Lipoprotein Particles	0.919	<0.001	0.914	<0.001	0.870	<0.001	0.994	0.840	0.925	<0.001	0.902	0.001
Phospholipids in HDL	0.889	<0.001	0.812	<0.001	0.919	0.001	0.933	0.034	0.998	0.887	0.923	0.013
Cholesteryl Esters in HDL	0.688	<0.001	0.753	<0.001	0.927	0.006	0.909	0.017	0.978	0.205	0.895	0.002
Free Cholesterol in HDL	0.784	<0.001	0.807	<0.001	0.948	0.048	0.953	0.190	1.017	0.328	0.948	0.120
Total Lipids in HDL	0.834	<0.001	0.796	<0.001	0.916	0.001	0.929	0.029	0.990	0.569	0.917	0.009
Total Concentration of Lipoprotein Particles	0.845	<0.001	0.814	<0.001	0.838	<0.001	0.925	0.020	0.900	<0.001	0.843	<0.001
Concentration of HDL Particles	0.854	<0.001	0.797	<0.001	0.846	<0.001	0.913	0.017	0.912	<0.001	0.846	<0.001
Phosphoglycerides	0.977	0.109	0.905	<0.001	0.874	<0.001	0.986	0.634	0.938	<0.001	0.913	0.002

Total Cholines	0.920	<0.001	0.893	<0.001	0.870	<0.001	0.982	0.564	0.931	<0.001	0.903	0.001
Phosphatidylcholines	0.950	0.001	0.898	<0.001	0.866	<0.001	0.978	0.498	0.926	<0.001	0.894	<0.001
Apolipoprotein A1	0.861	<0.001	0.796	<0.001	0.886	<0.001	0.920	0.020	0.960	0.014	0.890	<0.001
Concentration of Medium HDL Particles	0.874	<0.001	0.790	<0.001	0.893	<0.001	0.913	0.017	0.968	0.050	0.892	0.001
Total Lipids in Medium HDL	0.921	<0.001	0.809	<0.001	0.890	<0.001	0.919	0.017	0.965	0.023	0.894	<0.001
Phospholipids in Medium HDL	0.969	0.036	0.827	<0.001	0.895	<0.001	0.926	0.020	0.972	0.072	0.905	0.001
Cholesterol in Medium HDL	0.820	<0.001	0.772	<0.001	0.894	<0.001	0.905	0.016	0.960	0.014	0.878	<0.001
Cholesteryl Esters in Medium HDL	0.811	<0.001	0.768	<0.001	0.895	<0.001	0.900	0.016	0.958	0.011	0.876	<0.001
Free Cholesterol in Medium HDL	0.858	<0.001	0.794	<0.001	0.901	<0.001	0.926	0.026	0.977	0.168	0.900	0.002
Concentration of Small HDL Particles	0.968	0.026	0.873	<0.001	0.810	<0.001	0.931	0.020	0.852	<0.001	0.819	<0.001
Total Lipids in Small HDL	1.048	0.001	0.886	<0.001	0.825	<0.001	0.939	0.029	0.881	<0.001	0.847	<0.001
Phospholipids in Small HDL	1.063	<0.001	0.884	<0.001	0.837	<0.001	0.938	0.029	0.900	<0.001	0.859	<0.001
Cholesterol in Small HDL	0.940	<0.001	0.860	<0.001	0.811	<0.001	0.926	0.018	0.851	<0.001	0.815	<0.001
Cholesteryl Esters in Small HDL	0.918	<0.001	0.855	<0.001	0.810	<0.001	0.918	0.016	0.842	<0.001	0.804	<0.001

Free Cholesterol in Small HDL	1.015	0.278	0.898	<0.001	0.844	<0.001	0.963	0.204	0.905	<0.001	0.881	<0.001
Phospholipids to total lipids ratio in very small VLDL	1.345	<0.001	1.225	<0.001	1.222	<0.001	1.075	0.020	1.194	<0.001	1.286	<0.001
Phospholipids to total lipids ratio in IDL	1.085	<0.001	1.146	<0.001	1.168	<0.001	1.037	0.199	1.134	<0.001	1.172	<0.001

* Adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

† Data shown as per-SD change of metabolic biomarkers.

‡ *P* values were calculated through two-sided Wald tests followed by controlling FDR for multiple testing.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB= UK Biobank; HR = hazard ratio; FDR = false discovery rate; HDL = high-density lipoprotein; VLDL = very low-density lipoprotein; IDL = intermediate-density lipoprotein.

Table 1. The mediation effect of RNFLT metabolic states on the association between RNFLT and CMD risks in the UKB.

Outcomes *	**Base models †**		**Plus RNFLT metabolic states**		**Percentage §**
	HR (95% CI)	P_{FDR} **‡**	**HR (95% CI)**	P_{FDR} **‡**	
Type 2 diabetes	0.85 (0.81 to 0.89)	$\mathbf{2.73{\times}10^{-10}}$	0.94 (0.83 to 1.06)	0.391	63.69%
Myocardial infarction	0.86 (0.80 to 0.93)	$\mathbf{3.02{\times}10^{-4}}$	0.90 (0.77 to 1.06)	0.391	31.39%
Heart failure	0.79 (0.72 to 0.85)	$\mathbf{3.83{\times}10^{-8}}$	0.83 (0.70 to 0.99)	0.208	24.07%
Stroke	0.91 (0.82 to 1.01)	0.074	0.93 (0.75 to 1.15)	0.482	18.06%
All-cause mortality	0.87 (0.82 to 0.92)	$\mathbf{7.50{\times}10^{-6}}$	0.89 (0.79 to 1.00)	0.208	16.97%
Cardiovascular mortality	0.83 (0.73 to 0.95)	$\mathbf{9.42{\times}10^{-3}}$	0.87 (0.68 to 1.12)	0.391	23.10%

* Data are presented as per-SD change in RNFLT (4.24 μm).

† Base model was adjusted for age, sex, ethnicity, assessment centre, household income, Townsend deprivation index, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

‡ *P* values were calculated through two-sided Wald tests followed by controlling FDR for multiple testing. Bold indicates significance. Adjusted *P* were set identical when their step-up thresholds were truncated to a shared minimum.

§ Percent change in HR coefficient.

RNFLT = retinal nerve fibre layer thickness; CMD = cardiometabolic disease; UKB = UK Biobank; FDR = false positive rate.

Comment #14. More minor issues: Language in abstract is too expansive and not specific enough.

Response: We appreciate your comment. We have revised the ***Abstract*** to make it more specific. Vague terms such as "many" and "most" have been removed and replaced with concrete numerical results to provide a clearer summary of our findings. We have also removed the extended claim that "the retina is positioned as a valuable window into cardiometabolic health" and instead emphasize that further studies in broader populations are required to validate our findings (***Page 4, Line 98***).

Comment #15. Throughout, need to be more specific about the metabolite score (i.e. shouldn't call them fingerprints…fingerprints don't change over a lifetime like metabolites do).

Response: We fully agree with your suggestion and have reviewed the manuscript to modify the terminology throughout. Recognizing that "fingerprints" imply a lifelong characteristic whereas metabolites fluctuate over time, we have revised all references to "metabolic fingerprints" as "metabolic states" to more accurately reflect that these biomarkers capture a snapshot of health at a single time point.

Comment #16. Spell out all acronyms in main text first time they are used.

Response: We apologize for this oversight. We have reviewed the entire manuscript and have spelled out all acronyms the first time they are used. A list of acronyms has also been included at the beginning of the revised manuscript (***Page 3, Line 66***).

Comment #17. Should use "European ancestry" instead of "Caucasian ethnicity" (as the latter was not specifically addressed in UKB); shouldn't use "diabetic patients" (instead should use "patients with diabetes").

Response: We apologize for the oversight. We have reviewed the entire manuscript and replaced "Caucasian ethnicity" with "European ancestry" and changed "diabetic patients" to "patients with diabetes" as appropriate.

Comment #18. data use statement only covers UKB, not GDES.

Response: We apologize for this oversight. We have now added the data availability statement for the GDES (***Page 21, Line 603***).

Comment #19. Fig 2: all Y-axes need to be same range of values.

Response: We appreciate your suggestion regarding the Y-axes. We fully agree with the importance of using consistent Y-axis scales, as this helps to facilitate intuitive comparisons across outcomes. We have carefully considered this issue and experimented with three different plotting strategies, which are presented below for your easy reference:

(1) Uniform small scale (0–6%): As shown in ***Supplementary Fig. S5***, this scale clearly illustrates how RNFLT-associated metabolites stratify risk for low-incidence outcomes (e.g., stroke, cardiovascular mortality). However, the cumulative risk curves for high-incidence outcomes (e.g., T2D, myocardial infarction, all-cause mortality) are truncated. While the stratification of these outcomes is even more visually evident, this method fails to display the full risk curves.

(2) Uniform large scale (0–40%): As shown in ***Supplementary Fig. S6***, this scale clearly illustrates risk stratification for high-incidence outcomes. However, the cumulative risk curves for low-incidence outcomes become visually flattened and difficult to distinguish across the risk quintiles, which falsely suggests a lack of stratification. In fact, the low incidence of these outcomes (e.g., T2D incidence is 3.6 times higher than stroke) does not reflect a failure in stratification ability.

(3) Outcome-specific scales: As shown in ***Fig. 2***, we adjusted the Y-axis scales based on the cumulative risk range for each specific outcome to fully present the risk stratification ability across all outcomes.

To balance comparability and avoid misleading communication of results, we considered three main aspects:

(1) Cross-outcome comparability: Different CMD outcomes have significantly different incidence rates. While using a uniform scale may facilitate comparison across different outcomes, the overlapping risk curves for low-incidence outcomes may be misleading.

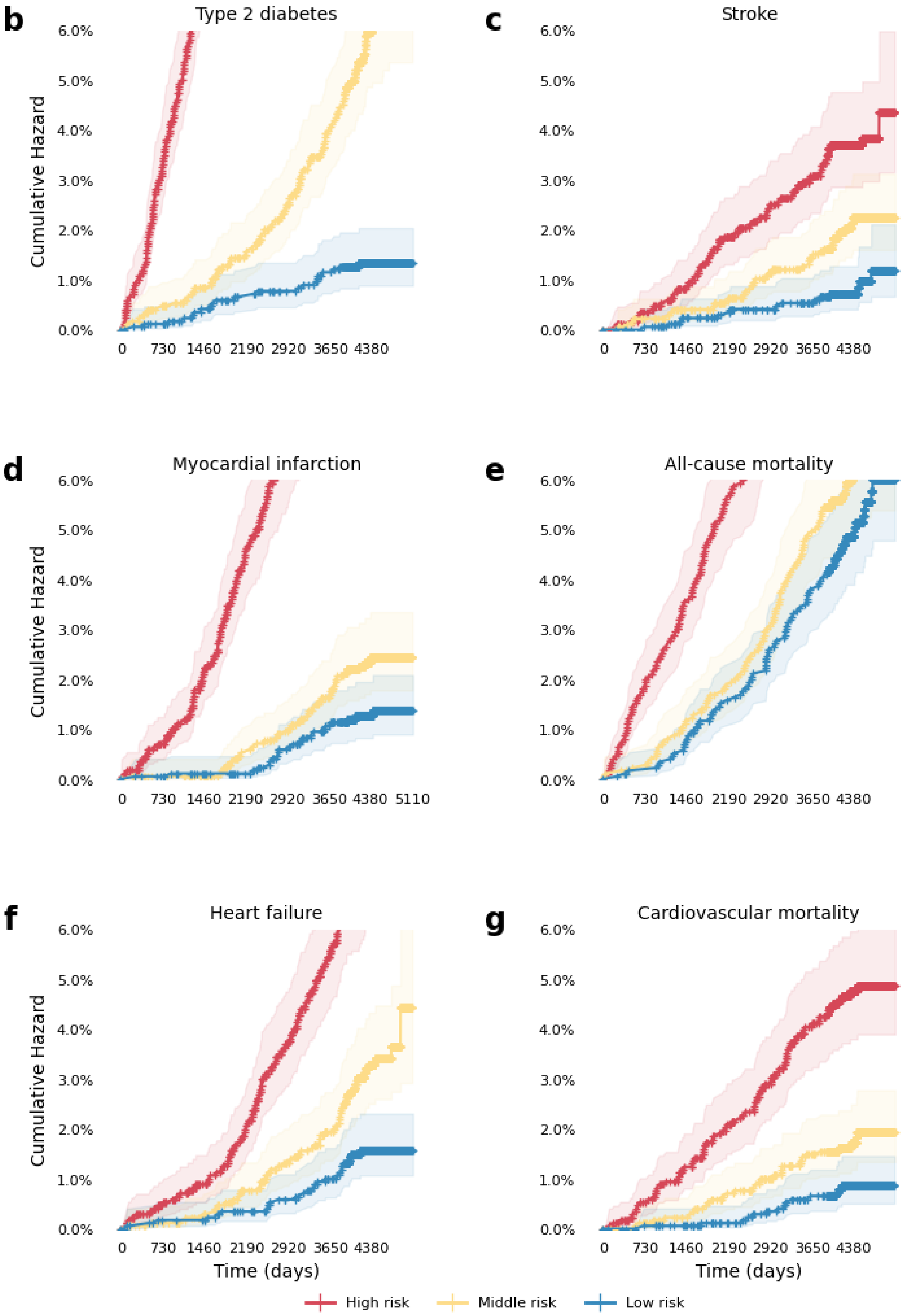
(2) Effective communication of results: The outcome-specific scales most clearly demonstrate the model's risk stratification ability for each outcome, regardless of incidence rate. However, cross-outcome comparison may be less intuitive.

(3) Clinical relevance: Healthcare providers are typically more interested in risk stratification for specific outcomes, rather than comparing outcomes directly. Outcome-specific scales better align with this clinical perspective.

In summary, to maintain comparability and effectively communicate our findings, we propose retaining the outcome-specific scales while also providing two versions with a unified scale for reference as supplementary. These two supplementary figures with uniform small and large scales are referenced in the figure legend and provided as ***Supplementary Figs. S5–S6***. We hope that this solution balances the benefits of each of these plotting strategies and effectively reflects the model's performance.

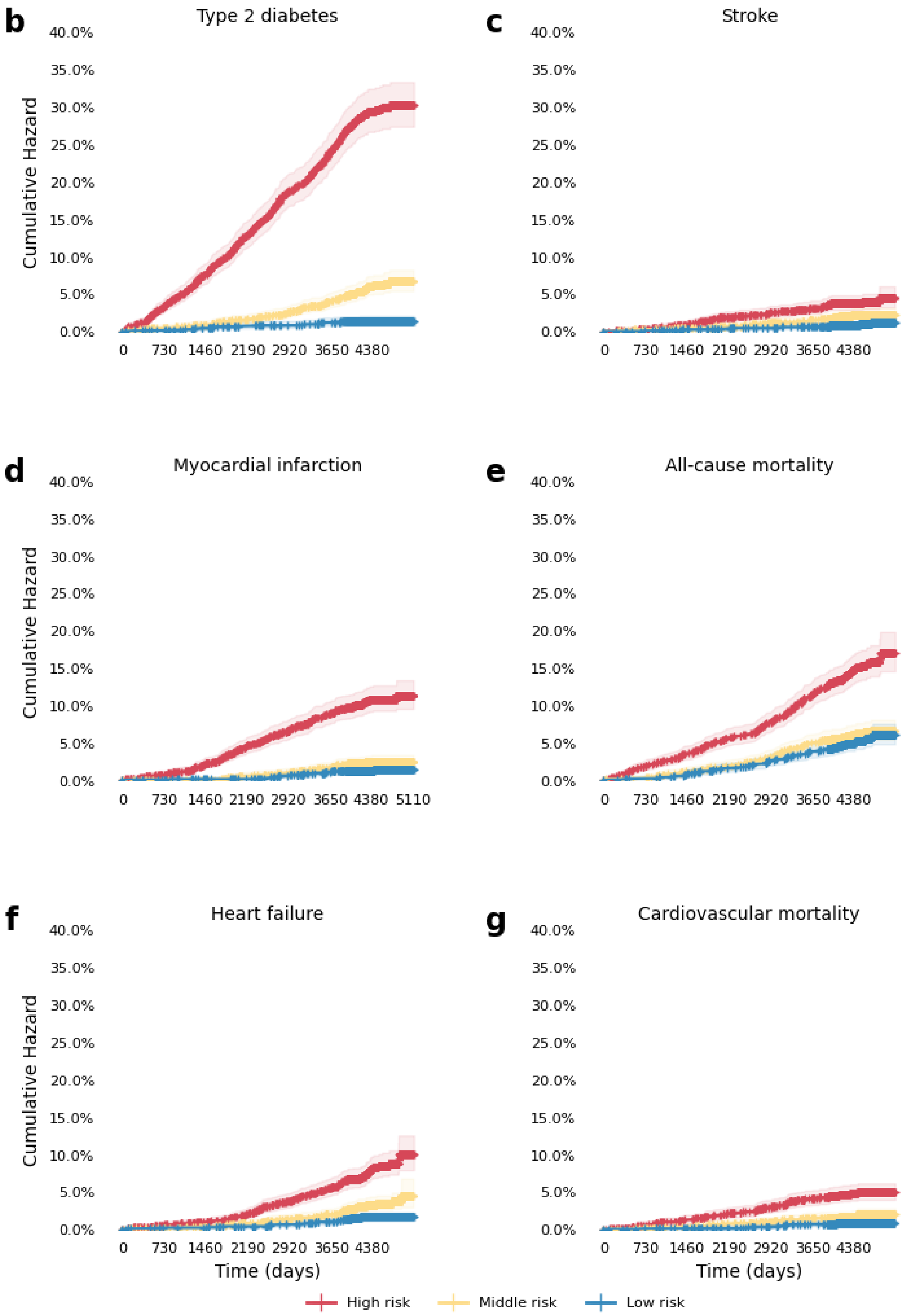
Supplementary Figure S5

Illustrations of uniform small scale (0–6%) of Fig.2 panel b–g.



Supplementary Figure S6

Illustrations of uniform large scale (0–30%) of Fig.2 panel b–g.



Comment #20. Table 2 should be a supplementary table.

Response: Thank you for your valuable suggestion. We have moved ***Table 2*** to the supplementary materials, now listed as ***Supplementary Table S14***.

Comment #21. need to adjust for multiple testing for incident CMD including the number of tests at the level of number of CMD tested.

Response: We greatly appreciate your suggestion. We have adjusted for multiple testing in all tables where appropriate using the Benjamini-Hochberg method. These changes have been implemented in ***Tables 1–3***, ***Supplementary Tables S2–S6***, ***S11–13***, ***S15***, ***S17–S18***, and ***S20–S26***.

Finally, please allow us to once again express our sincere gratitude for the valuable feedback you provided. Your insightful comments have been instrumental in refining our manuscript, and we deeply appreciate the time and effort you dedicated to reviewing our work. We have carefully addressed each of your specific comments, reviewed relevant prior literature, clarified methodologies, expanded discussions, and conducted a series of additional analyses including: (1) additionally adjusting for key confounders in all analyses; (2) incorporated more granular predictors in all baseline models; (3) conducted reclassification and decision analyses that reveals clinical relevance; and (4) integrated LC-MS metabolomics for broader metabolite coverage and expanded biological insights. All these findings have yielded robust results, further reinforcing our confidence in the findings presented in this manuscript. Additionally, we have reviewed the entire manuscript and made revisions based on your guidance, addressing the other minor suggestions you raised. We believe these improvements have enhanced the clarity, precision, and overall quality of the manuscript.

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Responds to the reviewers' comments

We would like to sincerely thank the Editors and Reviewers for their careful evaluation, thoughtful comments, and constructive suggestions, which have been invaluable in helping us improve the quality and clarity of our manuscript. We are truly grateful for the time and expertise each reviewer has devoted to our work.

We are especially thankful to **Reviewer #1** for the encouraging and supportive feedback. We are deeply honoured by your recognition that our revisions have adequately addressed your comments and that the manuscript has been substantially strengthened by the additional analyses.

We also extend our sincere gratitude to **Reviewer #2** for the rigorous and insightful feedback across both rounds of review. We are grateful that most concerns have been resolved in the previous revision. For your remaining comment regarding the interpretation of AUPRC, we believe there may be a potential misunderstanding, and we would like to humbly offer clarifications with the utmost respect.

Finally, we are deeply appreciative of **Reviewer #3** for the close and thoughtful reading of our manuscript. Your comments have prompted us to significantly refine both our conceptual framing and manuscript presentation. While a few concerns remain, we have made extensive revisions in response and provided detailed clarifications in the following point-by-point reply.

Below, we present our responses to each reviewer's comments. For ease of reference, the original comments are reproduced in **BLACK**, and our responses are highlighted in **BLUE**.

Responds to Reviewer #1

The authors have extensively revised the original submission and have adequately addressed the reviewer comments. The numerous supplementary tables provide critical analyses that strengthen their conclusions.

Response: We sincerely thank you for your generous and encouraging comments. We are truly grateful for the time and effort you devoted to reviewing our manuscript and for recognizing the extensive revisions we made. Your thoughtful feedback in the initial review round helped us greatly in improving the clarity, rigor, and overall quality of the work. It is a true honour to receive your support, and we deeply appreciate your contribution to the review process.

Responds to Reviewer #2

The authors have done a reasonable job addressing my comments. However, I am concerned that the new Supplementary Table S13 demonstrates very limited improvements compared to baseline. In fact, in 5 out of 6 diseases listed the baseline performance is within the confidence interval of the proposed models. Even if "significant" I am not sure how meaningful this is in the real world.

Response: We sincerely thank you for your encouraging and constructive comments. We are deeply grateful for the time and thought you invested in reviewing our work, and fully respect and appreciate your concern regarding the magnitude of the performance improvement originally reported in Supplementary Table S13. We would like to humbly offer some clarifications regarding the interpretation of the area under the precision-recall curve (AUPRC), as we believe there may be some important nuances that warrant mention.

First, unlike C-statistics, AUPRC has a dataset-specific baseline that depends directly on the prevalence of the positive class. As such, its absolute value can be misleading, especially when comparing across outcomes with varying base rates. **<u>Any interpretation of AUPRC values should therefore be made relative to</u>**

the outcome-specific baselines rather than on absolute magnitude alone. With this in mind, we respectfully point out that the performance improvements reported in the original Supplementary Table S13 are in fact substantial. Compared with the baseline models, the improvements reached 22.2% for stroke and 21.1% for CMD mortality. For type 2 diabetes and all-cause mortality, the gains were 11.0% and 8.8%, respectively. Even for heart failure and myocardial infarction, where gains were relatively smaller (6.8% and 6.0%), they remain meaningful given the low event rates and challenging prediction settings.

To incorporate your invaluable suggestions and contributions to the manuscript, we have now adopted AUPRC as the primary evaluation metric in the revised manuscript. We re-analysed all 30 model comparisons using AUPRC, updated ***Figure 3*** accordingly, and reported the full results in ***Supplementary Table S10*** (presented below for your kind reference). These results consistently demonstrate that incorporating RNFLT metabolic states led to marked and statistically significant improvements in predictive performance across all six CMD outcomes and across a broad range of benchmark models. Specifically, compared to minimal demographic models (Age & Sex), RNFLT metabolic states improved AUPRC by **21.6% to 117.2%** across different CMD outcomes. When benchmarked against more complex models, we observed improvements of **over 20%** for the SCORE2 models for type 2 diabetes (AUPRC% = 62.7%) and heart failure (AUPRC% = 24.8%), the WHO-CVD model for heart failure (AUPRC% = 24.8%), the FGCRS models for stroke (AUPRC% = 25.0%) and CMD mortality (AUPRC% = 21.4%), and the AHA/ASCVD models for stroke (AUPRC% = 24.9%) and CMD mortality (AUPRC% = 21.5%). Improvements of **over 10%** were also observed for the FGCRS, AHA/ASCVD and WHO-CVD models in the case of type 2 diabetes, myocardial infarction, stroke and CMD mortality.

Overall, the average improvements over all conventional models were:

- **43.8%** for type 2 diabetes,

- **16.8%** for myocardial infarction,
- **18.6%** for heart failure,
- **17.2%** for stroke,
- **10.6%** for all-cause mortality,
- **20.1%** for CMD mortality.

With utmost respect, we would also like to clarify a common but often misunderstood aspect of statistical inference related to confidence interval (CI) overlap. Visually comparing CIs is a rough heuristic appropriate only for **independent samples** and **normal distributions** (or at least approximate normality). In our case, however, AUPRCs before and after integration are calculated from the same dataset, and therefore, the two estimates are **NOT independent**. This constitutes a paired design, where inference should be based on the distribution of their difference rather than the marginal distributions of each estimate. In paired settings, the variance of the difference is not simply the sum of the individual variances but also depends on their covariance (joint distribution), which cannot be inferred by CI overlap alone. Therefore, **overlapping confidence intervals neither imply non-significance nor preclude a statistically meaningful difference**, unless the samples are independent (in which case the covariance term is null and can be ignored).

Moreover, the CI overlap heuristic relies on strict normality assumptions, which is also NOT valid for AUPRC. AUPRC is an **inherently bounded, skewed and non-normal metric**, particularly in highly imbalanced datasets such as ours (e.g. a ratio of 1:99 for stroke), where the skew is extreme. For these reasons, non-parametric paired bootstrapping is the appropriate and valid method for assessing the paired differences between the two AUPRCs. In this study, we employed 1,000 paired bootstrap resamples to estimate the sampling distribution of paired AUPRC differences between the baseline and integrated models. This approach directly estimates the sampling distribution of performance improvement while accounting for the covariance structure between the two models. All P values reported in ***Supplementary Table S10*** were obtained via this procedure and thus reflect the true significance of the observed performance differences, rather than heuristic inference based on individual CI overlap.

In conclusion, we respectfully emphasize that the performance improvements observed in our study are not only statistically valid but also clinically meaningful, **with most exceeding 10% and many surpassing 20%**. These improvements were consistently supported by robust, appropriate statistical methods, tailored for imbalanced datasets and paired comparisons. We truly hope that these clarifications are helpful in addressing your concerns, and we are deeply grateful for the opportunity to improve our manuscript based on your insightful comments. The key findings of this manuscript are summarised below for your perusal:

Key Findings

RNFL metabolic states

- mediate the connections between RNFL degeneration and CMD risk.
- improve risk prediction for incident CMD outcomes.

CMD outcomes	Effect mediated by RNFL metabolic states	Performance improved by RNFL metabolic states
Type 2 diabetes	63.7%	+ 43.8%
Myocardial infarction	44.7%	+ 16.8%
Heart failure	49.3%	+ 18.6%
Stroke	35.8%	+ 17.2%
All-cause mortality	17.0%	+ 10.6%
CMD mortality	38.8%	+ 20.1%

Supplementary Table S10

Incremental predictability of integrating RNFLT metabolic states for CMD outcome prediction beyond conventional models in the UKB population-II.

Outcomes *	AUPRC			Improvement ¶		P_{FDR} §
	No-skill baseline †	Integrated models ‡	Basic models	Absolute	Percent	
Type 2 diabetes						
Age&Sex	0.064	0.220 (0.199–0.241)	0.101 (0.093–0.110)	0.119	117.2%	<0.001
FGCRS	0.064	0.533 (0.503–0.562)	0.480 (0.448–0.512)	0.053	11.0%	<0.001
SCORE2	0.064	0.224 (0.203–0.246)	0.138 (0.125–0.150)	0.086	62.7%	<0.001
AHA/ASCVD	0.064	0.542 (0.511–0.570)	0.490 (0.457–0.520)	0.052	10.7%	<0.001
WHO-CVD	0.064	0.534 (0.504–0.562)	0.454 (0.423–0.483)	0.080	17.6%	<0.001
FRS for type 2 diabetes	0.064	0.623 (0.593–0.653)	0.568 (0.541–0.594)	0.055	9.7%	<0.001
Myocardial infarction						
Age&Sex	0.034	0.096 (0.089–0.105)	0.072 (0.068–0.076)	0.024	33.8%	<0.001
FGCRS	0.034	0.106 (0.099–0.115)	0.100 (0.093–0.108)	0.006	6.2%	<0.001
SCORE2	0.034	0.099 (0.092–0.107)	0.085 (0.079–0.092)	0.014	16.1%	<0.001
AHA/ASCVD	0.034	0.107 (0.099–0.115)	0.097 (0.090–0.104)	0.009	9.7%	<0.001
WHO-CVD	0.034	0.098 (0.091–0.106)	0.083 (0.077–0.089)	0.015	18.3%	<0.001
Heart failure						
Age&Sex	0.030	0.088 (0.081–0.096)	0.065 (0.061–0.069)	0.023	35.4%	<0.001
FGCRS	0.030	0.094 (0.087–0.102)	0.088 (0.081–0.095)	0.007	7.4%	<0.001
SCORE2	0.030	0.092 (0.085–0.100)	0.074 (0.068–0.079)	0.018	24.8%	<0.001
AHA/ASCVD	0.030	0.097 (0.090–0.106)	0.094 (0.086–0.103)	0.003	3.0%	0.028
WHO-CVD	0.030	0.100 (0.092–0.109)	0.082 (0.075–0.088)	0.018	22.2%	<0.001
Stroke						
Age&Sex	0.018	0.039 (0.035–0.042)	0.034 (0.032–0.037)	0.004	12.3%	0.006
FGCRS	0.018	0.044 (0.040–0.050)	0.036 (0.032–0.040)	0.009	25.0%	<0.001
SCORE2	0.018	0.040 (0.036–0.044)	0.035 (0.032–0.038)	0.005	14.7%	0.003
AHA/ASCVD	0.018	0.045 (0.040–0.050)	0.036 (0.033–0.039)	0.009	24.9%	<0.001
WHO-CVD	0.018	0.042 (0.039–0.047)	0.039 (0.035–0.042)	0.004	9.2%	0.018
All-cause mortality						
Age&Sex	0.074	0.171 (0.163–0.180)	0.141 (0.135–0.146)	0.030	21.6%	<0.001

FGCRS	0.074	0.185 (0.175–0.194)	0.170 (0.162–0.177)	0.015	8.8%	<0.001
SCORE2	0.074	0.181 (0.172–0.190)	0.166 (0.159–0.173)	0.015	9.1%	<0.001
AHA/ASCVD	0.074	0.186 (0.176–0.194)	0.171 (0.162–0.178)	0.015	8.8%	<0.001
WHO-CVD	0.074	0.183 (0.174–0.191)	0.175 (0.167–0.183)	0.008	4.6%	<0.001
CMD mortality						
Age&Sex	0.019	0.055 (0.049–0.061)	0.041 (0.038–0.045)	0.013	32.3%	<0.001
FGCRS	0.019	0.069 (0.062–0.078)	0.057 (0.051–0.063)	0.012	21.4%	<0.001
SCORE2	0.019	0.058 (0.052–0.065)	0.054 (0.049–0.061)	0.004	6.7%	0.010
AHA/ASCVD	0.019	0.070 (0.063–0.080)	0.058 (0.052–0.065)	0.012	21.5%	<0.001
WHO-CVD	0.019	0.063 (0.057–0.072)	0.054 (0.049–0.059)	0.010	18.6%	<0.001

* Basic models for CMD outcomes include minimal demographic characteristics (Age&Sex), Framingham General Cardiovascular Risk Score (FGCRS)1, Systematic Coronary Risk Evaluation 2 (SCORE2) algorithms2, American Heart Association/Atherosclerotic Cardiovascular Disease (AHA/ASCVD) algorithms3, and World Health Organization Cardiovascular Disease (WHO-CVD) risk charts4. Framingham risk scores for type 2 diabetes algorithms5-6 were additionally included for type 2 diabetes assessment.

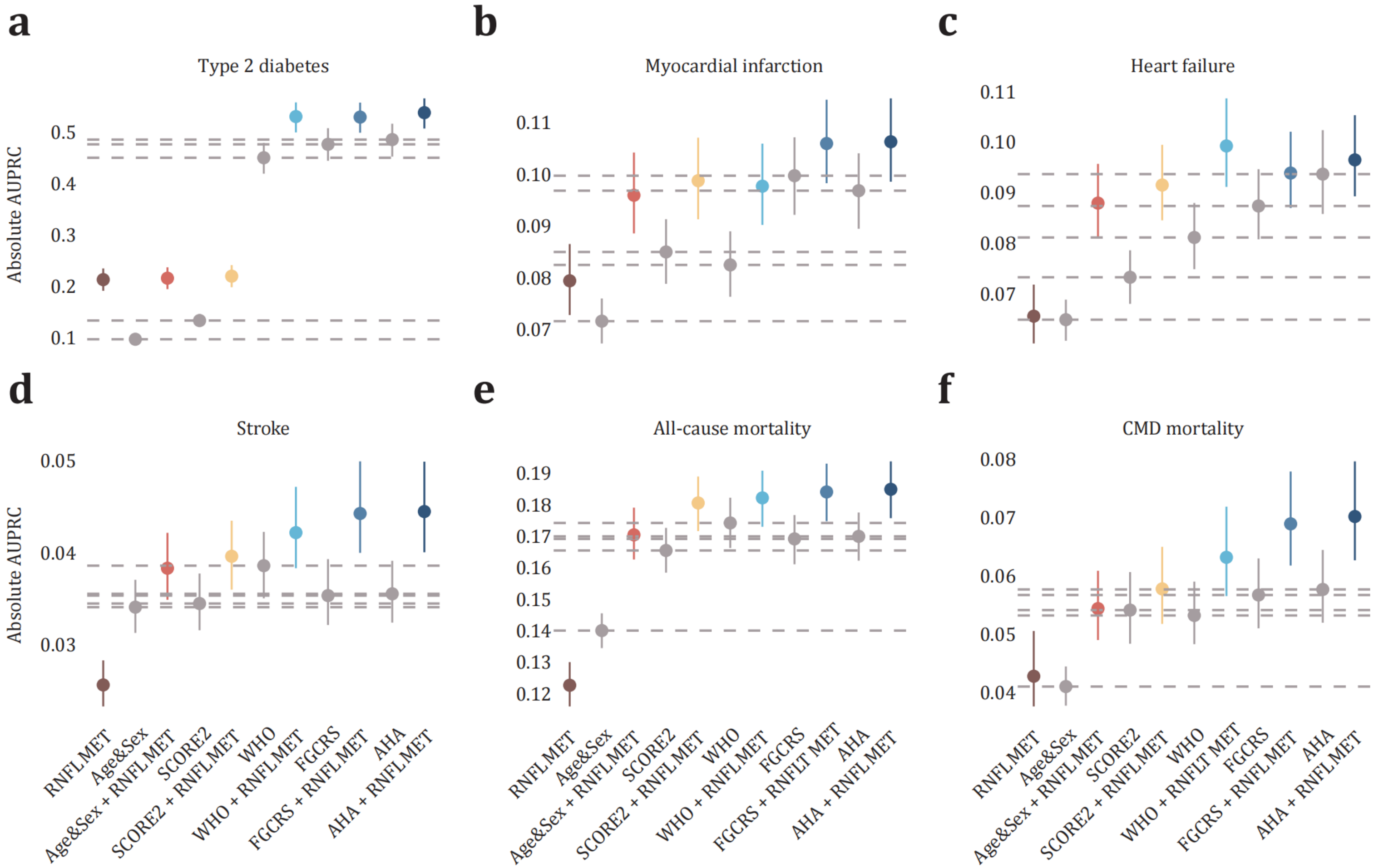
† The baseline (no-skill) AUPRC varies by outcome and corresponds to the event rates in the dataset (see Supplementary Table S7).

‡ Models that integrated RNFLT metabolic states.

¶ Improvement between integrated models and basic models. Interpretation of AUPRC values should be made relative to these outcome-specific baselines rather than on absolute magnitude alone.

§ P values were calculated through two-sided non-parametric bootstrap tests with 1,000 iterations followed by controlling FDR for multiple testing. Adjusted P were set identical when their step-up thresholds were truncated to a shared minimum.

Figure 3
Model performance in the UKB.



Responds to Reviewer #3

Comment #1. This manuscript details an analysis in UKB and the GDES cohort to evaluate association between metabolites (NMR Nightingale in UKB; more comprehensive MS metabolomics in GDES) and retinal nerve fiber layer thickness, and to determine if metabolites mediate the association between RNFL with incident cardiometabolic events. They find HDL parameters (more granular than composite HDL, as measured by NMR) are independently and incrementally associated with RNFL; mediate the association between RNFL with incident cardiometabolic diseases; and are themselves associated with incident CMD events. The authors attempted to address this reviewer's concerns but impact and interpretation concerns remain.

Response: We are sincerely grateful for your thoughtful and constructive comments. It is truly the dedication, rigor, and high standards of reviewers like you that uphold the excellence and integrity of research published in *Nature Communications*. We deeply appreciate your careful evaluation of our work and fully acknowledge your concerns regarding the interpretation and presentation of our findings. While some substantive issues may have been addressed in the previous revision, we humbly recognize that aspects of our manuscript, particularly the balance between primary and sensitivity analyses, the interpretation of AUPRC improvements, and the overall structural clarity, may have caused confusion. We sincerely apologize and have undertaken major revisions to the manuscript in response to each of your insightful comments.

In particular, we would like to respectfully clarify an important nuance regarding the interpretation of AUPRC that may not have been sufficiently emphasized in our prior response. Unlike C-statistics, the AUPRC has a dataset-specific baseline that directly depends on the prevalence of the positive class. Its absolute value can therefore be misleading, especially when comparing across outcomes with differing event rates. As such, all interpretations of AUPRC should be made relative to the outcome-specific baselines, rather than based on absolute magnitudes alone. We regret not explicitly presenting these relative improvements earlier, which may have led to the misunderstanding. To address this, and in line with Reviewer #2's helpful suggestion in the last round, we have now adopted AUPRC as the primary evaluation metric in the revised manuscript, given its robustness in highly imbalanced datasets. We have also re-visualized the main figures and tables and restructured relevant sections to more clearly highlight the relative improvements in predictive performance across outcomes and models.

Beyond this central clarification, we have made comprehensive, point-by-point revisions in response to your additional comments. We greatly admire your thoughtful attention to potential sources of confusion for readers, and we fully acknowledge that these stemmed from our own lack of clarity in presentation. With the utmost respect, we would be grateful for the opportunity to walk you through the specific revisions and clarifications we have made in response to your subsequent comments. The key findings of this manuscript are summarised below for your perusal:

Key Findings

RNFL metabolic states

- mediate the connections between RNFL degeneration and CMD risk.
- improve risk prediction for incident CMD outcomes.

CMD outcomes	Effect mediated by RNFL metabolic states	Performance improved by RNFL metabolic states
Type 2 diabetes	63.7%	+ 43.8%
Myocardial infarction	44.7%	+ 16.8%
Heart failure	49.3%	+ 18.6%
Stroke	35.8%	+ 17.2%
All-cause mortality	17.0%	+ 10.6%
CMD mortality	38.8%	+ 20.1%

Comment #2. I appreciate the careful inclusion of potential confounders across all the models, and that they continue to demonstrate statistically significant increases in c-statistic, newly added AUPRC, and in the mediation analyses. These appear to be well done, however, these results do not appear to be included as primary analyses but instead are buried as supplemental tables as sensitivity analyses. This will make it very confusing for the reader and as such, these fully adjusted models should be the primary analyses reported throughout.

Response: We are truly honoured and grateful for your recognition of the substantive improvements made in our previous revision. We fully acknowledge your concern that many of the newly added analyses, which are in fact central to our conclusions, were inappropriately presented as supplementary materials rather than as primary results.

To fully incorporate your valuable guidance and maximize its impact on the clarity and scientific rigor of the manuscript, we have now designated these more comprehensively adjusted models as the **primary analyses** throughout the revised manuscript, replacing the previously reported versions (***Page 7, Line 165***; ***Page 7, Line 186***; and ***Page 8, Line 198***). In addition, following Reviewer #2's helpful suggestion in the prior round, we have re-evaluated all model comparisons using AUPRC as the primary performance metric, given its robustness in imbalance datasets (***Page 9, Line 249***). We have also re-visualized and updated ***Figure 2***, ***Table 1***, and ***Table 2*** in the main text based on the revised primary analyses, and have correspondingly updated all relevant supplementary materials. We sincerely hope that these changes will improve clarity for readers and ensure the results are communicated in a more accessible and logically consistent way.

Comment #3. I continue to disagree with the interpretation of the robustness of the increase in c-statistics (and the newly included AUPRC)...while it may be true that they are comparable to other papers reporting new findings, that does not mean that they have clinical utility. The main increase is really around DM and not particularly robust for hard incident CVD outcomes. The addition of the NRI is, however, a major strength of the paper, but again highlight the results primarily around DM and not incident CVD. Further, the clinical models for incident DM discrimination should be compared to DM prediction models (not CVD prediction).

Response: We sincerely thank you for your thoughtful and constructive comments. We deeply respect your concern regarding the observed improvements in model performance. We would like to humbly offer some clarifications regarding the AUPRC interpretation, which may help address this concern.

First, we respectfully note that the absolute magnitude of AUPRC values can be misleading, as AUPRC has a dataset-specific baseline determined by the prevalence of the positive class. Therefore, **all interpretations of AUPRC should be made relative to outcome-specific baselines, rather than based on absolute magnitude alone**. In our previous revision, we only reported absolute AUPRCs without presenting the corresponding relative improvements, which may have caused the misunderstanding. In fact, when interpreted appropriately, the predictive gains are both substantial and not limited to T2D. For example, the relative improvement in stroke and CMD mortality exceeded 20%, both of which surpassed that for T2D. Although the absolute AUPRC for T2D may appear high, this reflects the high event rate rather than superior model performance. In response to Reviewer #2’s helpful suggestion, we have now adopted AUPRC as the primary evaluation metric for assessing model performance improvements, given its robustness in highly imbalanced datasets. We have now re-analysed all 30 model comparisons using this metric, re-visualized ***Figure 3*** based on AUPRC, and compiled the detailed results in ***Supplementary Table S10*** (presented below for your kind reference).

Our findings demonstrate that integrating RNFLT metabolic states led to considerable and significant improvements across all six CMD outcomes and across a broad range of benchmark models. Even when excluding T2D, the addition of RNFLT metabolic states improved AUPRC for all "hard" CVD outcomes by **12.3% to 35.4%** relative to the Age & Sex model. When benchmarked against more complex models, we observed improvements of **more than 20%** for heart failure over the SCORE2 model (AUPRC% = 24.8%), stroke over the FGCRS model (AUPRC% = 25.0%) and AHA/ASCVD model (AUPRC% = 24.9%), and CMD mortality over the FGCRS model (AUPRC% = 21.4%) and AHA/ASCVD model (AUPRC% = 21.5%). Improvements of **more than 10%** were also observed for myocardial infarction over the SCORE2 model (AUPRC% = 16.1%) and WHO-CVD model (AUPRC% = 18.3%), stroke over the SCORE2 model (AUPRC% = 14.7%), and CMD mortality over the WHO-CVD model (AUPRC% = 18.6%).

Overall, the average improvements over conventional models for hard CVD outcomes were:

- **16.8%** for myocardial infarction,
- **18.6%** for heart failure,
- **17.2%** for stroke,
- **10.6%** for all-cause mortality,
- **20.1%** for CMD mortality.

These results are now clearly described in the main text (***Page 9, Line 249***) and visually summarized in the updated ***Figure 3*** and ***Supplementary Table S10***.

To address your insightful comment regarding benchmarking for T2D, we have also compared our model against an established diabetes-specific prediction model (FRS for T2D). Though the most substantial improvement seems to be observed for T2D over CVD models (117.2% improvement over Age&Sex model and an average improvement of 43.8% over all CVD models), the improvement became comparable to that observed for hard CVD outcomes when benchmarking against the T2D model (ΔAUPRC% = 9.7%). This result has been described in the main text (***Page 10, Line 257***).

In summary, integrating RNFLT metabolic states resulted in meaningful predictive improvements across all CMD outcomes, including both T2D and hard incident CVD endpoints, **with most exceeding 10% and many surpassing 20%**. While the relative gain for T2D appeared larger when compared to CVD models, direct benchmarking against a T2D-specific model demonstrated comparable performance to that of hard CVD outcomes. As noted in our previous response, the magnitude of improvement is consistent with or greater than what has been reported in other landmark studies published in top-tier journals, such as *Nature Medicine* and *Nature Aging*.[1-4] To avoid confusion for readers, we now present these results as primary analyses and retain the C-statistics as a complementary metric to facilitate intuitive interpretability and cross-outcome comparison given differing event rates. We sincerely hope that these clarifications and updates address your concerns, and we are deeply grateful for your invaluable feedback.

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3. Liu, Y. et al. Integration of polygenic and gut metagenomic risk prediction for common diseases. *Nat Aging*, (2024).
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Supplementary Table S10

Incremental predictability of integrating RNFLT metabolic states for CMD outcome prediction beyond conventional models in the UKB population-II.

Outcomes *	AUPRC			Improvement ¶		P_{FDR} §
	No-skill baseline †	Integrated models ‡	Basic models	Absolute	Percent	
Type 2 diabetes						
Age&Sex	0.064	0.220 (0.199–0.241)	0.101 (0.093–0.110)	0.119	117.2%	<0.001
FGCRS	0.064	0.533 (0.503–0.562)	0.480 (0.448–0.512)	0.053	11.0%	<0.001
SCORE2	0.064	0.224 (0.203–0.246)	0.138 (0.125–0.150)	0.086	62.7%	<0.001
AHA/ASCVD	0.064	0.542 (0.511–0.570)	0.490 (0.457–0.520)	0.052	10.7%	<0.001
WHO-CVD	0.064	0.534 (0.504–0.562)	0.454 (0.423–0.483)	0.080	17.6%	<0.001
FRS for type 2 diabetes	0.064	0.623 (0.593–0.653)	0.568 (0.541–0.594)	0.055	9.7%	<0.001
Myocardial infarction						
Age&Sex	0.034	0.096 (0.089–0.105)	0.072 (0.068–0.076)	0.024	33.8%	<0.001
FGCRS	0.034	0.106 (0.099–0.115)	0.100 (0.093–0.108)	0.006	6.2%	<0.001
SCORE2	0.034	0.099 (0.092–0.107)	0.085 (0.079–0.092)	0.014	16.1%	<0.001
AHA/ASCVD	0.034	0.107 (0.099–0.115)	0.097 (0.090–0.104)	0.009	9.7%	<0.001
WHO-CVD	0.034	0.098 (0.091–0.106)	0.083 (0.077–0.089)	0.015	18.3%	<0.001
Heart failure						
Age&Sex	0.030	0.088 (0.081–0.096)	0.065 (0.061–0.069)	0.023	35.4%	<0.001
FGCRS	0.030	0.094 (0.087–0.102)	0.088 (0.081–0.095)	0.007	7.4%	<0.001
SCORE2	0.030	0.092 (0.085–0.100)	0.074 (0.068–0.079)	0.018	24.8%	<0.001
AHA/ASCVD	0.030	0.097 (0.090–0.106)	0.094 (0.086–0.103)	0.003	3.0%	0.028
WHO-CVD	0.030	0.100 (0.092–0.109)	0.082 (0.075–0.088)	0.018	22.2%	<0.001
Stroke						
Age&Sex	0.018	0.039 (0.035–0.042)	0.034 (0.032–0.037)	0.004	12.3%	0.006
FGCRS	0.018	0.044 (0.040–0.050)	0.036 (0.032–0.040)	0.009	25.0%	<0.001
SCORE2	0.018	0.040 (0.036–0.044)	0.035 (0.032–0.038)	0.005	14.7%	0.003
AHA/ASCVD	0.018	0.045 (0.040–0.050)	0.036 (0.033–0.039)	0.009	24.9%	<0.001
WHO-CVD	0.018	0.042 (0.039–0.047)	0.039 (0.035–0.042)	0.004	9.2%	0.018
All-cause mortality						
Age&Sex	0.074	0.171 (0.163–0.180)	0.141 (0.135–0.146)	0.030	21.6%	<0.001

FGCRS	0.074	0.185 (0.175–0.194)	0.170 (0.162–0.177)	0.015	8.8%	<0.001
SCORE2	0.074	0.181 (0.172–0.190)	0.166 (0.159–0.173)	0.015	9.1%	<0.001
AHA/ASCVD	0.074	0.186 (0.176–0.194)	0.171 (0.162–0.178)	0.015	8.8%	<0.001
WHO-CVD	0.074	0.183 (0.174–0.191)	0.175 (0.167–0.183)	0.008	4.6%	<0.001
CMD mortality						
Age&Sex	0.019	0.055 (0.049–0.061)	0.041 (0.038–0.045)	0.013	32.3%	<0.001
FGCRS	0.019	0.069 (0.062–0.078)	0.057 (0.051–0.063)	0.012	21.4%	<0.001
SCORE2	0.019	0.058 (0.052–0.065)	0.054 (0.049–0.061)	0.004	6.7%	0.010
AHA/ASCVD	0.019	0.070 (0.063–0.080)	0.058 (0.052–0.065)	0.012	21.5%	<0.001
WHO-CVD	0.019	0.063 (0.057–0.072)	0.054 (0.049–0.059)	0.010	18.6%	<0.001

* Basic models for CMD outcomes include minimal demographic characteristics (Age&Sex), Framingham General Cardiovascular Risk Score (FGCRS)1, Systematic Coronary Risk Evaluation 2 (SCORE2) algorithms2, American Heart Association/Atherosclerotic Cardiovascular Disease (AHA/ASCVD) algorithms3, and World Health Organization Cardiovascular Disease (WHO-CVD) risk charts4. Framingham risk scores for type 2 diabetes algorithms5-6 were additionally included for type 2 diabetes assessment.

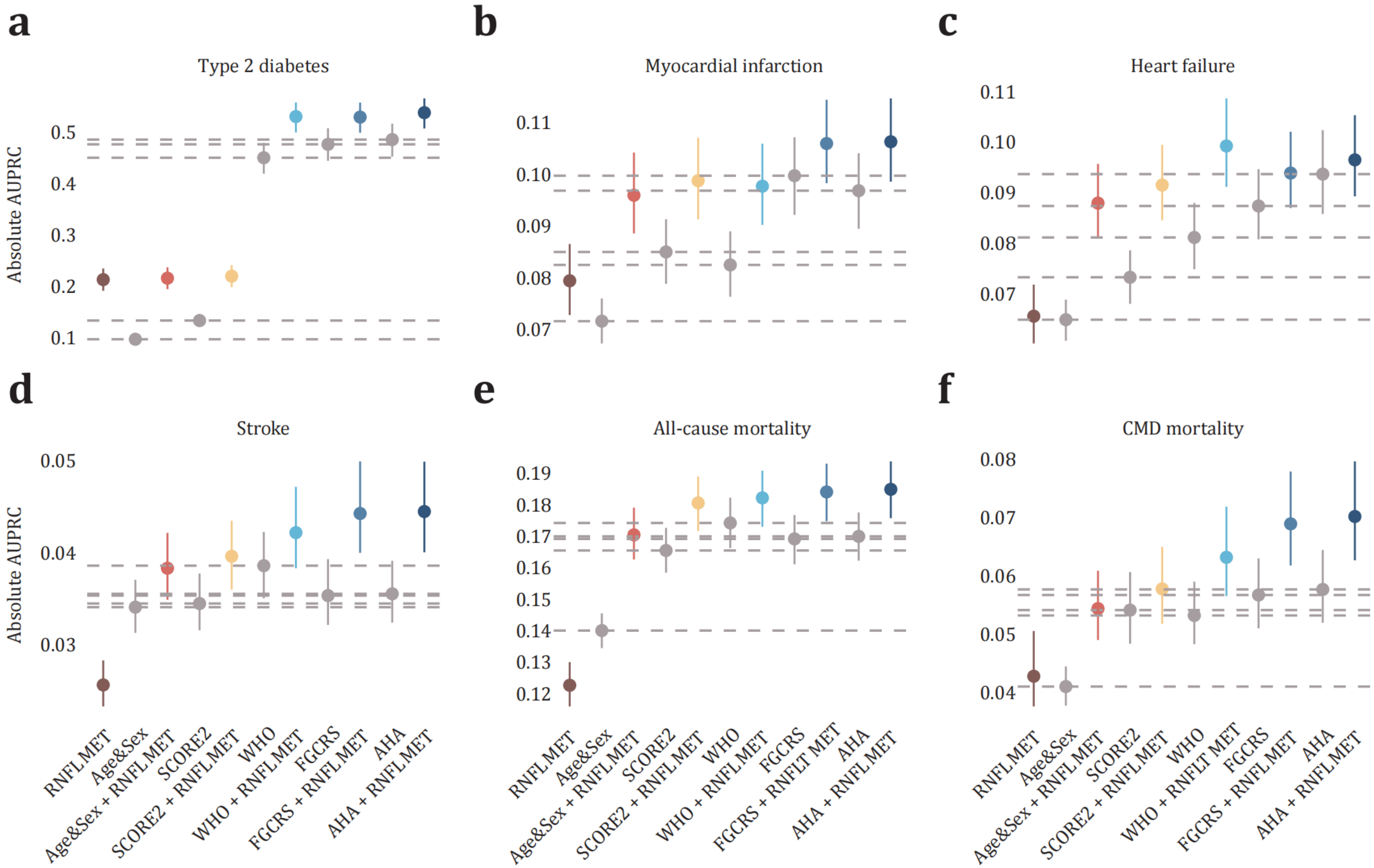
† The baseline (no-skill) AUPRC varies by outcome and corresponds to the event rates in the dataset (see Supplementary Table S7).

‡ Models that integrated RNFLT metabolic states.

¶ Improvement between integrated models and basic models. Interpretation of AUPRC values should be made relative to these outcome-specific baselines rather than on absolute magnitude alone.

§ P values were calculated through two-sided non-parametric bootstrap tests with 1,000 iterations followed by controlling FDR for multiple testing. Adjusted P were set identical when their step-up thresholds were truncated to a shared minimum.

Figure 3
Model performance in the UKB.



Comment #4. Relatedly, the authors argue that the paper is about metabolic mechanisms linking RNFL and CMD but most of the paper is devoted to assessing clinical utility, and not really digging into these mechanisms (in fact the abstract doesn't even mention that the metabolite states are reporting primarily on HDL subclasses).

Response: We sincerely thank you for this important and insightful comment. It is with great regret that we acknowledge the confusion caused by the structure and layout of the original manuscript, and we are truly grateful for the opportunity to clarify and improve the presentation of our study. In response to your concerns, we have made substantial revisions throughout the manuscript to better articulate the mechanistic framework of our study. These include: (1) explicitly highlighting the metabolic findings in the ***Abstract***, (2) clarifying the study design in the ***Introduction***, (3) elaborating on the mediation and enrichment analyses in the ***Results***, and (4) expanded the mechanistic interpretation in the ***Discussion***. We respectfully request a moment of your time to clarify the study's conceptual framework and the specific revisions we have made.

The central hypothesis of this study is that **a specific metabolic basis may underlie the connection between RNFL thinning and CMD risk, and that characterizing these intrinsic metabolic states may provide both biological insight and early risk stratification across CMD outcomes**. Our study was thus structured around a three-step logic: (1) systematically identify metabolites associated with RNFL thickness; (2) validate that these metabolites are also linked to incident CMD outcomes and evaluate their mediating role in the RNFL–CMD connections; (3) construct RNFL metabolic states based on these metabolites and assess their predictive value across diverse CMD endpoints.

This metabolic discovery and mediation framework aims to provide a potential molecular explanation for the observed long-range systemic connection between a retinal phenotype and CMD risk. Elucidating the molecular mediators of this link is crucial not only for establishing RNFL as a biologically valid biomarker but also for revealing early systemic perturbations that may be actionable. Indeed, if these RNFL-linked metabolites truly reflect mechanistically relevant, early-stage biological changes (rather than statistical artifacts or confounding), then they should also identify individuals at elevated future risk, particularly within populations where traditional risk factors are known to underperform (e.g., women, individuals of low socioeconomic status, or those with limited educational attainment).

In this context, **the predictive modelling serves not as a standalone objective but as a functional validation of the mechanistic hypothesis**. Beyond establishing predictive utility across CMD outcomes, it tests whether the identified RNFL metabolic state meaningfully captures early pathophysiological processes. As expected, the metabolic findings not only mediate the RNFL–CMD associations, but also provide independent predictive value that complements conventional models, particularly in socially disadvantaged populations. This suggests that the RNFL metabolic state may capture mechanistic relevant processes not reflected in traditional risk factors. Nonetheless, we fully acknowledge your concern that our previous version placed disproportionate emphasis on model performance. To address this, we have revised the manuscript to ensure a more balanced presentation. In particular, we have moved the subgroup stratification analyses by social determinants of health (SDoH) to the ***Supplementary Materials***.

We have also expanded the mediation analysis in the ***Results*** section and directly compared the mediating effects of RNFL-associated metabolites to those of traditional risk factors (***Page 8, Line 198***). These findings are summarized in the updated ***Table 2*** (presented below for your kind reference). Specifically, after accounting for comprehensive set of socioeconomic, lifestyle, and clinical risk factors, most associations between RNFLT and CMD outcomes remained significant, suggesting that these factors explained only a proportion of the observed RNFL–CMD link (e.g., 19.4% for myocardial infarction and 7.7% for all-cause mortality). In contrast, when RNFL-associated metabolites were included in the models, the direct associations between RNFL and CMD outcomes were substantially attenuated toward the null. This change suggested that these metabolic biomarkers accounted for a major proportion of the relationship between RNFL and CMD risk. **The estimated proportions were 63.7% for T2D, 44.7% for myocardial infarction, 49.3% for heart failure, 35.8% for stroke, 17.0% for all-cause mortality, and 38.8% for CMD mortality.**

To further strengthen the biological interpretability of our findings, we conducted a pathway enrichment analysis and expanded the ***Discussion*** on implicated pathways. Several biologically relevant pathways were identified, such as aminoacyl-tRNA biosynthesis, arginine and phenylalanine metabolism, glutathione metabolism, as well as nicotinate and pantothenate metabolism. Phenylalanine and arginine metabolism are among the top pathways enriched. Confirmed in apoA-I knockout mice,[1] disruptions in phenylalanine metabolism compromises apolipoprotein expression,[2] while arginine modulates endothelial function and supports HDL-mediated anti-inflammation and cholesterol transport.[3] Glutathione maintains HDL stability by supporting HDL-associated enzymes and the glyoxalase system.[4] Its depletion has been implicated in HDL structural contraction and particle shrinkage,[5] consistent with our NMR findings on shifts in HDL subclasses. In addition, enrichment of linoleic acid and vitamin metabolism indicates broader disturbances in membrane lipid composition and coenzyme biosynthesis, both crucial for HDL integrity and phospholipid exchange. These findings provide complementary biochemical insights that converge on HDL remodelling, positioning RNFL as a downstream reflection of systemic perturbations in early CMD pathogenesis (***Page 16, Line 446***).

In summary, **the mediation analysis and predictive modelling in this study represent two interconnected components of the same conceptual framework**. The incremental predictive value of RNFL metabolic states, particularly among individuals underrepresented or underserved by traditional models, supports their role as early, mechanistically relevant indicators of CMD risk. We have now made extensive revisions across the manuscript to more clearly reflect this conceptual framework and sincerely hope these changes address your concerns.

References

1. Liu, J. et al. LC-MS-Based Metabolomics and Lipidomics Study of High-Density-Lipoprotein-Modulated Glucose Metabolism with an apoA-I Knockout Mouse Model. *J. Proteome Res.* **18**, 48-56 (2019).
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Table 2

The mediation effect of traditional risk factor and RNFLT-associated metabolites on the association between RNFLT and CMD risks.

Outcomes *	**Base models †**		**Base + traditional risk factors ‡**			**Base + RNFLT metabolites**		
	HR (95% CI)	***P$_{adj}$* §**	**HR (95% CI)**	***P$_{adj}$* §**	**% ¶**	**HR (95% CI)**	***P$_{adj}$* §**	**% ¶**
Type 2 diabetes	0.85 (0.81 to 0.89)	**2.0×10^{-10}**	0.92 (0.87 to 0.97)	**0.012**	49.2	0.94 (0.83 to 1.06)	0.879	63.7
Myocardial infarction	0.83 (0.77 to 0.89)	**1.3×10^{-6}**	0.86 (0.80 to 0.93)	**0.001**	19.4	0.90 (0.77 to 1.06)	0.879	44.7
Heart failure	0.70 (0.63 to 0.77)	**1.5×10^{-12}**	0.74 (0.61 to 0.90)	**0.012**	16.4	0.83 (0.70 to 0.99)	0.221	49.3
Stroke	0.89 (0.80 to 0.98)	**0.022**	0.91 (0.82 to 1.01)	0.074	21.6	0.93 (0.75 to 1.15)	0.879	35.8
All-cause mortality	0.87 (0.82 to 0.92)	**1.1×10^{-5}**	0.88 (0.78 to 0.98)	0.051	7.7	0.89 (0.78 to 1.01)	0.346	17.0
CMD mortality	0.79 (0.71 to 0.89)	**2.5×10^{-4}**	0.83 (0.73 to 0.95)	**0.024**	20.4	0.87 (0.68 to 1.12)	0.879	38.8

* Data are presented as per-SD change in RNFLT (4.24 μm).

† Base model was adjusted for age, sex, ethnicity, and assessment centre.

‡ Traditional risk factors include household income, Townsend deprivation, education attainment, smoking, alcohol, body-mass index, use of lipid-lowering medications, enzymatic assay-based HDL cholesterol, systolic blood pressure, and HbA1c level.

§ Adjusted P were calculated through two-sided Wald tests followed by controlling FDR for multiple testing. Bold indicates significance. Adjusted P were set identical when their step-up thresholds were truncated to a shared minimum.

¶ Percent change in HR coefficient. The proportion mediated by traditional risk factors and RNFL-associated metabolites may sum up to over 100% as these two proportions were calculated independently in mutually adjusted models.

Comment #5. Thank you for clarifying the difference between this manuscript and your prior manuscript published in the Br J. Ophtho, with the primary difference between using OCT techniques in this paper. Having "split" the ocular imaging analyses into separate papers also reduces impact of this paper, and further, there is no comparison of the two papers to provide context to the reader.

Response: We deeply admire your broad familiarity with the literature and sincerely thank you for this thoughtful and constructive comment. It is truly an honour to receive such careful and expert guidance. In our previous revision, we included a detailed comparison between this manuscript and our prior work published in *BJO* (2024)[1] in the supplementary materials. However, we humbly acknowledge that this content may have been overshadowed by the extensive supplemental analyses, and we sincerely apologize for the resulting lack of clarity. We have now revised the main text and incorporated a concise yet informative comparison in both the ***Introduction*** and ***Discussion***, referencing our related work in *BJO* (2024)[1] and *Nature Communications* (2023)[2]. While the difference in imaging modality is indeed one key distinction, we would be grateful for the opportunity to clarify in greater depth the conceptual differences between these studies.

In our earlier work, **we introduced the concept of "oculomics" to systematically explore the relationship between distinct ocular phenotypes and systemic health**[3]—a concept that has since inspired the emergence of a new integrative field. However, despite the surge of studies correlating ocular phenotypes with systemic conditions within this framework, the biological basis underpinning these associations remain underexplored. Elucidating the molecular mediators of this link is crucial not only for establishing ocular phenotypes as valid biomarkers but also for revealing early systemic perturbations that may be actionable. To this end, we have conducted several studies to explore the metabolic underpinnings of retinal phenotypes associated with systemic diseases, each focused on a different theme and yielded domain-specific discoveries. Our previous work published in *Nature Communications* (2023)[2] identified **retinal pigment epithelium (RPE) thinning** as an early biomarker of diabetes and its complications, and uncovered metabolic pathways linking RPE structure to nephropathy and retinopathy. Building on this framework, our subsequent study in *BJO* (2024)[1] focused on **retinal aging** and its metabolic associations with systemic age-related diseases (ARDs).

We respectfully clarify the connections and differences between the *BJO* (2024) study and the current manuscript. The *BJO* study addressed a completely distinct biological and clinical domain—**gerontology**. In that work, we proposed and developed **a deep learning-based retinal aging clock** from fundus photographs, using the difference between predicted and chronological age (commonly termed the retinal age gap, or RAG) to infer accelerated ocular aging. We had previously shown RAG to correlate with aging-related biomarkers[4,5] and multiple systemic diseases, including Parkinson's disease,[6] kidney failure[7], stroke[8], obesity[9], multimorbidity[10], and mortality[11]. Building on these findings, we applied our metabolomics framework to this aging signal and identified metabolites linking accelerated retinal aging to multi-organ ARDs. By contrast, the current study extends our framework to a distinct retinal anatomical structure—**axons of the retinal ganglion cells**—and to a different disease domain—**cardiometabolic medicine**.

We also respectfully emphasize the fundamental differences between the ocular phenotypes and mechanistic signals studied in the two papers. The *BJO* study utilized **fundus photographs**, capturing **macroscopic visual patterns** such as vessel texture, pigmentation, and macular reflex. The deep learning model extracted complex, non-localized signals influenced by a combination of these factors. In addition, minor media opacities—such as lens or vitreous changes—significantly affect deep learning predictions, as they are tightly associated with aging. The resulting RAG was a non-structural, time-based physiological signal, serving as a proxy for aging. In contrast, RNFL thickness is a **well-defined, quantitative anatomical measurement** obtained via **OCT**, representing the axons of retinal ganglion cells—an extension of the central nervous system. RNFL thinning reflects neurodegeneration, axonal loss, metabolic dysregulation, and microvascular insufficiency. While RAG reflects a **global aging-driven signal**, RNFL maps to a **metabolic–vascular dysfunction–driven signal** more directly linked to cardiometabolic pathology. Although these pathways may partially intersect—given the interplay between aging and metabolism—they are mechanistically distinct and were studied under separate domains and hypotheses.

To summarize, our prior work published in *BJO* (2024) focused on aging biology, investigating how ocular aging correlates with multi-organ ARDs. The phenotype studied was a temporal signal derived from macroscopic image-based features, and the study operated under an aging-driven hypothesis. While cardiovascular disease was included among the outcomes, it was interpreted within the framework of systemic aging. In contrast, the current manuscript provides a biologically interpretable, anatomically derived biomarker relevant to early cardiometabolic dysfunction. The ocular phenotype is a microstructural measurement (RNFL thickness), the outcomes are focused exclusively on CMD, and the underlying hypothesis centres on metabolic–vascular pathway disruption. **The two studies therefore differ across multiple dimensions:**

- **Ocular phenotype**: macroscopic, deep learning–based RAG vs. microscopic, anatomically defined RNFL thickness
- **Outcome domain**: multi-organ ARDs vs. CMDs
- **Biological basis**: global aging vs. metabolic–vascular dysfunction
- **Imaging modality**: fundus photograph vs. OCT
- **Public health focus**: aging research vs. cardiometabolic medicine

We believe that this diversity of inquiry, driven by distinct biological hypotheses applied to distinct ocular signals, is precisely what lends the oculomics framework its translational relevance and potential for cross-disciplinary discovery.[3] In the revised manuscript, we have now summarized and cited both prior studies in the ***Introduction*** and ***Discussion***, articulating how the current work conceptually builds upon, yet is mechanistically and clinically distinct from, the earlier ones. We sincerely hope these clarifications and revisions address your concerns, and we are once again grateful for your insightful feedback.

All in all, we are sincerely grateful for your rigorous, insightful, and constructive feedback throughout the entire review process. Your comments reflect deep expertise in both cardiometabolic epidemiology and biomarker evaluation, and have greatly enhanced the clarity, rigor, and conceptual framing of our study. It has truly been an honour to learn from your thoughtful critiques. Thanks to your guidance, the manuscript has been substantially improved in clarity, coherence, and scientific robustness. We would like to once again express our heartfelt gratitude for your time, insight, and dedication to advancing high-quality research, and we sincerely hope that the current version has addressed your concerns to your satisfaction.

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Responds to the reviewer's comments:

We would like to express our sincere gratitude to the Editor and the Reviewers for their careful evaluation of our revised manuscript. We deeply appreciate the time, effort, and insightful feedback provided throughout the review process, which have greatly enhanced the clarity, rigor, and overall quality of our work. We are delighted that Reviewer #1 and Reviewer #2 found the revision satisfactory and had no further concerns. We also sincerely thank Reviewer #3 for the remaining constructive guidance.

Replies to Reviewer #2

Comment: The authors have adequately responded to my comments. Refocusing the manuscript on AUPRC as the primary metric has made this a stronger article.

Response: We sincerely thank the reviewer for the positive evaluation and for recognizing the strengthened focus of the manuscript. We are deeply grateful for your insightful comments and constructive suggestions throughout the review process, which have greatly improved the clarity and rigor of our work. We are delighted that the revised version has met your expectations.

Replies to Reviewer #3

Comment: The authors have addressed many of the previous concerns including having fully adjusted models as part of the primary manuscript, adding more mechanistic insights including pathway analyses, and clarifying differences between this and their prior manuscripts. However, throughout, for at least two reviewers, they double down on the importance of the clinical interpretation of the increase in c-statistic and AUPRC, noting the magnitude of the increase the AUPRC. The addition of the FRS-DM score as the clinical model was vital (not sure why any CVD model was used for T2DM at all), but while they state that that was a good thing because it led to a comparable effect to CVD, it actually significantly decreased the % change in AUPRC to only 9%. To further frame the clinical relevance of the change in models (which really is what the results are focused on despite the argument that this was a mechanistic manuscript), the authors should at the minimum provide the NRI data as primary tables, make sure they include the FRS-DM score as a model for comparison in the NRI (currently not included in Supplement 15), and place these results in careful context with a softening of the interpretations throughout about clinical utility (the NRIs are modest at best).

Response: We would like to express our sincere gratitude for your thoughtful and constructive feedback, and we are delighted that you found the revisions addressing previous concerns substantially improved the manuscript. We also appreciate the time and effort you dedicated to reviewing our work.

In response to your current guidance, we have now moved the NRI results from the Supplementary Information to the main manuscript as a primary table (now Table 3). We have also ensured that the FRS-T2D score is included as a comparison model. In addition, the interpretations have been softened to emphasize the moderate yet consistent improvements across all outcomes.

Once again, we sincerely thank you for your insightful comments and kind recognition, which have greatly helped us refine both the analytical presentation and interpretative framing of our study.